HAM RADIO TECHNICIAN

VISUAL CRAM MAP 2026–2030

A diagram-first guide to the FCC Element 2 concepts, rules, and calculations beginners mix up

NORTHFIELD SIGNAL GUIDES

Independent study aid

This book is not affiliated with or endorsed by the Federal Communications Commission, NCVEC, ARRL, any Volunteer Examiner Coordinator, or any examination provider.

The official NCVEC question pool controls question wording and accepted answers. FCC Part 97 controls the rules. This edition uses the corrected pool released February 19, 2026, effective July 1, 2026 through June 30, 2030. Check NCVEC for later errata or withdrawals.

No pass, score, or study-time result is promised. This is an exam companion, not an operating handbook, engineering reference, or substitute for current rules and safe local practice.

How the book works

LEARN → PRACTISE → DIAGNOSE → RETURN

THE WORKING RULE

Learn one reusable model here. Practise with a free randomized tool. When you miss an item, use its official ID to return to the model. This book deliberately does not imitate an exam simulator.

Keep the task narrow: identify the relationship, choose the safe or lawful action, and connect the item to its official ID.

Read an ID

T5D06 is an address

THE WORKING RULE

T5 identifies electrical principles. D identifies the group. 06 identifies the question. The address remains stable inside this pool cycle and makes every concept traceable.

Keep the task narrow: identify the relationship, choose the safe or lawful action, and connect the item to its official ID.

What the exam samples

35 questions from 10 subelements

THE WORKING RULE

The exam uses a fixed blueprint across rules, procedures, propagation, practices, electrical principles, components, circuits, signals, antennas, and safety. You need 26 correct answers to pass.

Keep the task narrow: identify the relationship, choose the safe or lawful action, and connect the item to its official ID.

Contents

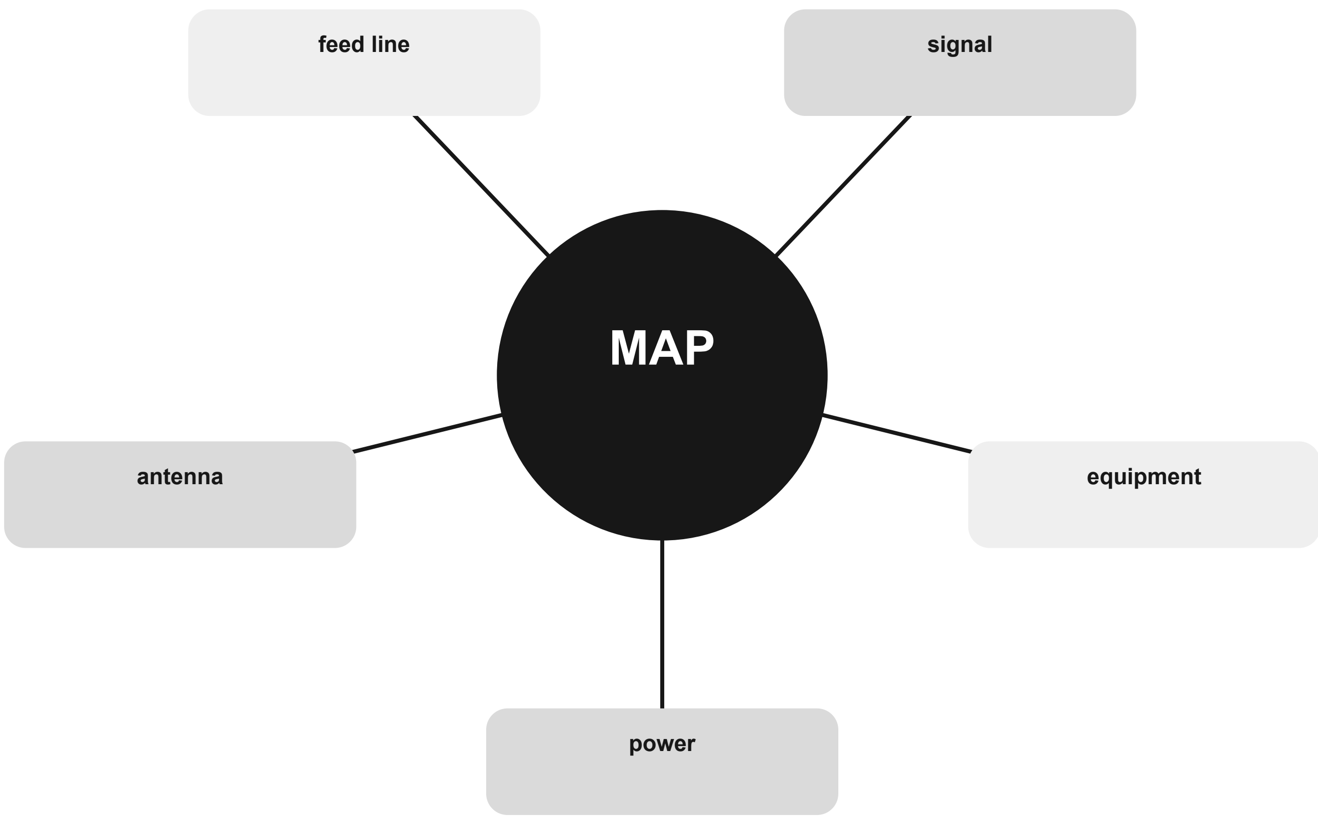
Chapters and reference

Keep the task narrow: identify the relationship, choose the safe or lawful action, and connect the item to its official ID.

The whole system

One station, many views

Electrical power drives equipment. Equipment creates and receives signals. Feed lines and antennas connect the station to a propagation path. Operating rules and safety constrain every step.



Before exam day

Refresh and practise

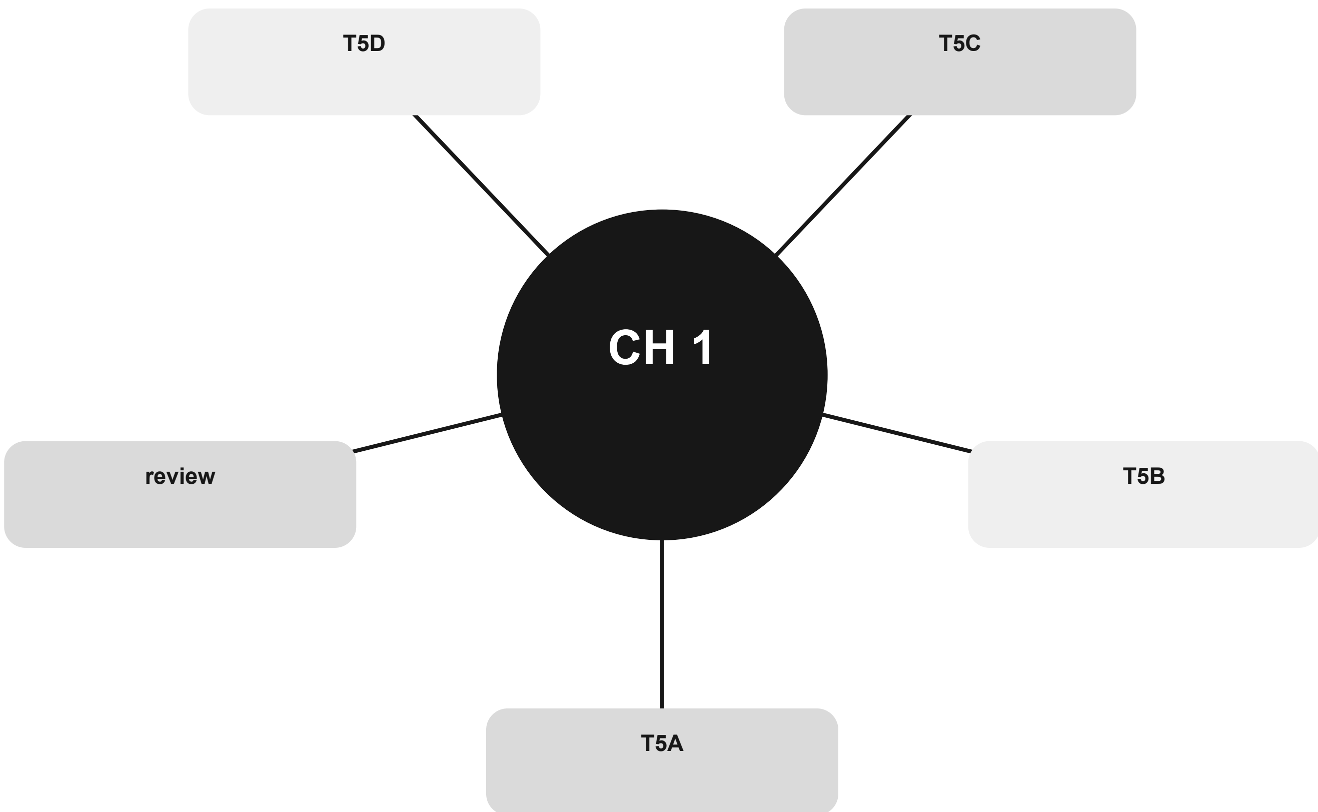
THE WORKING RULE

Check the current NCVEC pool page, use randomized practice, and verify exam arrangements with the chosen examination provider. Do not rely on an old pool or a memorized answer if an erratum changes it.

Keep the task narrow: identify the relationship, choose the safe or lawful action, and connect the item to its official ID.

Electricity without intimidation

Turn units and relationships into a small set of dependable moves.



Official groups: T5A, T5B, T5C, T5D

Your route through this chapter

T5A The electrical vocabulary

Distinguish current, voltage, power, frequency, conductors, insulators, AC, and DC.

T5B Metric prefixes and decibels

Move between common units and read gain or loss without arithmetic panic.

T5C Power, reactance, and radio-frequency units

Choose the right unit and use the power relationship at exam depth.

T5D Ohm's law and circuit paths

Choose the missing quantity and recognize series versus parallel.

After the last group, use the official-pool anchors to identify weak spots. Do randomized exam practice outside this book.

The electrical vocabulary

Distinguish current, voltage, power, frequency, conductors, insulators, AC, and DC.

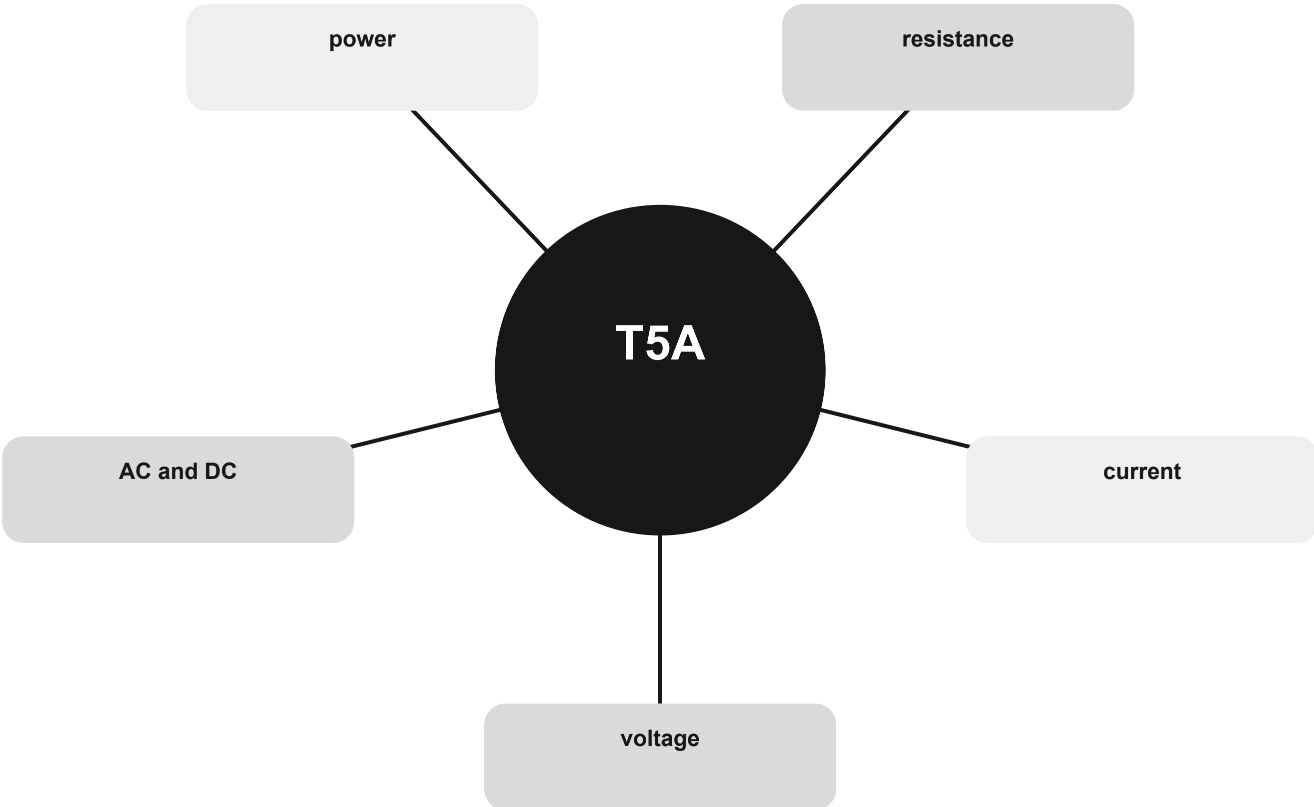
THE MENTAL MODEL

Picture voltage as the difference that can drive charge, current as charge flow, and resistance as opposition. Power is the rate at which electrical energy is used or delivered.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are voltage, current, resistance, power, AC and DC.

The electrical vocabulary concept map



Read from the center outward. Each node is a cue that should lead back to the same working model.

Recognize, then connect

RECOGNIZE	CONNECT
voltage Use the relationship, not an isolated word.	**power** Use the relationship, not an isolated word.
current Use the relationship, not an isolated word.	**AC and DC** Use the relationship, not an isolated word.
resistance Use the relationship, not an isolated word.	

The exam often places related words close together. Do not choose by familiarity. Identify the requested job, path, unit, or rule first.

Official-pool anchors

T5A01 Electrical current is measured in which of the following units: Amperes.

T5A02 Electrical power is measured in which of the following units: Watts.

T5A03 What is the term for the flow of electrons in an electric circuit: Current.

T5A04 What term describes the number of times per second that an alternating current makes a complete cycle: Frequency.

T5A05 A difference in which of the following causes electron flow: Voltage.

T5A06 What is the unit of frequency: Hertz.

T5A07 Why are metals generally good conductors of electricity: They have many free electrons.

T5A08 Which of the following is a good electrical insulator: Glass.

T5A09 Which of the following describes alternating current: Current that alternates between positive and negative directions.

T5A10 Which term describes the rate at which electrical energy is used: Power.

T5A11 What type of current flow is opposed by resistance: All these choices are correct.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

Metric prefixes and decibels

Move between common units and read gain or loss without arithmetic panic.

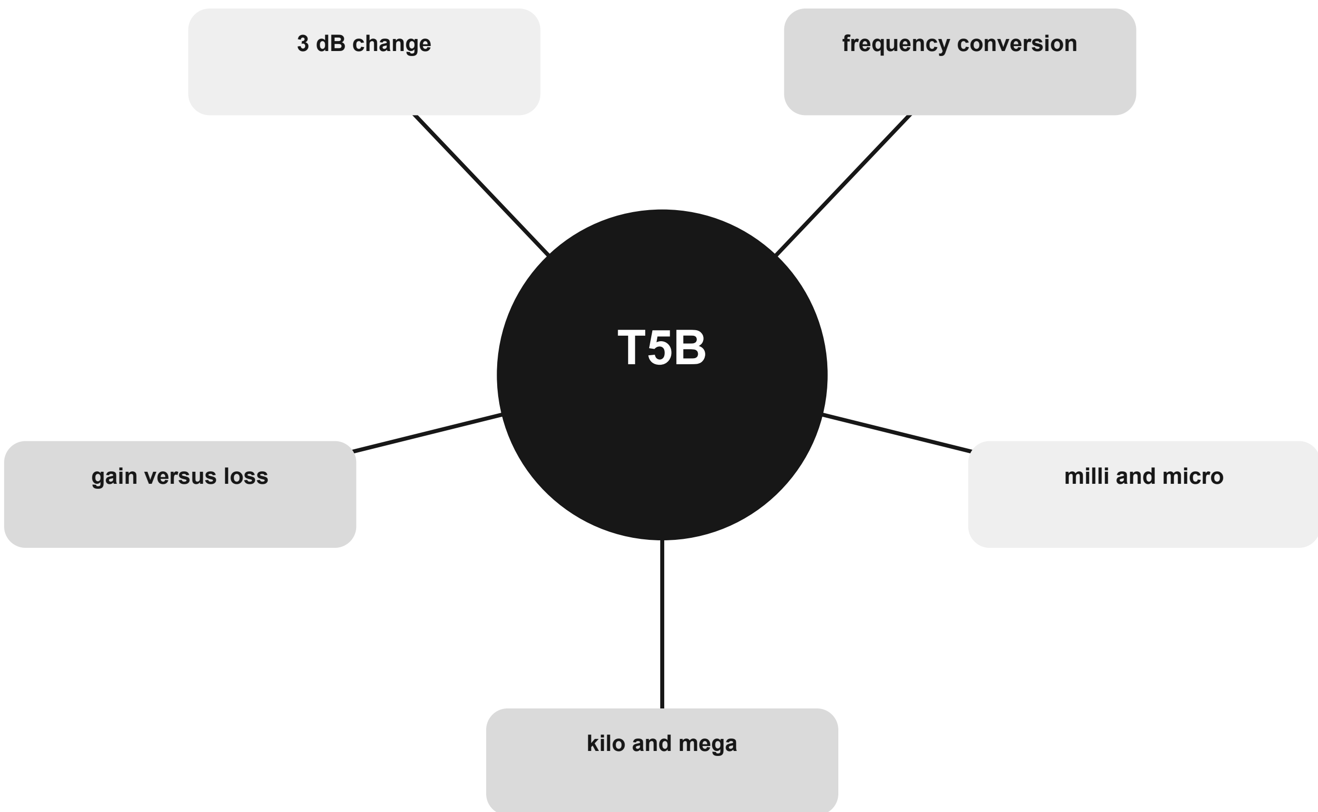
THE MENTAL MODEL

Prefixes move the decimal point: kilo is one thousand, mega is one million, milli is one thousandth, and micro is one millionth. A positive decibel value is gain; a negative value is loss.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are kilo and mega, milli and micro, frequency conversion, 3 dB change, gain versus loss.

Metric prefixes and decibels concept map



Read from the center outward. Each node is a cue that should lead back to the same working model.

Recognize, then connect

RECOGNIZE

kilo and mega

Use the relationship, not an isolated word.

milli and micro

Use the relationship, not an isolated word.

frequency conversion

Use the relationship, not an isolated word.

CONNECT

3 dB change

Use the relationship, not an isolated word.

gain versus loss

Use the relationship, not an isolated word.

The exam often places related words close together. Do not choose by familiarity. Identify the requested job, path, unit, or rule first.

Official-pool anchors

T5B01 How many milliamperes is 1.5 amperes: 1500 milliamperes.

T5B02 Which is equal to 1,500,000 hertz: 1500 kHz.

T5B03 Which is equal to one kilovolt: One thousand volts.

T5B04 Which is equal to one microvolt: One one-millionth of a volt.

T5B05 Which is equal to 500 milliwatts: 0.5 watts.

T5B06 Which is equal to 3000 milliamperes: 3 amperes.

T5B07 Which is equal to 3.525 MHz: 3525 kHz.

T5B08 Which is equal to 1,000,000 picofarads: 1 microfarad.

T5B09 Which decibel value most closely represents a power increase from 5 watts to 10 watts: 3 dB.

T5B10 Which decibel value most closely represents a power decrease from 12 watts to 3 watts: -6 dB.

T5B11 Which decibel value represents a power increase from 20 watts to 200 watts: 10 dB.

T5B12 Which is equal to 28400 kHz: 28.400 MHz.

T5B13 Which is equal to 2425 MHz: 2.425 GHz.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

Power, reactance, and radio-frequency units

Choose the right unit and use the power relationship at exam depth.

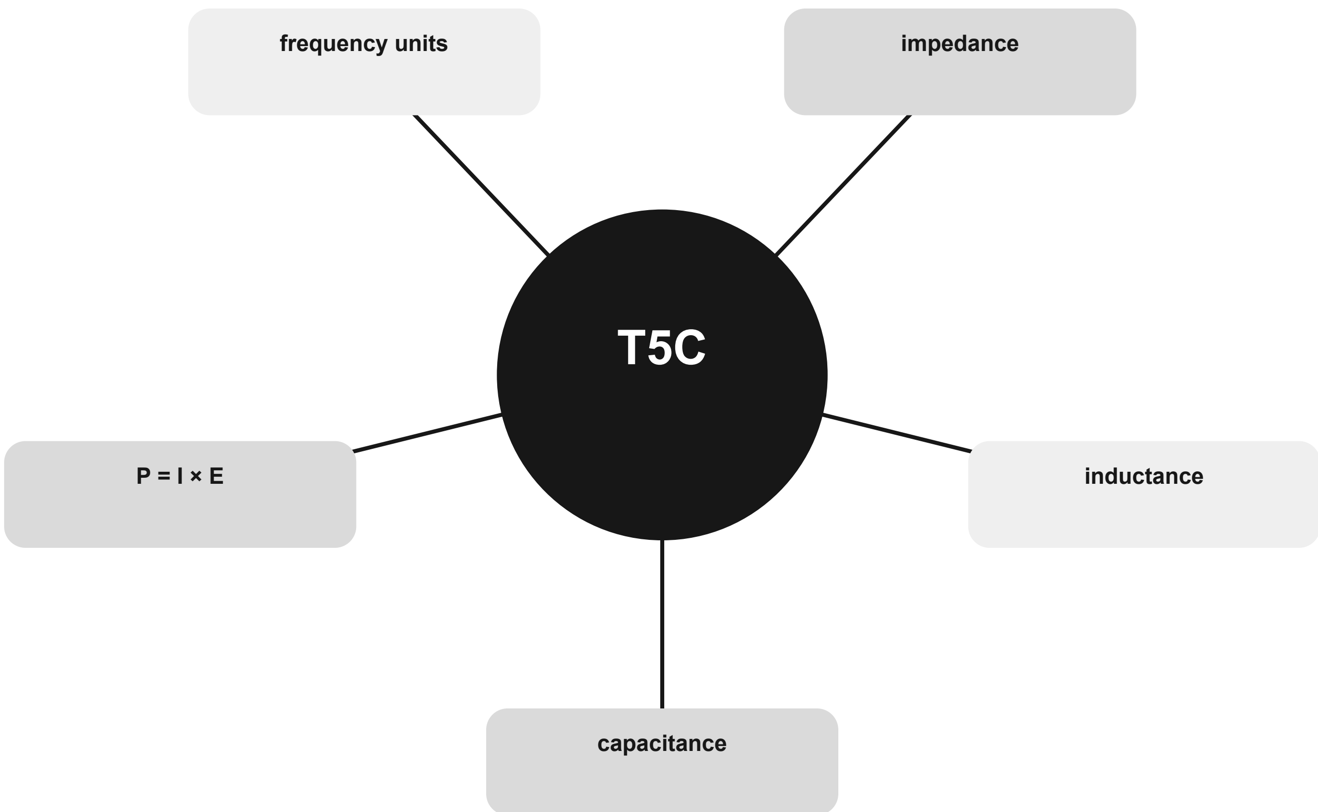
THE MENTAL MODEL

Capacitance and inductance store energy in different fields. Impedance is the broader opposition an AC signal sees. Electrical power follows $P = I \times E$.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are capacitance, inductance, impedance, frequency units, $P = I \times E$.

Power, reactance, and radio-frequency units concept map



Read from the center outward. Each node is a cue that should lead back to the same working model.

Recognize, then connect

RECOGNIZE	CONNECT
capacitance Use the relationship, not an isolated word.	**frequency units** Use the relationship, not an isolated word.
inductance Use the relationship, not an isolated word.	**P = I × E** Use the relationship, not an isolated word.
impedance Use the relationship, not an isolated word.	

The exam often places related words close together. Do not choose by familiarity. Identify the requested job, path, unit, or rule first.

Official-pool anchors

T5C01 What describes the ability to store energy in an electric field: Capacitance.

T5C02 What is the unit of capacitance: Farad.

T5C03 What describes the ability to store energy in a magnetic field: Inductance.

T5C04 What is the unit of inductance: Henry.

T5C05 What is the unit of impedance: Ohm.

T5C06 What is the abbreviation for kilohertz: kHz.

T5C07 What is the abbreviation for megahertz: MHz.

T5C08 What formula is used to calculate electrical power (P) in a DC circuit: P = I x E.

T5C09 How much power is delivered by a voltage of 13.8 volts DC and a current of 10 amperes: 138 watts.

T5C10 How much power is delivered by a voltage of 12 volts DC and a current of 2.5 amperes: 30 watts.

T5C11 How much current is required to deliver 120 watts at a voltage of 12 volts DC: 10 amperes.

T5C12 What is impedance: The opposition to AC current flow.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

Ohm's law and circuit paths

Choose the missing quantity and recognize series versus parallel.

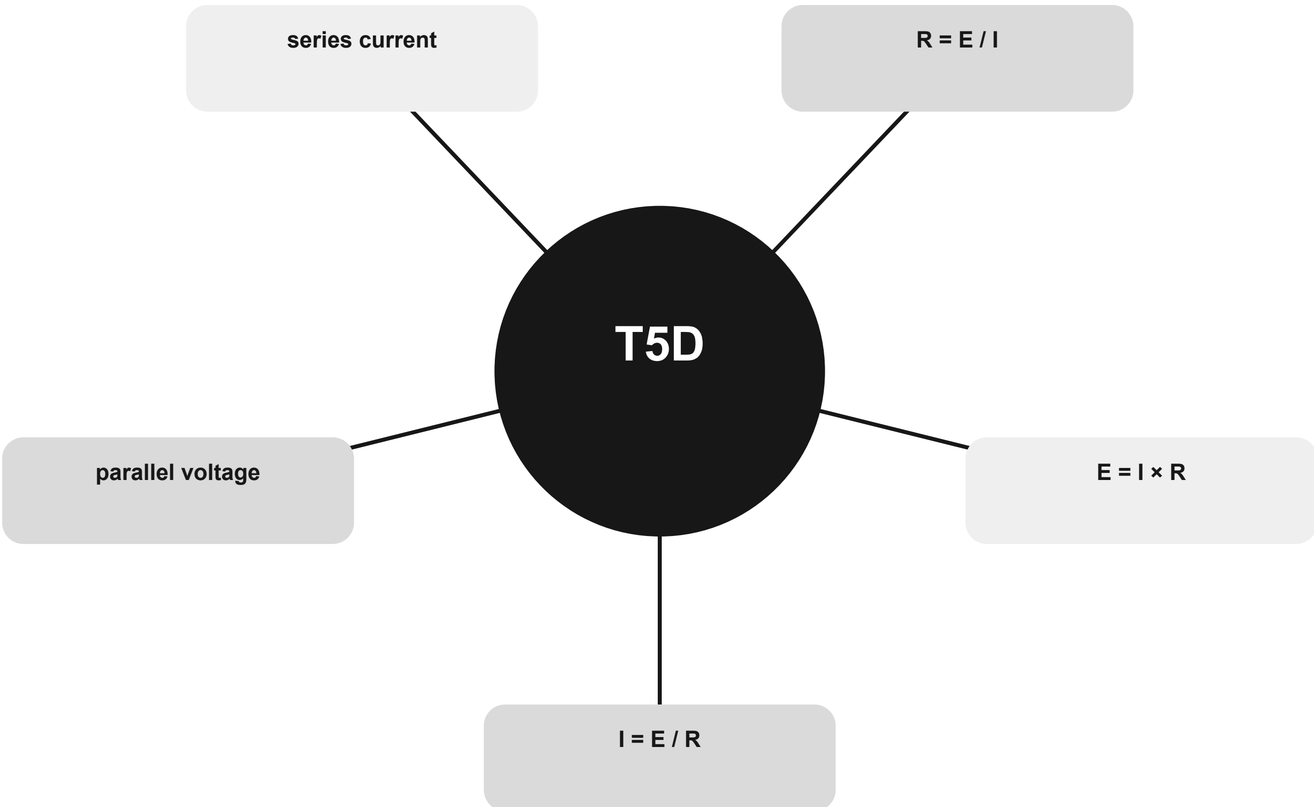
THE MENTAL MODEL

Use the requested unit to select the formula: amperes mean current, volts mean voltage, and ohms mean resistance. Series means one current path; parallel branches share two endpoints.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are I = E / R, E = I × R, R = E / I, series current, parallel voltage.

Ohm's law and circuit paths concept map



Read from the center outward. Each node is a cue that should lead back to the same working model.

Recognize, then connect

RECOGNIZE	CONNECT
$I = E / R$ Use the relationship, not an isolated word.	**series current** Use the relationship, not an isolated word.
$E = I \times R$ Use the relationship, not an isolated word.	**parallel voltage** Use the relationship, not an isolated word.
$R = E / I$ Use the relationship, not an isolated word.	

The exam often places related words close together. Do not choose by familiarity. Identify the requested job, path, unit, or rule first.

Official-pool anchors

T5D01 What formula is used to calculate current in a circuit: I = E / R.

T5D02 What formula is used to calculate voltage in a circuit: E = I x R.

T5D03 What formula is used to calculate resistance in a circuit: R = E / I.

T5D04 What is the resistance of a circuit in which a current of 3 amperes flows when connected to 90 volts: 30 ohms.

T5D05 What is the resistance of a circuit for which the applied voltage is 12 volts and the current flow is 1.5 amperes: 8 ohms.

T5D06 What is the resistance of a circuit that draws 4 amperes from a 12-volt source: 3 ohms.

T5D07 What is the current in a circuit with an applied voltage of 120 volts and a resistance of 80 ohms: 1.5 amperes.

T5D08 What is the current through a 100-ohm resistor connected across 200 volts: 2 amperes.

T5D09 What is the current through a 24-ohm resistor connected across 240 volts: 10 amperes.

T5D10 What is the voltage across a 2-ohm resistor if a current of 0.5 amperes flows through it: 1 volt.

T5D11 What is the voltage across a 10-ohm resistor if a current of 1 ampere flows through it: 10 volts.

T5D12 What is the voltage across a 10-ohm resistor if a current of 2 amperes flows through it: 20 volts.

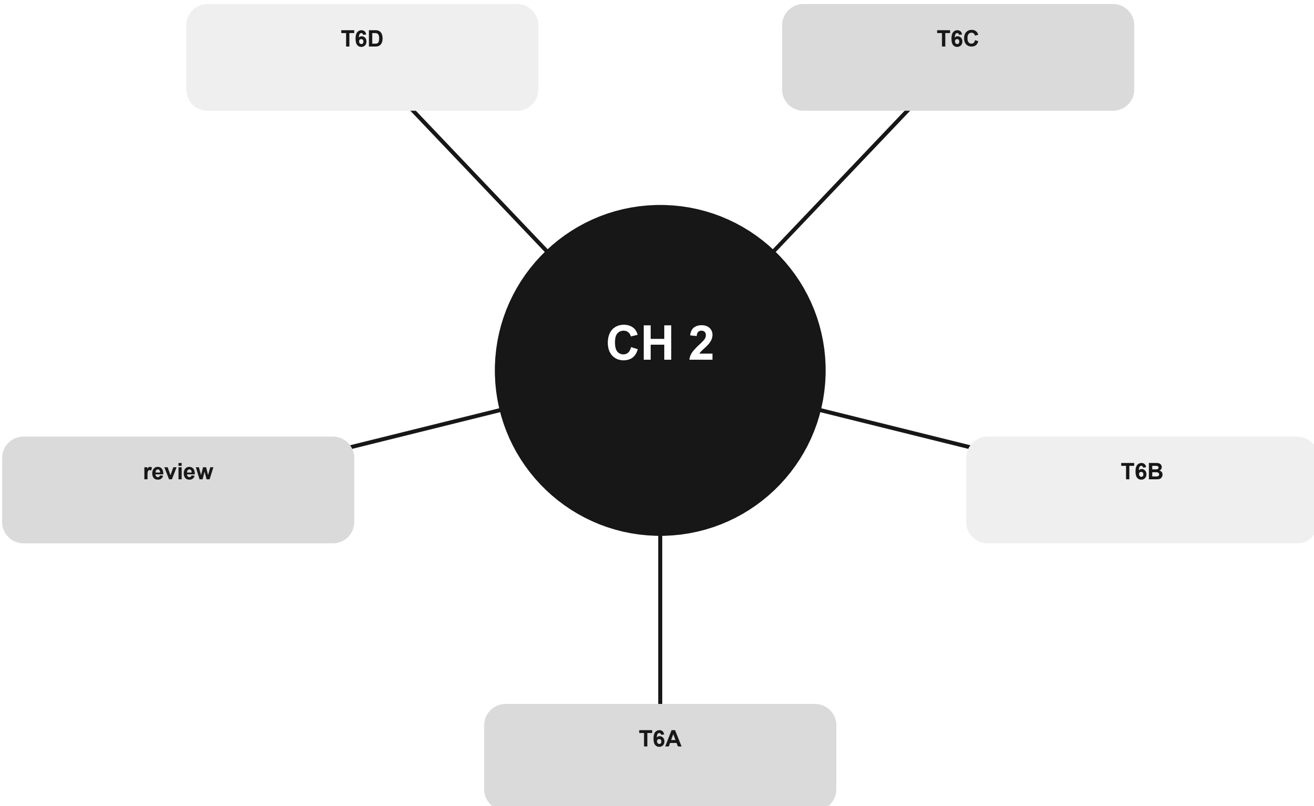
T5D13 In which type of circuit is the current always the same through all components: Series.

T5D14 In which type of circuit is the voltage always the same across all components: Parallel.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

Components as jobs, not symbols

Recognize parts by the job they perform, then connect the job to the symbol.



Official groups: T6A, T6B, T6C, T6D

Your route through this chapter

T6A Passive parts, switches, and batteries

Recognize common components by what they oppose, store, switch, or supply.

T6B Diodes and transistors

Tell one-way devices from control and amplification devices.

T6C Reading schematic symbols

Translate a drawing into components and connections.

T6D Functional building blocks

Match a component or circuit to the job it performs.

After the last group, use the official-pool anchors to identify weak spots. Do randomized exam practice outside this book.

Passive parts, switches, and batteries

Recognize common components by what they oppose, store, switch, or supply.

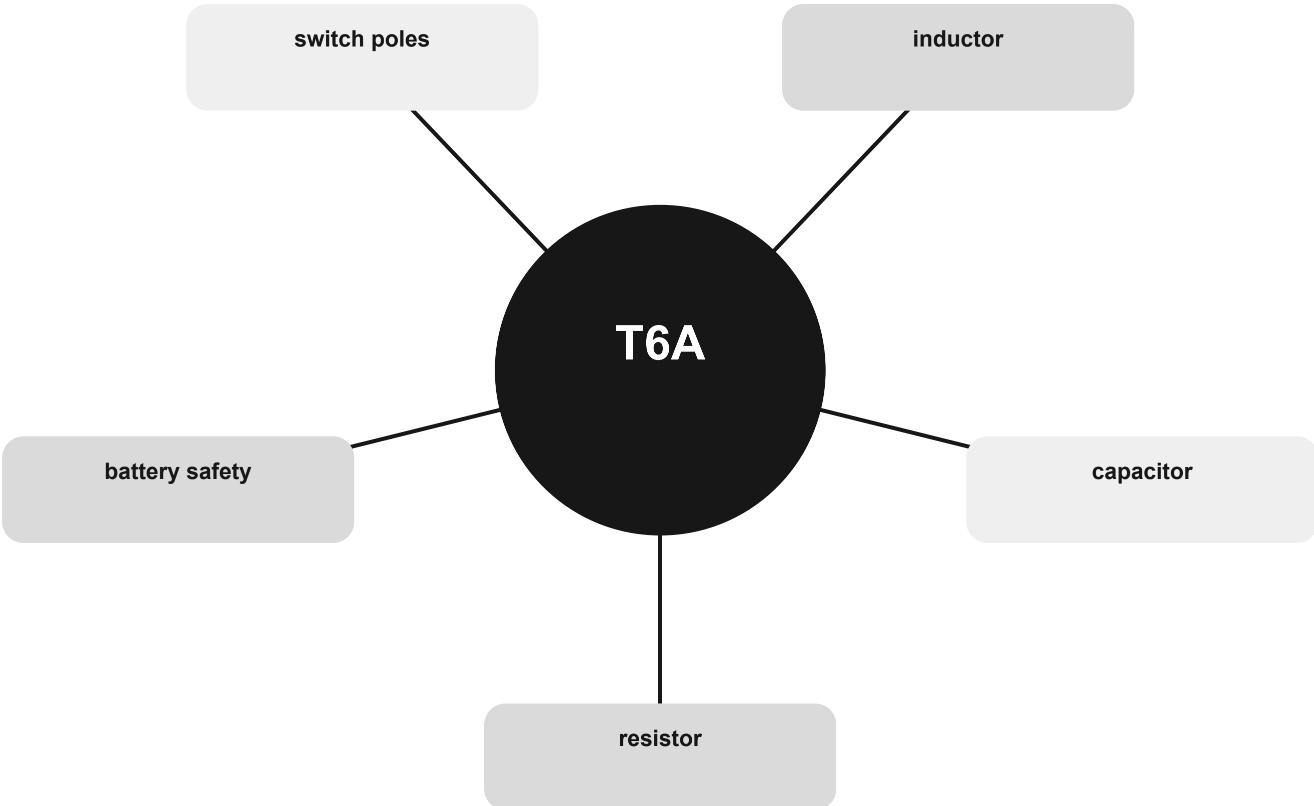
THE MENTAL MODEL

A resistor limits current, a capacitor stores energy in an electric field, and an inductor stores energy in a magnetic field. Switch names describe poles and throws.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are resistor, capacitor, inductor, switch poles, battery safety.

Passive parts, switches, and batteries concept map



Read from the center outward. Each node is a cue that should lead back to the same working model.

Recognize, then connect

RECOGNIZE

resistor

Use the relationship, not an isolated word.

capacitor

Use the relationship, not an isolated word.

inductor

Use the relationship, not an isolated word.

CONNECT

switch poles

Use the relationship, not an isolated word.

battery safety

Use the relationship, not an isolated word.

The exam often places related words close together. Do not choose by familiarity. Identify the requested job, path, unit, or rule first.

Official-pool anchors

T6A01 What electrical component opposes the flow of current in a DC circuit: Resistor.

T6A02 What type of component is often used as an adjustable volume control: Potentiometer.

T6A03 What electrical parameter is controlled by a potentiometer: Resistance.

T6A04 What electrical component stores energy in an electric field: Capacitor.

T6A05 What type of electrical component consists of conductive surfaces separated by an insulator: Capacitor.

T6A06 What type of electrical component stores energy in a magnetic field: Inductor.

T6A07 What electrical component is typically constructed as a coil of wire: Inductor.

T6A08 What is the function of an SPDT switch: A single circuit is switched between one of two other circuits.

T6A09 What type of switch is represented by component 3 in figure T-2: Single-pole single-throw.

T6A10 Which of the following battery chemistries is rechargeable: All these choices are correct.

T6A11 Which of the following battery chemistries is not rechargeable: Carbon-zinc.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

Diodes and transistors

Tell one-way devices from control and amplification devices.

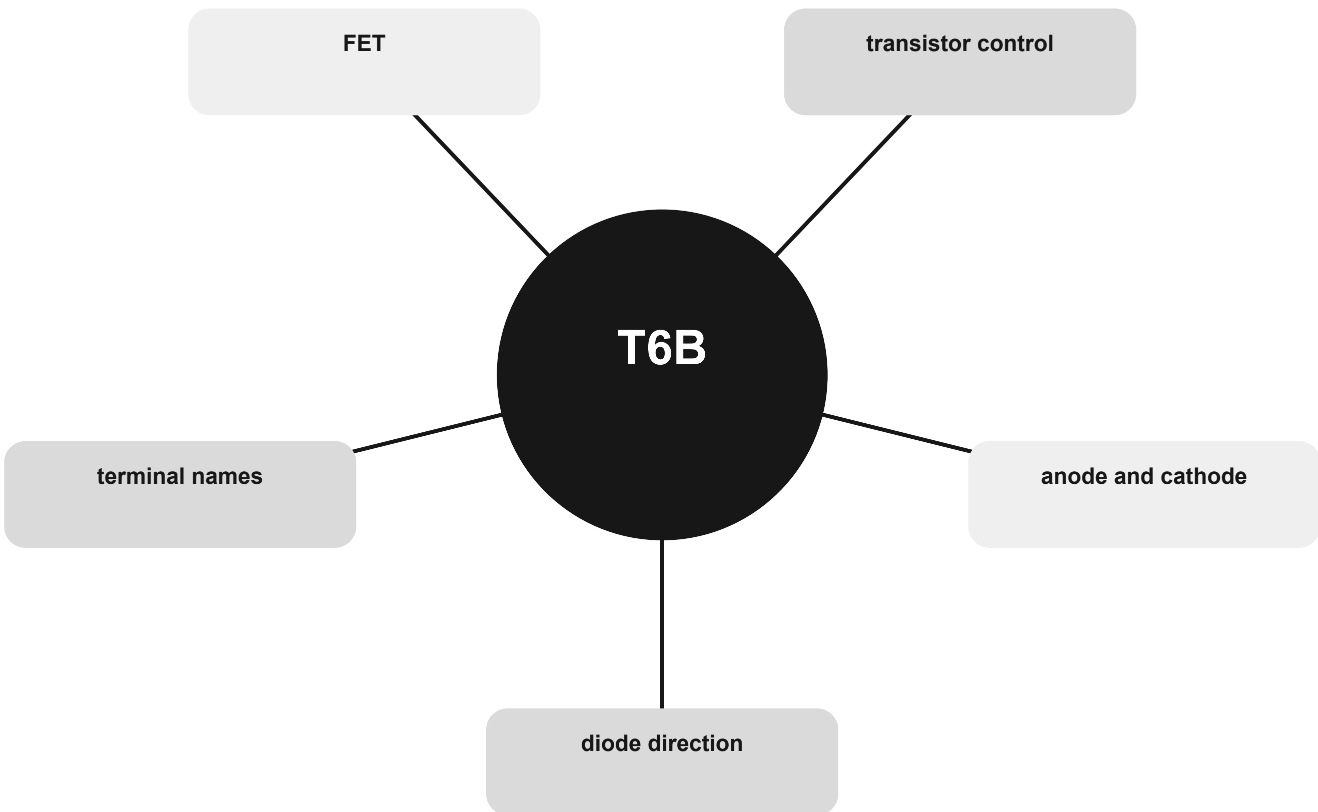
THE MENTAL MODEL

A diode favors current in one direction. A transistor uses one signal to control another and can act as an amplifier or switch. Terminal names identify the device family.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are diode direction, anode and cathode, transistor control, FET, terminal names.

Diodes and transistors concept map



Read from the center outward. Each node is a cue that should lead back to the same working model.

Recognize, then connect

RECOGNIZE	CONNECT
diode direction Use the relationship, not an isolated word.	**FET** Use the relationship, not an isolated word.
anode and cathode Use the relationship, not an isolated word.	**terminal names** Use the relationship, not an isolated word.
transistor control Use the relationship, not an isolated word.	

The exam often places related words close together. Do not choose by familiarity. Identify the requested job, path, unit, or rule first.

Official-pool anchors

T6B01 Which is true about forward voltage drop in a diode: It is lower in some diode types than in others.

T6B02 What electronic component allows current to flow in only one direction: Diode.

T6B03 Which of these components can be used as an electronic switch: Transistor.

T6B04 Which of the following components can consist of three regions of semiconductor material: Transistor.

T6B05 What type of transistor has a gate, drain, and source: Field-effect.

T6B06 How is the cathode lead of a semiconductor diode often marked on the package: With a stripe.

T6B07 What causes a light-emitting diode (LED) to emit light: Forward current.

T6B08 What does the abbreviation FET stand for: Field Effect Transistor.

T6B09 What are the names for the electrodes of a diode: Anode and cathode.

T6B10 Which of the following can provide power gain: Transistor.

T6B11 What does the term gain mean in amplifiers: All these choices are correct.

T6B12 What are the names of the electrodes of a bipolar junction transistor: Emitter, base, collector.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

Reading schematic symbols

Translate a drawing into components and connections.

THE MENTAL MODEL

A schematic shows electrical relationships, not physical placement. Read one symbol at a time, then follow the connecting lines to reconstruct the circuit's jobs.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are schematic, symbols, connections, variable parts, antenna symbol.

Reading schematic symbols concept map

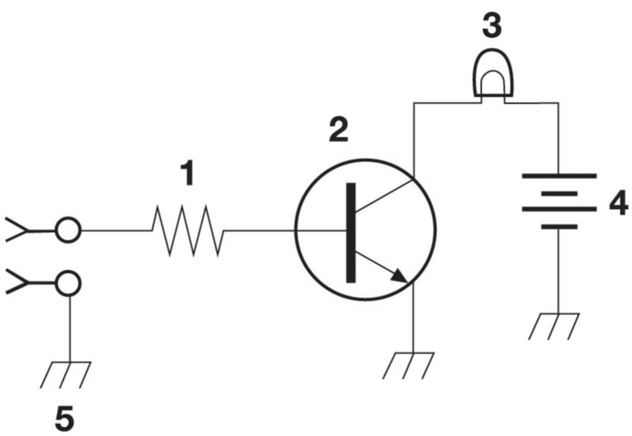


Figure T-1

OFFICIAL FIGURE T-1 · NCVEC public-domain pool

Recognize, then connect

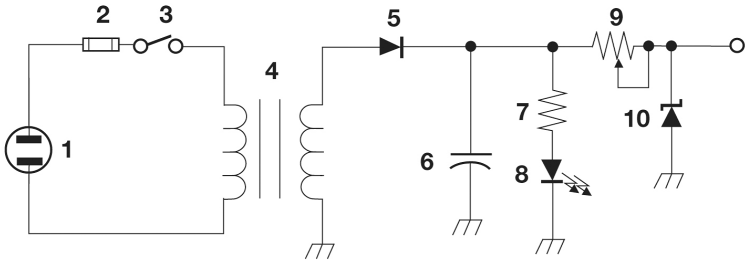


Figure T-2

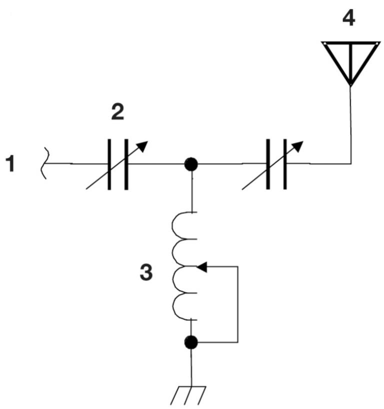


Figure T-3

OFFICIAL FIGURES T-2 AND T-3 · NCVEC public-domain pool

Official-pool anchors

T6C01 What is an electrical diagram using standard component symbols called: Schematic.

T6C02 What is component 1 in figure T-1: Resistor.

T6C03 What is component 2 in figure T-1: Transistor.

T6C04 What is component 3 in figure T-1: Lamp.

T6C05 What is component 4 in figure T-1: Battery.

T6C06 What is component 6 in figure T-2: Capacitor.

T6C07 What is component 8 in figure T-2: Light emitting diode.

T6C08 What is component 9 in figure T-2: Variable resistor.

T6C09 What is component 4 in figure T-2: Transformer.

T6C10 What is component 3 in figure T-3: Variable inductor.

T6C11 What is component 4 in figure T-3: Antenna.

T6C12 Which of the following is accurately represented in electrical schematics: Component connections.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

Functional building blocks

Match a component or circuit to the job it performs.

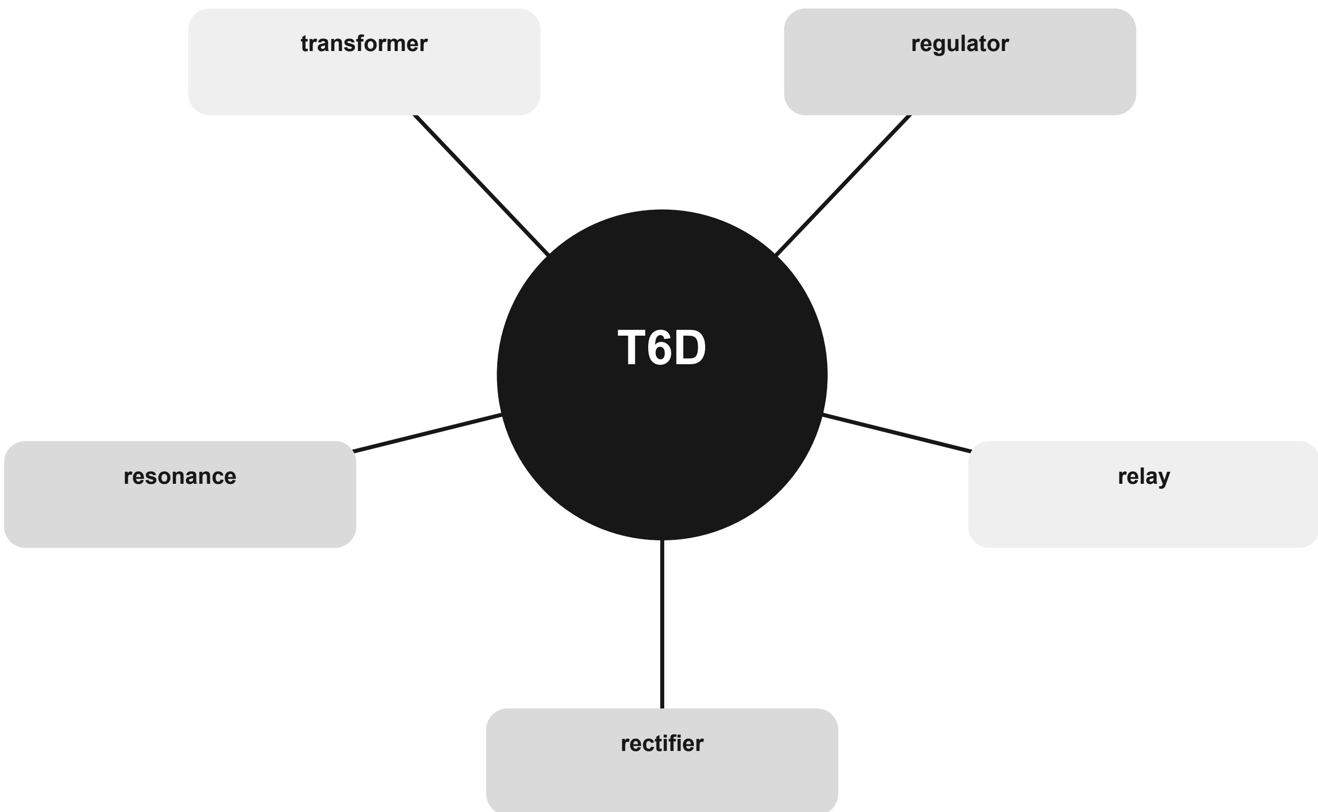
THE MENTAL MODEL

Rectifiers change AC toward DC, regulators hold voltage steady, transformers transfer energy between windings, and resonant circuits select frequencies.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are rectifier, relay, regulator, transformer, resonance.

Functional building blocks concept map



Read from the center outward. Each node is a cue that should lead back to the same working model.

Recognize, then connect

RECOGNIZE	CONNECT
rectifier Use the relationship, not an isolated word.	**transformer** Use the relationship, not an isolated word.
relay Use the relationship, not an isolated word.	**resonance** Use the relationship, not an isolated word.
regulator Use the relationship, not an isolated word.	

The exam often places related words close together. Do not choose by familiarity. Identify the requested job, path, unit, or rule first.

Official-pool anchors

T6D01 Which of the following devices or circuits changes an alternating current into a varying direct current signal: Rectifier.

T6D02 What is a relay: An electrically-controlled switch.

T6D03 Which of the following is a reason to use shielded wire: To prevent coupling of unwanted signals to or from the wire.

T6D04 Which of the following displays an electrical quantity as a numeric value: Meter.

T6D05 What type of circuit controls the amount of voltage from a power supply: Regulator.

T6D06 What component changes 120 V AC power to a lower AC voltage for other uses: Transformer.

T6D07 Which of the following is commonly used as a visual indicator: LED.

T6D08 Which of the following is combined with an inductor to make a resonant circuit: Capacitor.

T6D09 What is the name of a device that combines several semiconductors and other components into one package: Integrated circuit.

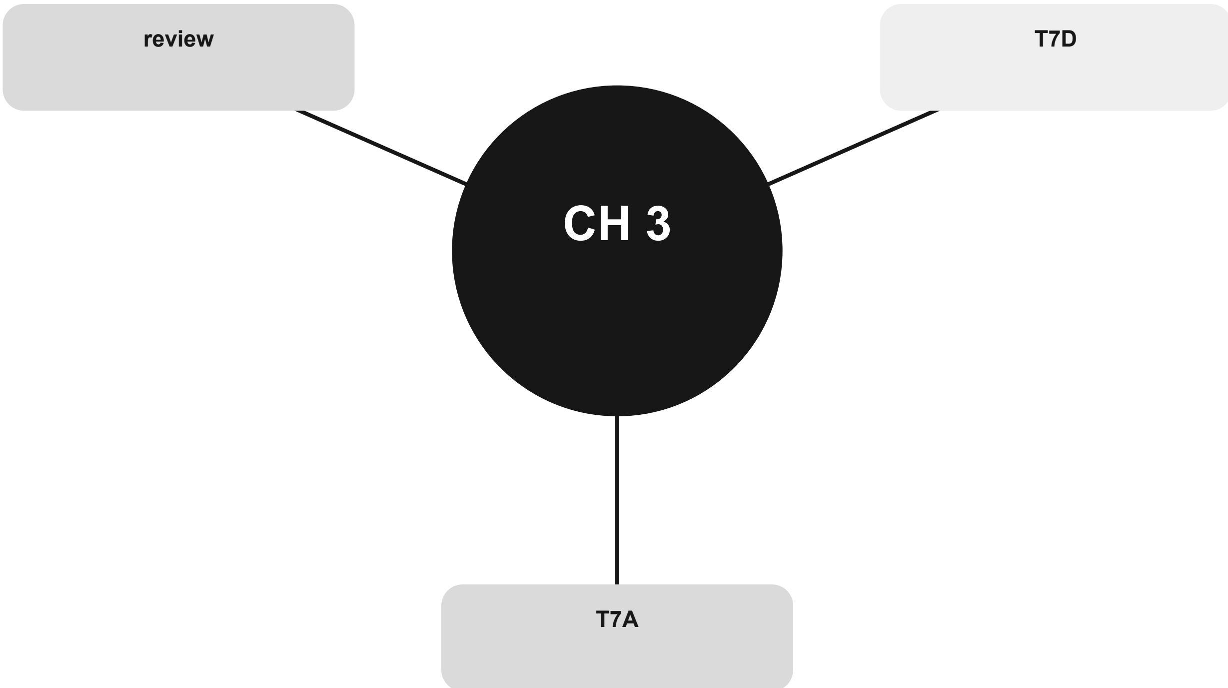
T6D10 What is the function of component 2 in figure T-1: Control the flow of current.

T6D11 Which of the following is a resonant or tuned circuit: An inductor and a capacitor in series or parallel.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

Inside a radio and on the bench

Follow signal flow and connect test instruments without guessing.



Official groups: T7A, T7D

Your route through this chapter

T7A Radio blocks and signal flow

Follow a signal through receiver, transmitter, and frequency-conversion stages.

T7D Meters and safe measurements

Connect meters correctly and recognize safe bench practice.

After the last group, use the official-pool anchors to identify weak spots. Do randomized exam practice outside this book.

Radio blocks and signal flow

Follow a signal through receiver, transmitter, and frequency-conversion stages.

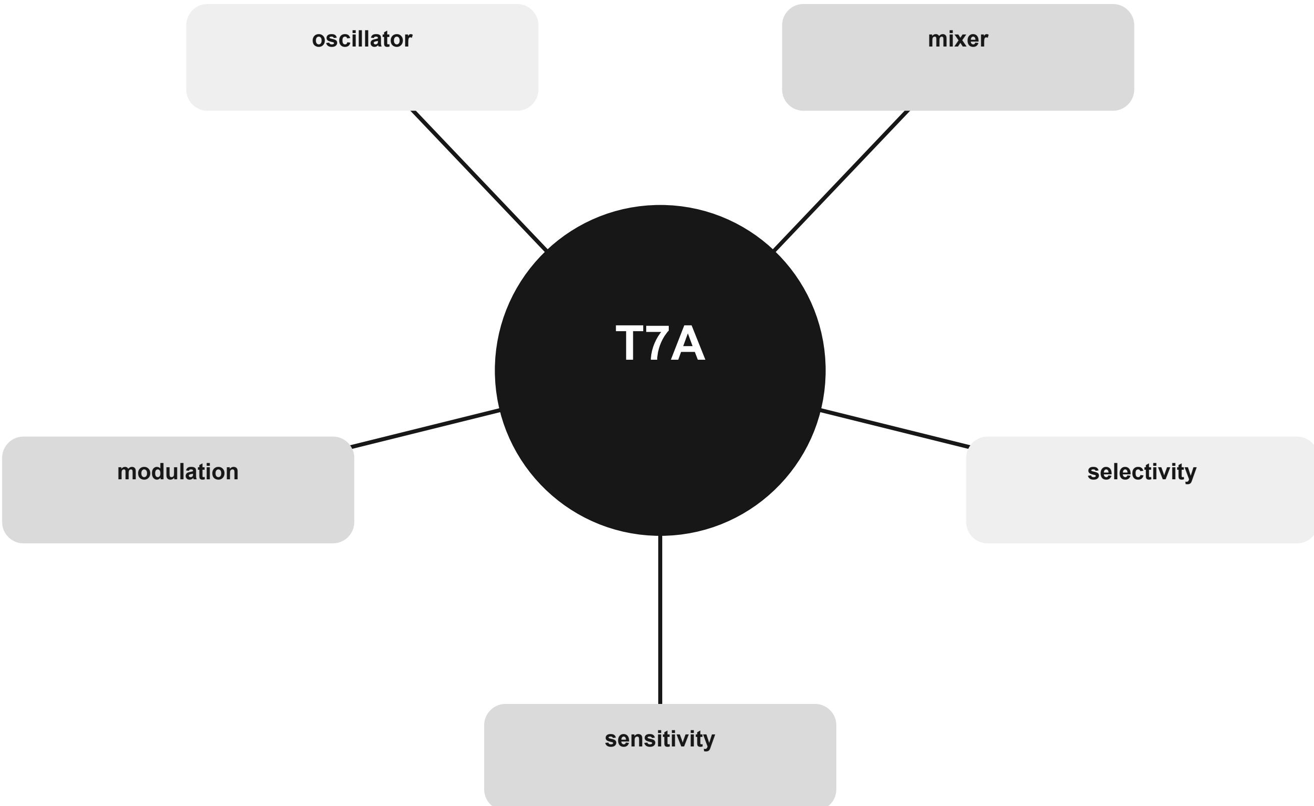
THE MENTAL MODEL

Sensitivity is about weak signals; selectivity is about separating nearby signals. Mixers combine frequencies, oscillators create a signal, and modulation places information on a carrier.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are sensitivity, selectivity, mixer, oscillator, modulation.

Radio blocks and signal flow concept map



Read from the center outward. Each node is a cue that should lead back to the same working model.

Recognize, then connect

RECOGNIZE	CONNECT
sensitivity Use the relationship, not an isolated word.	**oscillator** Use the relationship, not an isolated word.
selectivity Use the relationship, not an isolated word.	**modulation** Use the relationship, not an isolated word.
mixer Use the relationship, not an isolated word.	

The exam often places related words close together. Do not choose by familiarity. Identify the requested job, path, unit, or rule first.

Official-pool anchors

T7A01 Which term describes the ability of a receiver to detect the presence of a signal: Sensitivity.

T7A02 What is a transceiver: A device that combines a receiver and transmitter.

T7A03 Which of the following is used to convert a signal from one frequency to another: Mixer.

T7A04 Which term describes the ability of a receiver to discriminate between multiple signals: Selectivity.

T7A05 What is the name of a circuit that generates a signal at a specific frequency: Oscillator.

T7A06 What device converts the RF input and output of a transceiver to another band: Transverter.

T7A07 What is the function of a transceiver’s PTT input: Switches transceiver from receive to transmit when grounded.

T7A08 Which of the following describes combining speech with an RF carrier signal: Modulation.

T7A09 What is the function of the switch which selects either SSB or CW-FM on some VHF power amplifiers: Set the amplifier for proper operation in the selected mode.

T7A10 What can be added to the output of a transceiver to increase the transmitted output power: An RF power amplifier.

T7A11 What is the function of the Variable Frequency Oscillator (VFO) circuit in a transceiver: Set the receive and transmit frequency.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

Meters and safe measurements

Connect meters correctly and recognize safe bench practice.

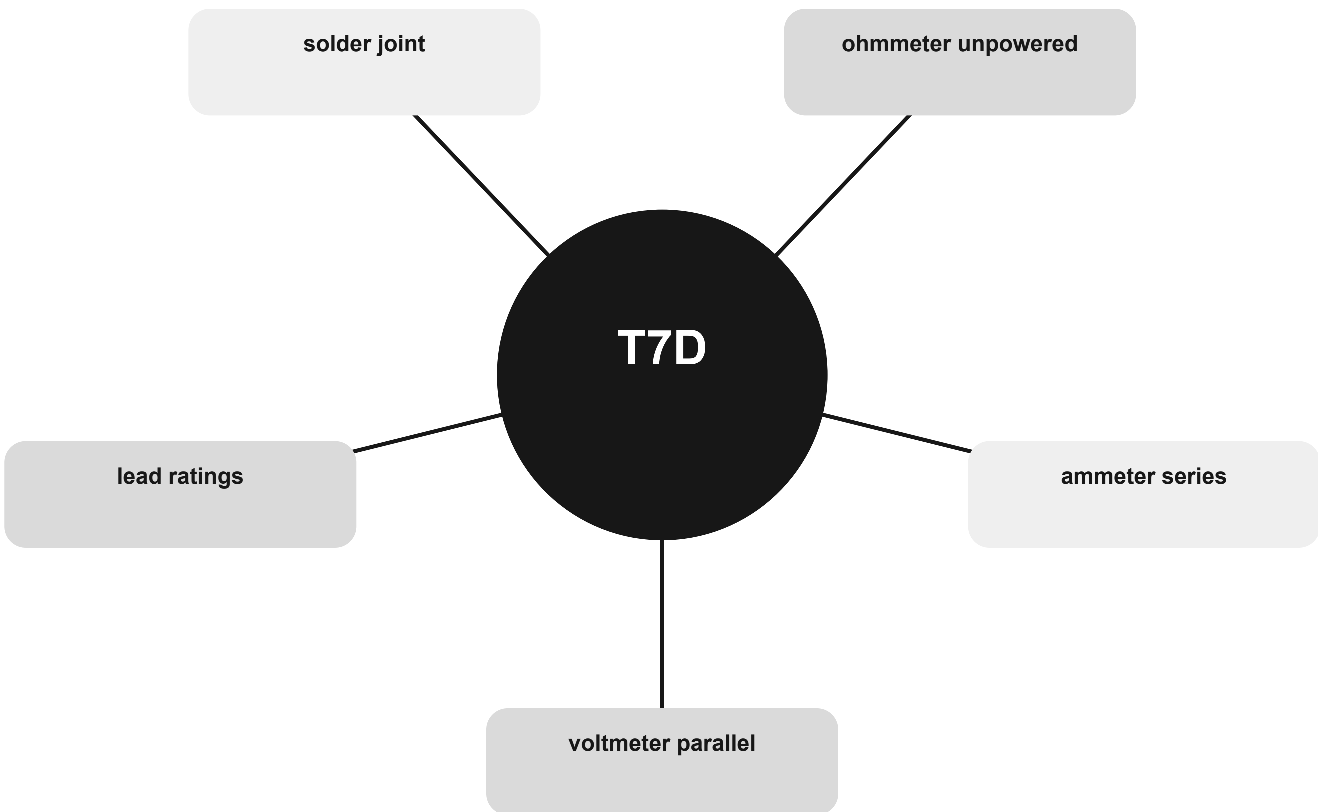
THE MENTAL MODEL

A voltmeter looks across two points; an ammeter goes through the current path. An ohmmeter applies a small internal current, so the circuit must be unpowered.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are voltmeter parallel, ammeter series, ohmmeter unpowered, solder joint, lead ratings.

Meters and safe measurements concept map



Read from the center outward. Each node is a cue that should lead back to the same working model.

Recognize, then connect

RECOGNIZE

voltmeter parallel

Use the relationship, not an isolated word.

ammeter series

Use the relationship, not an isolated word.

ohmmeter unpowered

Use the relationship, not an isolated word.

CONNECT

solder joint

Use the relationship, not an isolated word.

lead ratings

Use the relationship, not an isolated word.

The exam often places related words close together. Do not choose by familiarity. Identify the requested job, path, unit, or rule first.

Official-pool anchors

T7D01 Which instrument would you use to measure electric potential: A voltmeter.

T7D02 How is a voltmeter connected to a component to measure applied voltage: In parallel.

T7D03 When configured to measure current, how is a multimeter connected to a component: In series.

T7D04 Which instrument is used to measure electric current: An ammeter.

T7D05 How does an ohmmeter measure the resistance of a circuit or component: By applying a small current and measuring the resulting voltage.

T7D06 Which of the following can damage a multimeter: Attempting to measure voltage when using the resistance setting.

T7D07 Which of the following measurements are made using a multimeter: Voltage and resistance.

T7D08 Which of the following types of solder should not be used for radio and electronic applications: Acid-core solder.

T7D09 What is the characteristic appearance of a cold tin-lead solder joint: A rough or lumpy surface.

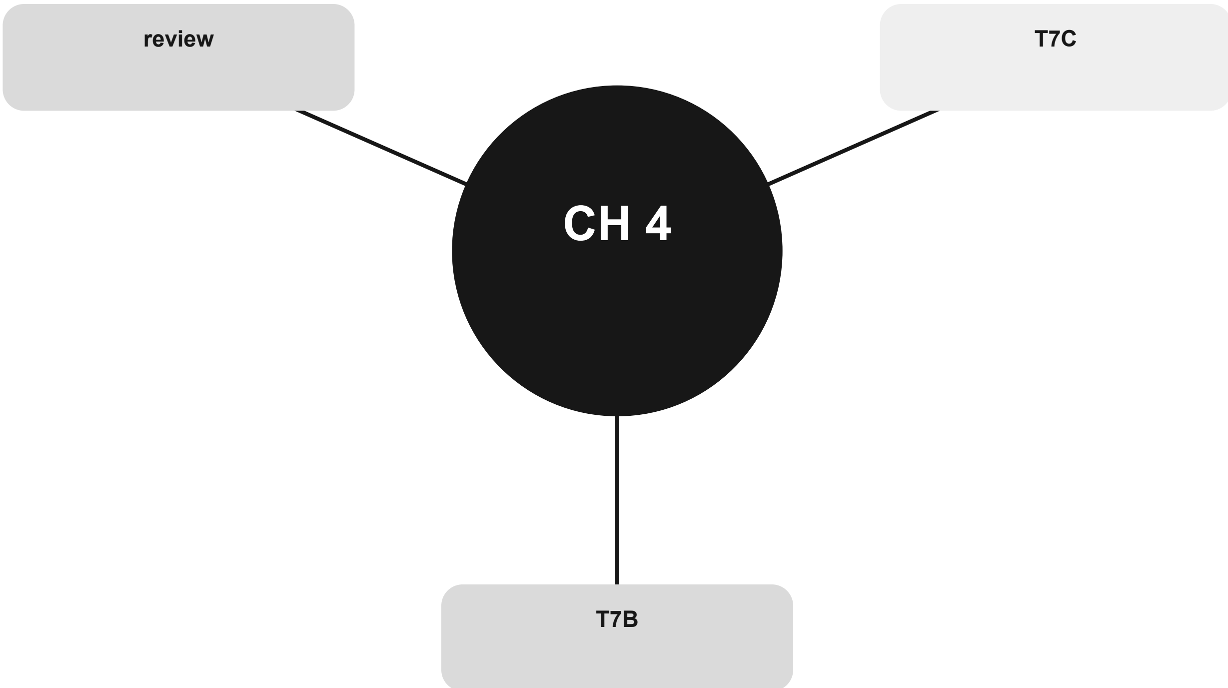
T7D10 What reading indicates that an ohmmeter is connected across a large, discharged capacitor: Increasing resistance with time.

T7D11 Which of the following precautions should be taken when measuring in-circuit resistance with an ohmmeter: Ensure that the circuit is not powered.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

Troubleshoot by symptom, cause, and safe action

Use visible symptoms to choose the first sensible and safe check.



Official groups: T7B, T7C

Your route through this chapter

T7B Interference and distorted audio

Move from symptom to likely cause before changing equipment.

T7C SWR, feed lines, and test loads

Recognize mismatch, loss, and safe transmitter testing.

After the last group, use the official-pool anchors to identify weak spots. Do randomized exam practice outside this book.

Interference and distorted audio

Move from symptom to likely cause before changing equipment.

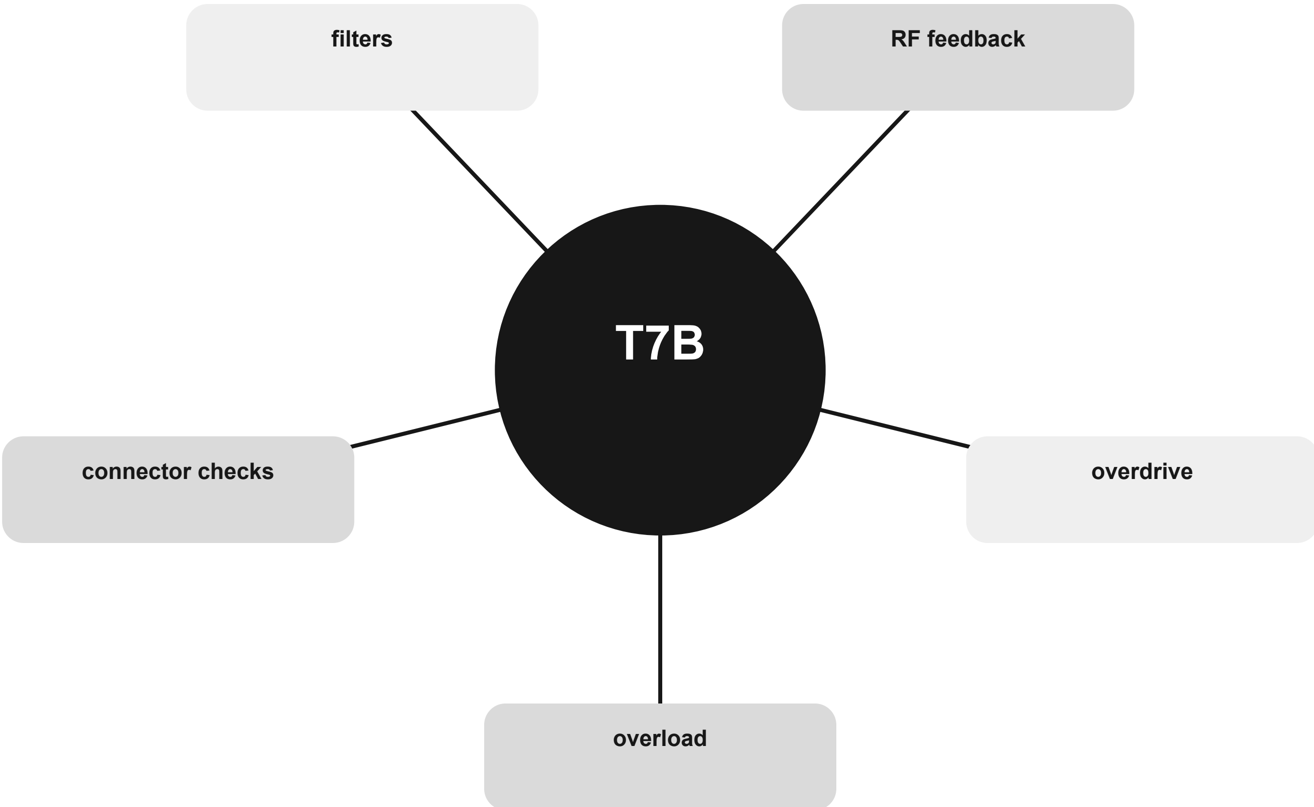
THE MENTAL MODEL

Strong nearby signals can overload receivers. Too much microphone gain or RF feedback distorts transmitted audio. Start with correct station operation and targeted filtering.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are overload, overdrive, RF feedback, filters, connector checks.

Interference and distorted audio concept map



Read from the center outward. Each node is a cue that should lead back to the same working model.

Recognize, then connect

RECOGNIZE

overload

Use the relationship, not an isolated word.

overdrive

Use the relationship, not an isolated word.

RF feedback

Use the relationship, not an isolated word.

CONNECT

filters

Use the relationship, not an isolated word.

connector checks

Use the relationship, not an isolated word.

The exam often places related words close together. Do not choose by familiarity. Identify the requested job, path, unit, or rule first.

Official-pool anchors

T7B01 What can you do if you are told your FM handheld or mobile transceiver is over-deviating: Talk farther away from the microphone.

T7B02 What would cause a broadcast AM or FM radio to receive an amateur radio transmission unintentionally: The receiver is unable to reject strong signals outside the AM or FM band.

T7B03 Which of the following can cause radio frequency interference: All these choices are correct.

T7B04 Which of the following might be the cause of low RF power output from a solid-state transceiver: High SWR.

T7B05 Which of the following might reduce interference by an amateur station to a non-amateur over-the-air radio receiver: Block the amateur signal with a filter at the antenna input of the affected receiver.

T7B06 Which of the following actions should you take if a neighbor tells you that your station's transmissions are interfering with their radio or TV reception: Make sure that your station is functioning properly and that it does not cause interference to your own radio or television when it is tuned to the same channel.

T7B07 Which of the following can reduce interference to a 2-meter band transceiver from a nearby commercial FM station: Installing a band-reject filter.

T7B08 What should you do if something in a neighbor's home is causing harmful interference to your amateur station: All these choices are correct.

T7B09 What should be the first step to resolve non-fiber optic cable TV interference caused by your amateur radio transmission: Be sure all TV feed line coaxial connectors are installed properly.

T7B10 What might be a problem if you receive a report that your audio signal through an FM repeater is distorted or unintelligible: All these choices are correct.

T7B11 Which of the following can eliminate distorted voice transmissions: Adding a clip-on ferrite "choke" to the microphone cable to prevent the transmitted signal from feeding back into the transmitter.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

SWR, feed lines, and test loads

Recognize mismatch, loss, and safe transmitter testing.

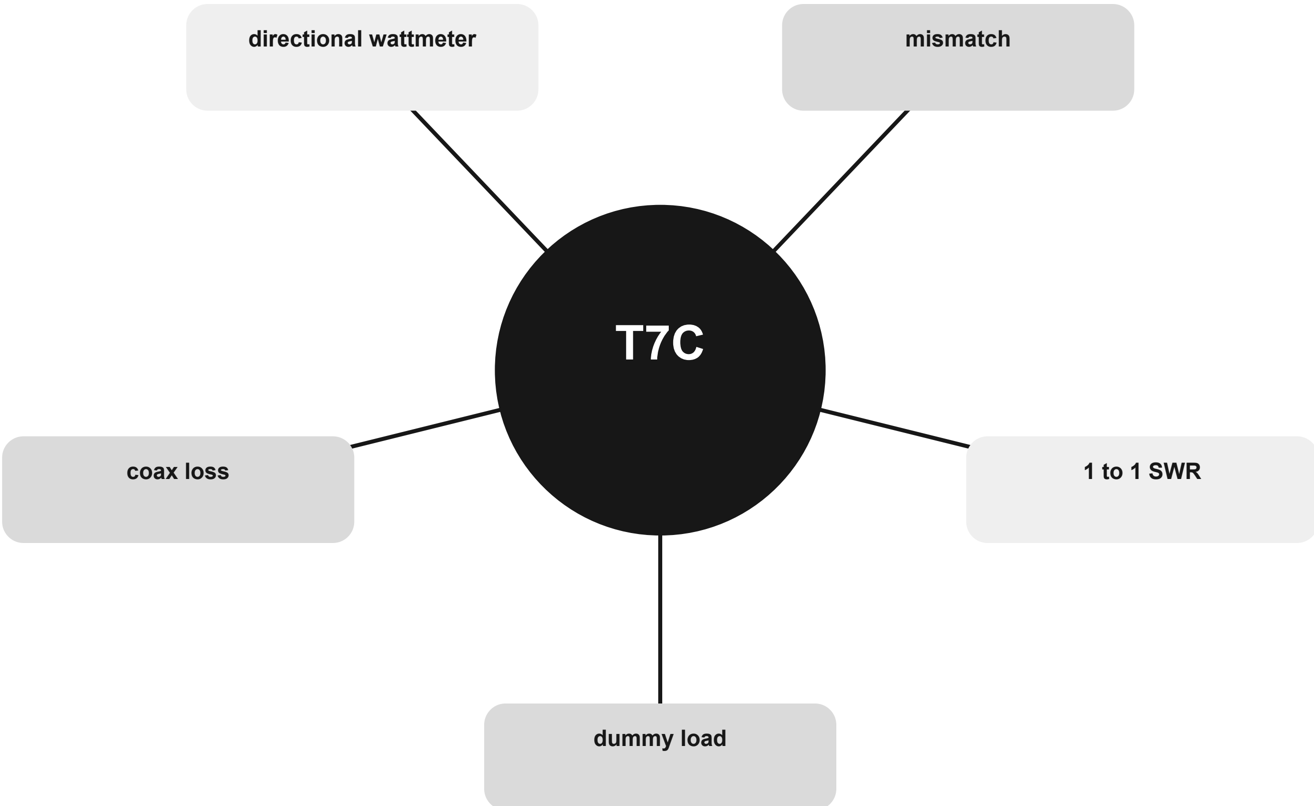
THE MENTAL MODEL

A dummy load lets a transmitter operate without radiating. SWR indicates how well the load matches the line. Reflected energy becomes heat and high SWR can stress an output stage.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are dummy load, 1 to 1 SWR, mismatch, directional wattmeter, coax loss.

SWR, feed lines, and test loads concept map



Read from the center outward. Each node is a cue that should lead back to the same working model.

Recognize, then connect

RECOGNIZE

dummy load

Use the relationship, not an isolated word.

1 to 1 SWR

Use the relationship, not an isolated word.

mismatch

Use the relationship, not an isolated word.

CONNECT

directional wattmeter

Use the relationship, not an isolated word.

coax loss

Use the relationship, not an isolated word.

The exam often places related words close together. Do not choose by familiarity. Identify the requested job, path, unit, or rule first.

Official-pool anchors

T7C01 What is the primary purpose of a dummy load: To prevent transmitting signals over the air when making tests.

T7C02 Which of the following is used to determine if an antenna is resonant at the desired operating frequency: An antenna analyzer.

T7C03 What does a typical RF dummy load consist of: A 50-ohm non-inductive resistor mounted on a heat sink.

T7C04 What reading on an SWR meter indicates a perfect impedance match between the antenna and the feed line: 1:1.

T7C05 Why do most solid-state transmitters reduce output power as SWR increases beyond a certain level: To protect the RF output amplifier transistors.

T7C06 What does an SWR reading of 4:1 indicate: Impedance mismatch.

T7C07 What happens to power lost in a feed line: It is converted into heat.

T7C08 Which instrument can be used to determine SWR: Directional wattmeter.

T7C09 Which of the following causes failure of coaxial cables: Moisture contamination.

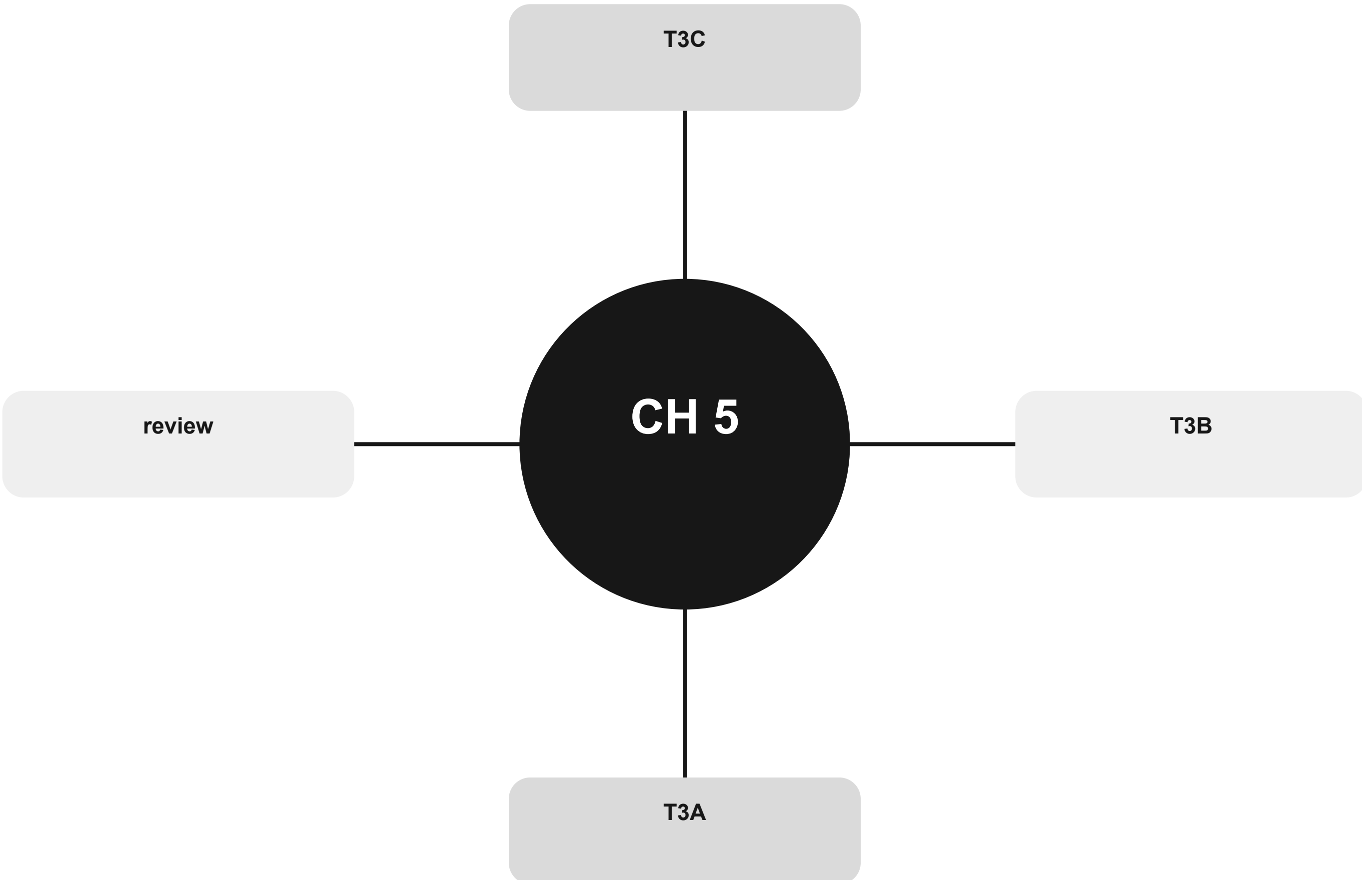
T7C10 Why should the outer jacket of coaxial cable be resistant to ultraviolet light: Ultraviolet light can damage the jacket and allow water to enter the cable.

T7C11 What is an advantage of foam-dielectric versus solid-dielectric coaxial cable: It has less loss per foot.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

Waves, frequency, and propagation

Picture what a radio wave is doing before memorizing a propagation name.



Official groups: T3A, T3B, T3C

Your route through this chapter

T3A What happens along the path

Recognize polarization, obstruction, multipath, and weather effects.

T3B Frequency and wavelength

Relate fields, speed, frequency, and wavelength.

T3C Propagation modes

Match a path shape and condition to its propagation name.

After the last group, use the official-pool anchors to identify weak spots. Do randomized exam practice outside this book.

What happens along the path

Recognize polarization, obstruction, multipath, and weather effects.

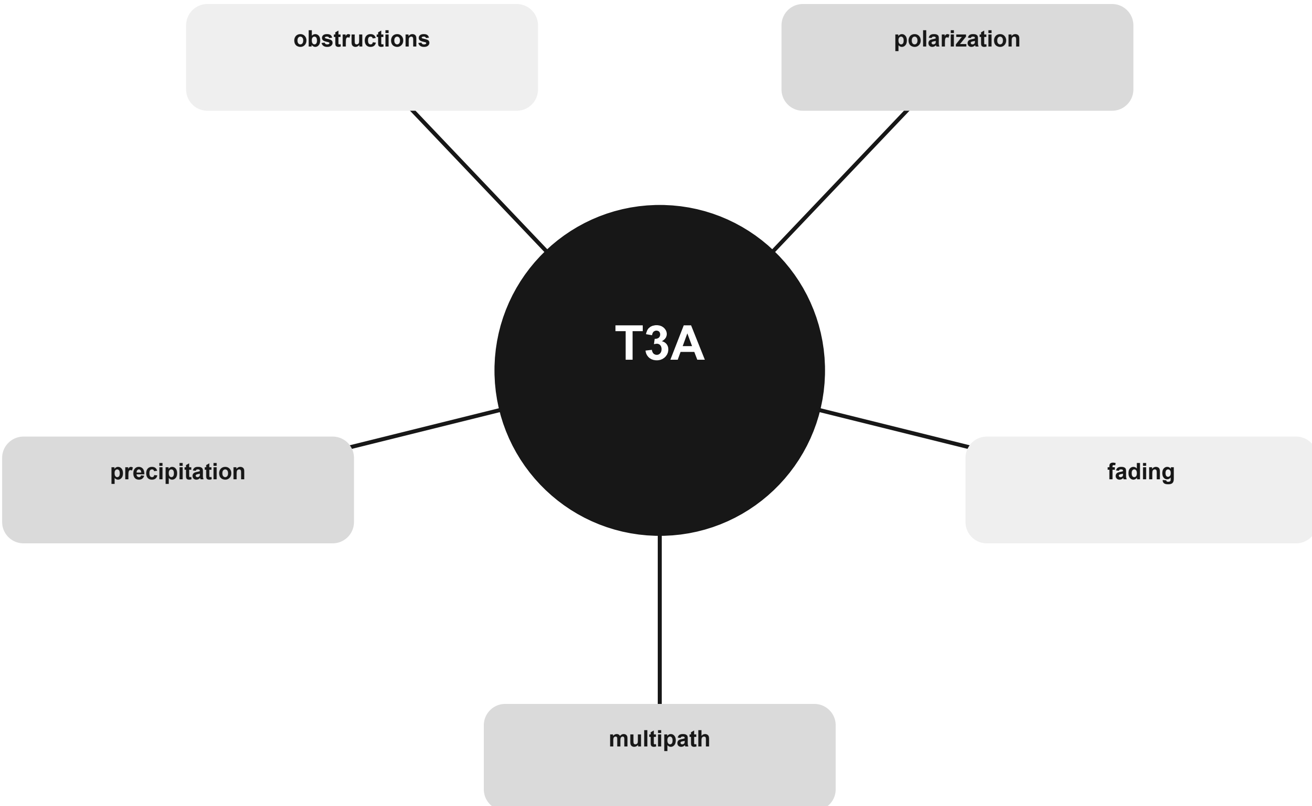
THE MENTAL MODEL

A received signal may arrive by more than one path. Those copies can reinforce or cancel, producing fading and flutter. Antenna polarization and local obstructions matter.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are multipath, fading, polarization, obstructions, precipitation.

What happens along the path concept map



Read from the center outward. Each node is a cue that should lead back to the same working model.

Recognize, then connect

RECOGNIZE

multipath

Use the relationship, not an isolated word.

fading

Use the relationship, not an isolated word.

polarization

Use the relationship, not an isolated word.

CONNECT

obstructions

Use the relationship, not an isolated word.

precipitation

Use the relationship, not an isolated word.

The exam often places related words close together. Do not choose by familiarity. Identify the requested job, path, unit, or rule first.

Official-pool anchors

T3A01 Why do VHF signal strengths sometimes vary greatly when the antenna is moved only a few feet: Multipath propagation cancels or reinforces signals.

T3A02 How does vegetation affect UHF and microwave signals: Absorbs signals, leading to poor reception of weak signals.

T3A03 What antenna polarization is normally used for long-distance CW and SSB contacts on the VHF and UHF bands: Horizontal.

T3A04 What is the effect of antenna cross-polarization over a line-of-sight VHF or UHF path: Received signal strength is reduced.

T3A05 When using a directional antenna, how might your station be able to communicate with a distant repeater if buildings or obstructions are blocking the direct line of sight path: Try to find a path that reflects signals to the repeater.

T3A06 What is the meaning of the term “picket fencing”: Rapid flutter on mobile signals due to multipath propagation.

T3A07 What weather condition might decrease range at microwave frequencies: Precipitation.

T3A08 What is a likely cause of irregular fading of signals propagated by the ionosphere: Random combining of signals arriving via different paths.

T3A09 Which of the following results from the fact that signals propagated by the ionosphere are elliptically polarized: Either vertically or horizontally polarized antennas may be used for transmission or reception.

T3A10 What effect does multi-path propagation have on data transmissions: Error rates are likely to increase.

T3A11 Which region of the atmosphere can reflect HF radio waves: The ionosphere.

T3A12 What effect does fog or rain have on 10-meter and 6-meter band signals: Little effect.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

Frequency and wavelength

Relate fields, speed, frequency, and wavelength.

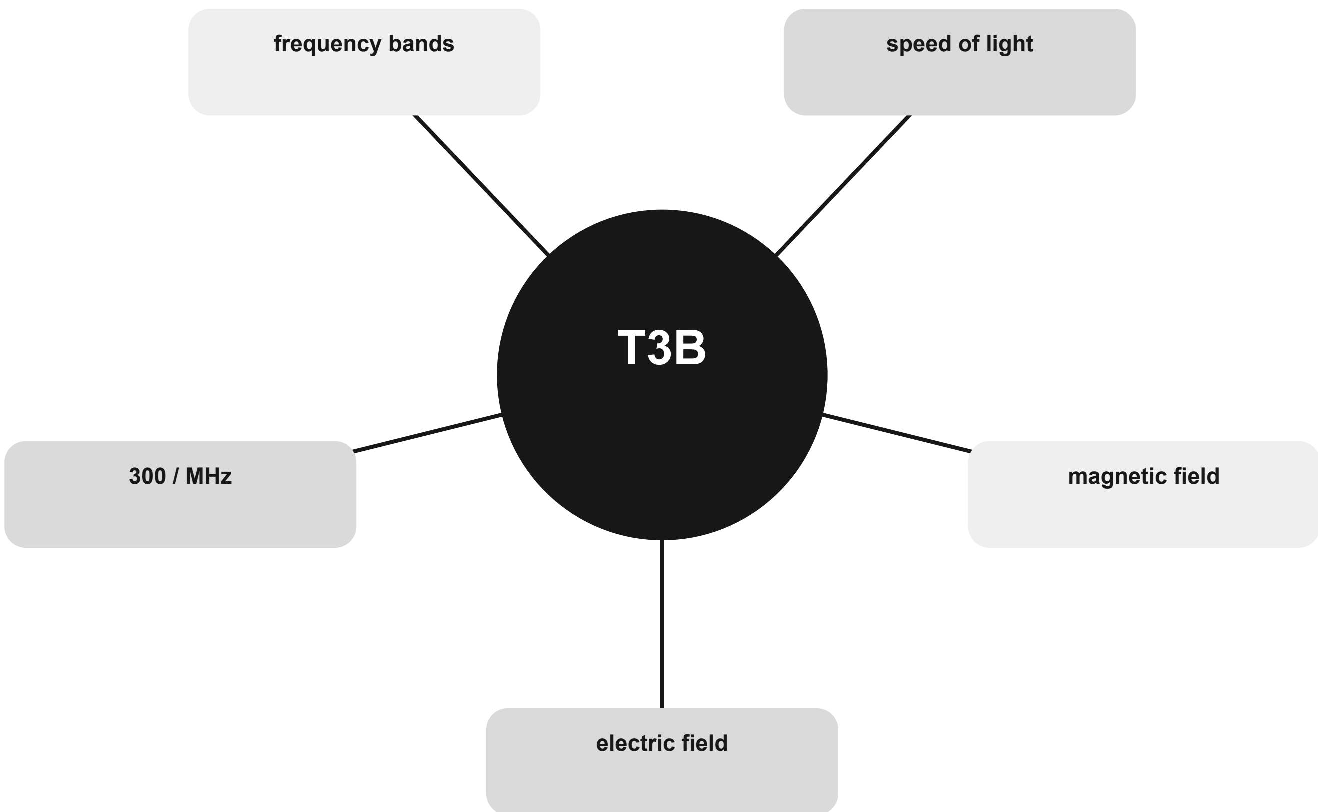
THE MENTAL MODEL

Radio waves have electric and magnetic fields at right angles. In free space they travel at the speed of light. Higher frequency means shorter wavelength; wavelength in meters is approximately 300 divided by megahertz.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are electric field, magnetic field, speed of light, frequency bands, 300 / MHz.

Frequency and wavelength concept map



Read from the center outward. Each node is a cue that should lead back to the same working model.

Recognize, then connect

RECOGNIZE

electric field

Use the relationship, not an isolated word.

magnetic field

Use the relationship, not an isolated word.

speed of light

Use the relationship, not an isolated word.

CONNECT

frequency bands

Use the relationship, not an isolated word.

300 / MHz

Use the relationship, not an isolated word.

The exam often places related words close together. Do not choose by familiarity. Identify the requested job, path, unit, or rule first.

Official-pool anchors

T3B01 What is the relationship between the electric and magnetic fields of an electromagnetic wave: They are at right angles.

T3B02 What property of a radio wave defines its polarization: The orientation of the electric field.

T3B03 What are the two components of a radio wave: Electric and magnetic fields.

T3B04 What is the velocity of a radio wave traveling through free space: Speed of light.

T3B05 What is the relationship between wavelength and frequency: Wavelength gets shorter as frequency increases.

T3B06 What is the formula for converting frequency to approximate wavelength in meters: Wavelength in meters equals 300 divided by frequency in megahertz.

T3B07 In addition to frequency, which of the following is used to identify amateur radio bands: The approximate wavelength in meters.

T3B08 What frequency range is referred to as VHF: 30 MHz to 300 MHz.

T3B09 What frequency range is referred to as UHF: 300 to 3000 MHz.

T3B10 What frequency range is referred to as HF: 3 to 30 MHz.

T3B11 What is the approximate velocity of a radio wave in free space: 300,000,000 meters per second.

T3B12 Which of these frequencies travels at the highest velocity in free space: All radio frequencies travel at the same velocity.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

Propagation modes

Match a path shape and condition to its propagation name.

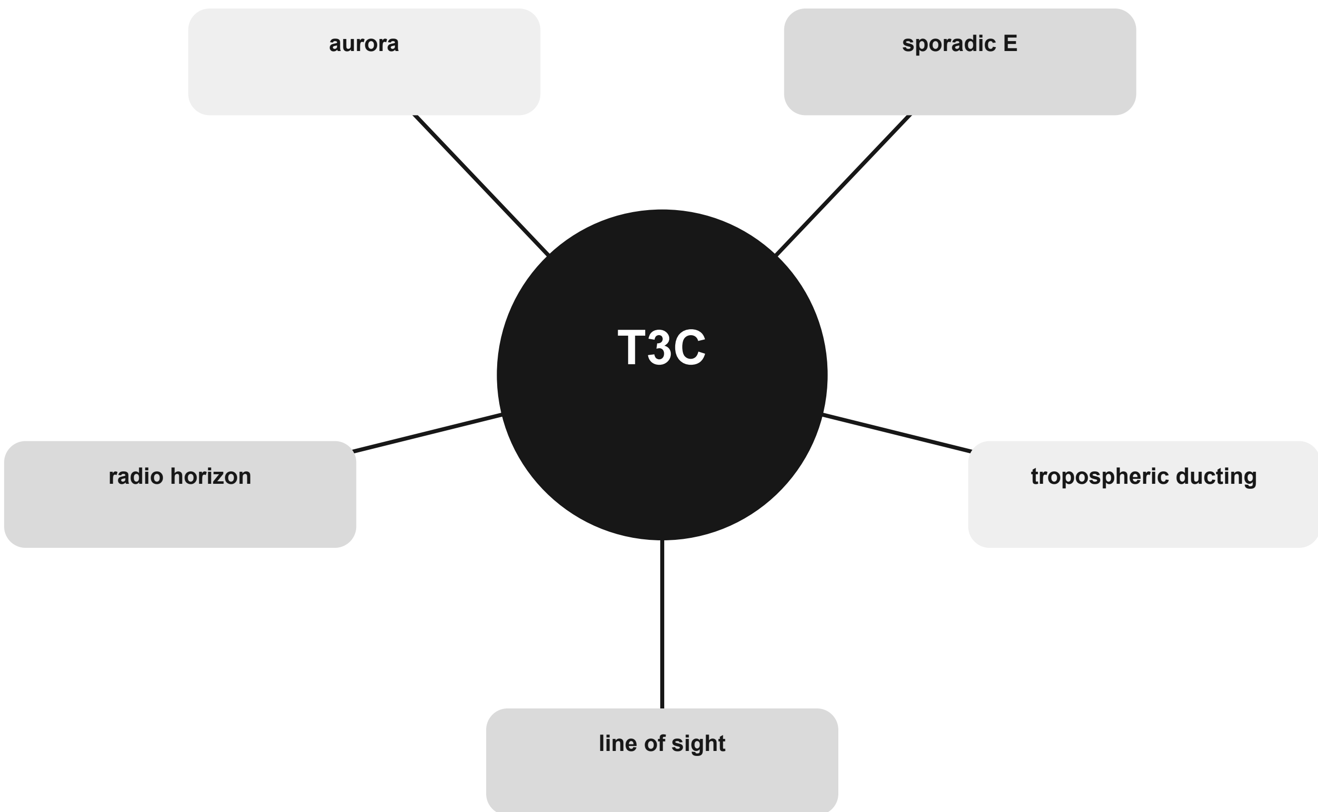
THE MENTAL MODEL

VHF and UHF are often line-of-sight, but atmosphere, terrain, aurora, meteors, and sporadic ionization can extend or distort the path. HF ionospheric propagation is much more common.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are line of sight, tropospheric ducting, sporadic E, aurora, radio horizon.

Propagation modes concept map



Read from the center outward. Each node is a cue that should lead back to the same working model.

Recognize, then connect

RECOGNIZE

line of sight

Use the relationship, not an isolated word.

tropospheric ducting

Use the relationship, not an isolated word.

sporadic E

Use the relationship, not an isolated word.

CONNECT

aurora

Use the relationship, not an isolated word.

radio horizon

Use the relationship, not an isolated word.

The exam often places related words close together. Do not choose by familiarity. Identify the requested job, path, unit, or rule first.

Official-pool anchors

T3C01 Why are simplex UHF signals rarely heard beyond their radio horizon: UHF signals are usually not propagated by the ionosphere.

T3C02 What is a characteristic of HF communication compared with communications on VHF and higher frequencies: Long-distance ionospheric propagation is far more common on HF.

T3C03 What is one characteristic of VHF signals received via auroral backscatter: They are distorted, with a characteristic raspy sound.

T3C04 Which of the following types of propagation is most commonly associated with occasional strong signals on the 10-, 6-, and 2-meter bands from beyond the radio horizon: Sporadic E.

T3C05 Which of the following effects may allow radio signals to travel beyond obstructions between the transmitting and receiving stations: Knife-edge diffraction.

T3C06 What type of propagation is responsible for allowing over-the-horizon VHF and UHF communications to ranges of approximately 300 miles on a regular basis: Tropospheric ducting.

T3C07 What band is best suited for communicating via meteor scatter: 6 meters.

T3C08 What causes tropospheric ducting: Temperature inversions in the atmosphere.

T3C09 What is generally the best time for long-distance 10-meter band propagation via the F region: From dawn to shortly after sunset during periods of high sunspot activity.

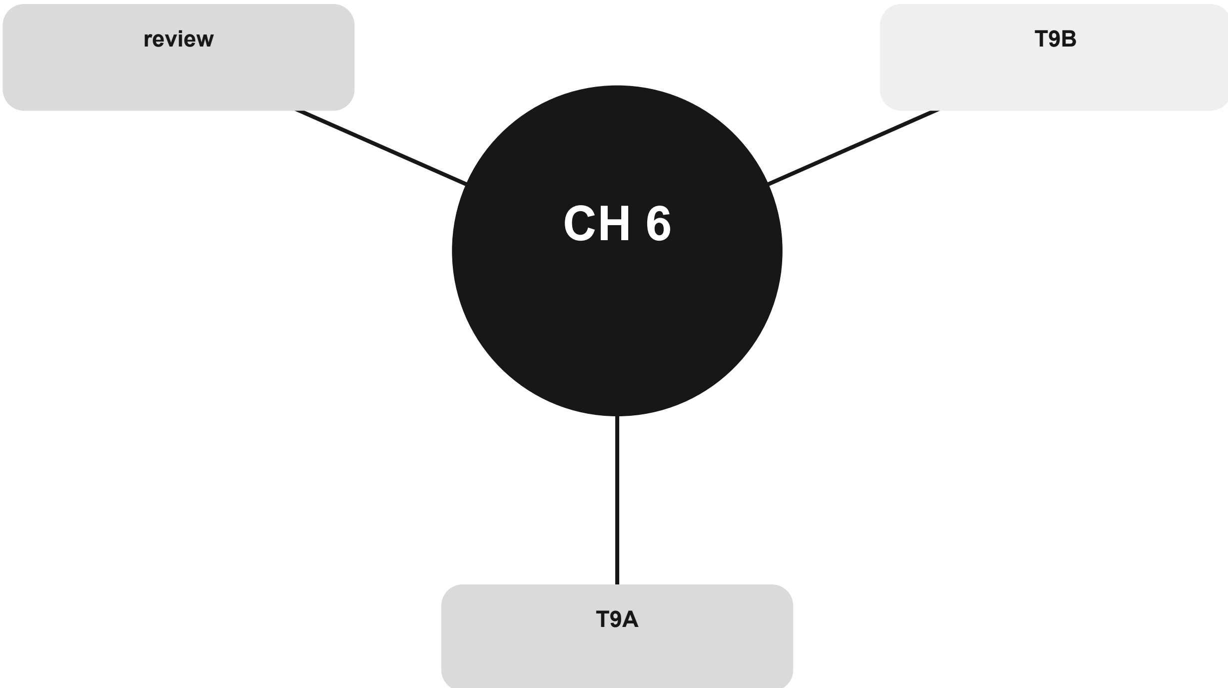
T3C10 Which of the following bands may provide long-distance communications via the ionosphere’s F region during the peak of the sunspot cycle: 6 and 10 meters.

T3C11 Why is the radio horizon for VHF and UHF signals more distant than the visual horizon: The atmosphere refracts radio waves slightly.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

Antennas, feed lines, and SWR

See the complete path from transmitter to feed line to antenna and outward.



Official groups: T9A, T9B

Your route through this chapter

T9A Antenna shape, direction, and length

Connect polarization, gain, resonance, and radiation pattern.

T9B The feed-line chain

See impedance, loss, connectors, weather, and matching as one system.

After the last group, use the official-pool anchors to identify weak spots. Do randomized exam practice outside this book.

Antenna shape, direction, and length

Connect polarization, gain, resonance, and radiation pattern.

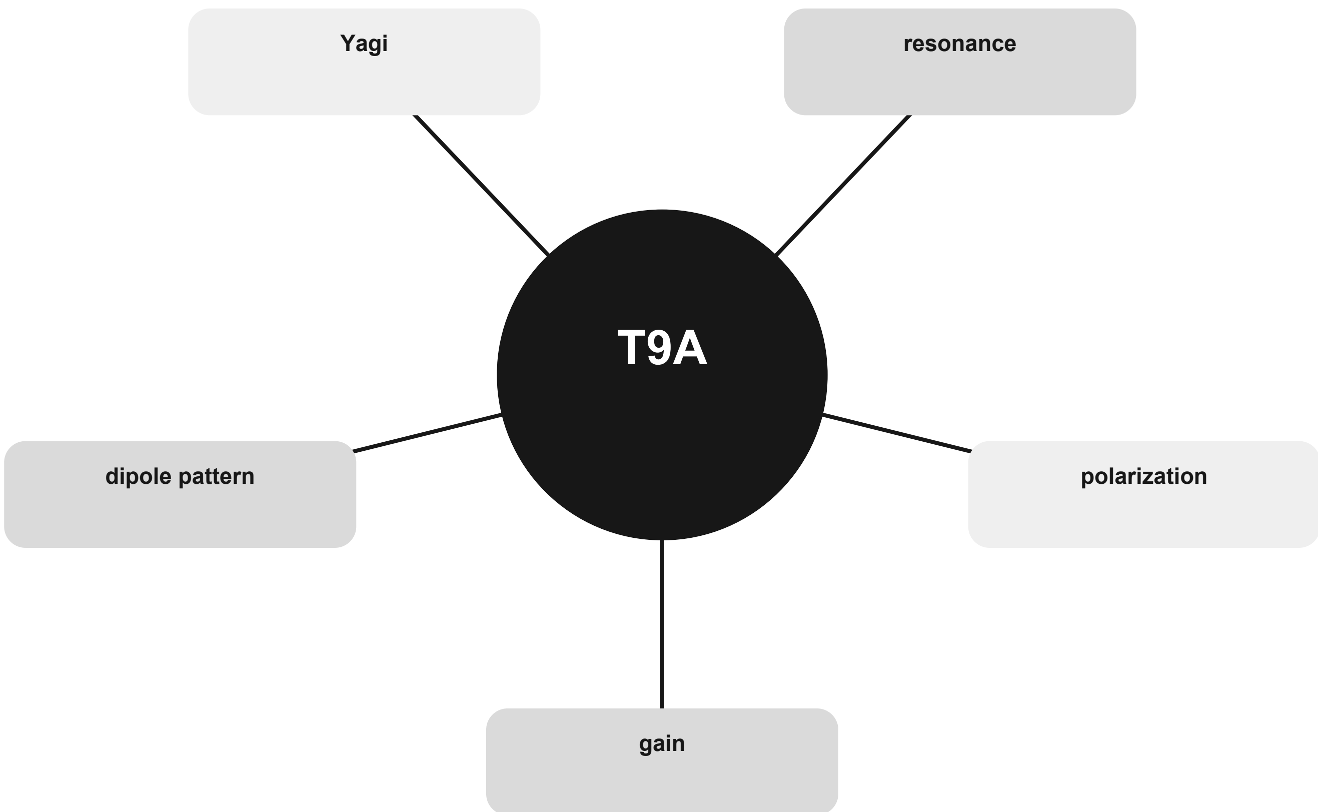
THE MENTAL MODEL

An antenna does not create power; gain concentrates radiation in favored directions. Resonant length changes inversely with frequency, and orientation determines polarization.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are gain, polarization, resonance, Yagi, dipole pattern.

Antenna shape, direction, and length concept map



Read from the center outward. Each node is a cue that should lead back to the same working model.

Recognize, then connect

RECOGNIZE

gain

Use the relationship, not an isolated word.

polarization

Use the relationship, not an isolated word.

resonance

Use the relationship, not an isolated word.

CONNECT

Yagi

Use the relationship, not an isolated word.

dipole pattern

Use the relationship, not an isolated word.

The exam often places related words close together. Do not choose by familiarity. Identify the requested job, path, unit, or rule first.

Official-pool anchors

T9A01 What is a beam antenna: An antenna that concentrates signals in one direction.

T9A02 Which of the following describes a type of antenna loading: Electrically lengthening by inserting inductors in radiating elements.

T9A03 How is the polarization of an antenna described: By the orientation of the electric field.

T9A04 What is a disadvantage of a handheld radio transceiver’s short flexible antenna compared to a full-sized quarter- wave antenna: It has low efficiency.

T9A05 Which of the following increases the resonant frequency of a dipole antenna: Shortening it.

T9A06 Which of the following types of antennas offers the greatest gain: Yagi.

T9A07 What is a potential drawback of using a handheld VHF transceiver inside a vehicle that lacks an externally mounted antenna: Signal strength is reduced due to the shielding effect of the vehicle.

T9A08 Why is a 19-inch-long vertical antenna often used on 2 meters: It is a resonant quarter-wave.

T9A09 What is an advantage of a 5/8-wavelength whip antenna for VHF or UHF mobile service compared to a 1/4-wave antenna: It has more gain.

T9A10 In which direction does a half-wave dipole antenna radiate the strongest signal: Broadside to the antenna.

T9A11 What is antenna gain: The increase in signal strength in a specified direction compared to a reference antenna.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

The feed-line chain

See impedance, loss, connectors, weather, and matching as one system.

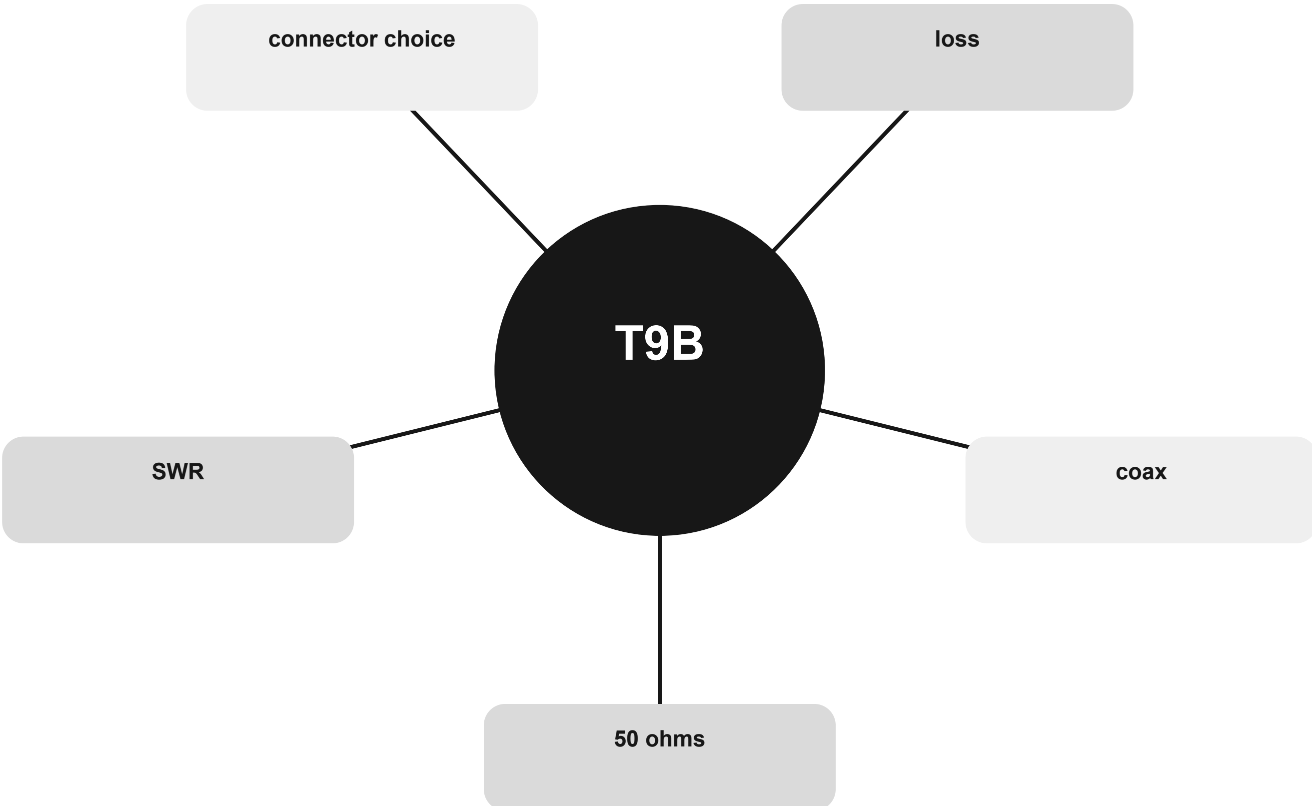
THE MENTAL MODEL

Coaxial cable is convenient and commonly 50 ohms in amateur systems. Loss rises with frequency. An antenna tuner changes the impedance match seen by the transmitter; it does not remove feed-line loss.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are 50 ohms, coax, loss, connector choice, SWR.

The feed-line chain concept map



Read from the center outward. Each node is a cue that should lead back to the same working model.

Recognize, then connect

RECOGNIZE	CONNECT
50 ohms Use the relationship, not an isolated word.	**connector choice** Use the relationship, not an isolated word.
coax Use the relationship, not an isolated word.	**SWR** Use the relationship, not an isolated word.
loss Use the relationship, not an isolated word.	

The exam often places related words close together. Do not choose by familiarity. Identify the requested job, path, unit, or rule first.

Official-pool anchors

T9B01 Which of the following connectors should be carefully taped for weather protection when used outdoors: All these choices are correct.

T9B02 What is the most common impedance of coaxial cables used in amateur radio: 50 ohms.

T9B03 Why is coaxial cable the most common feed line for amateur radio antenna systems: It is easy to use and requires few special installation considerations.

T9B04 What is the major function of an antenna tuner (antenna coupler): It matches the antenna system impedance to the transceiver's output impedance.

T9B05 What happens as the frequency of a signal in coaxial cable is increased: The loss increases.

T9B06 Which of the following connector types is most suitable as an RF connector for frequencies above 400 MHz: Type N.

T9B07 Which of the following is true of PL-259 type coax connectors: They are commonly used at HF and VHF frequencies.

T9B08 Which of the following is a source of loss in coaxial feed line: All these choices are correct.

T9B09 What can cause erratic changes in SWR: Loose connection in the antenna or feed line.

T9B10 What is the electrical difference between RG-58 and RG-213 coaxial cable: RG-213 cable has less loss at a given frequency.

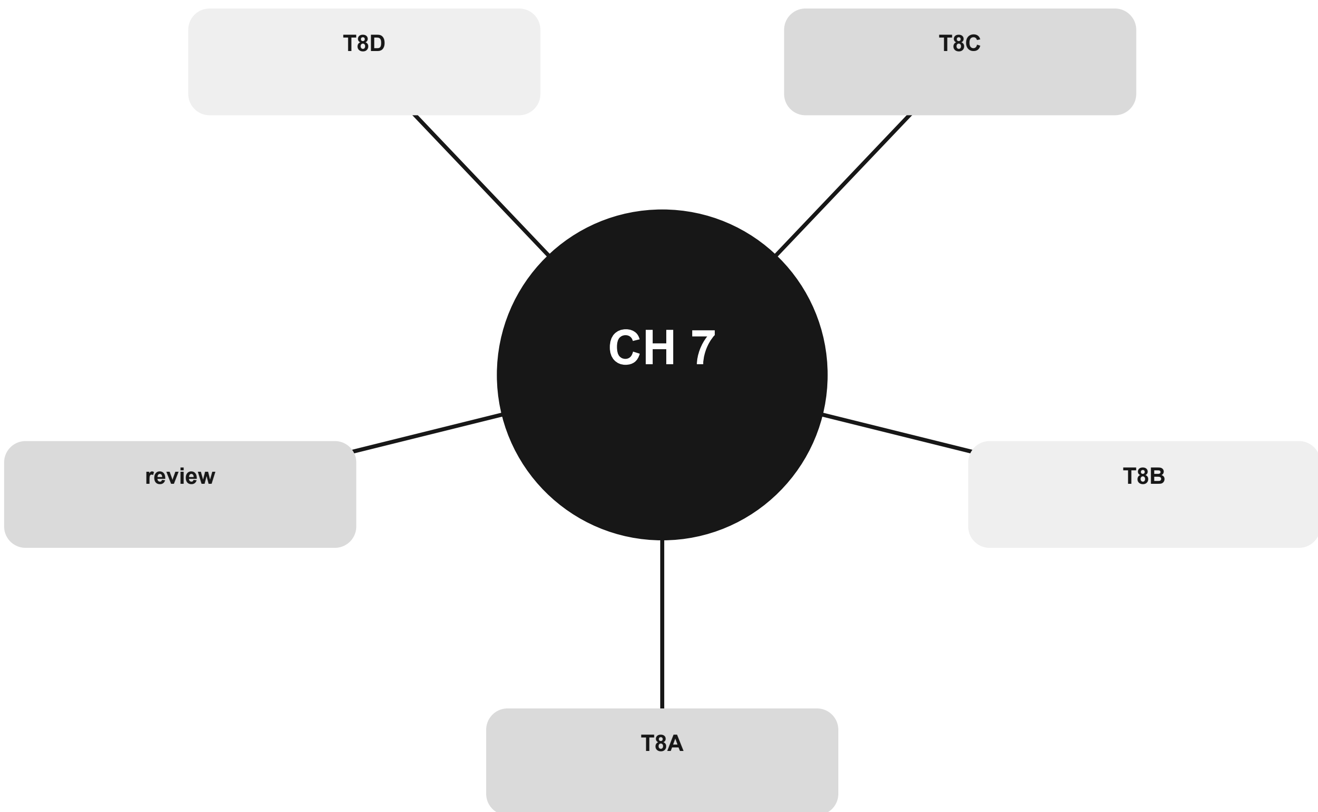
T9B11 Which of the following types of feed line has the lowest loss: Air-insulated hardline.

T9B12 What is standing wave ratio (SWR): A measure of how well a load is matched to a transmission line.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

Signals and ways hams communicate

Match signal type and operating method to purpose, bandwidth, and path.



Official groups: T8A, T8B, T8C, T8D

Your route through this chapter

T8A Modes and bandwidth

Match FM, SSB, CW, and television signals to typical bandwidth and use.

T8B Satellite paths

Separate uplink, downlink, orbit data, Doppler shift, and power discipline.

T8C Operating activities and internet linking

Recognize direction finding, contest exchange, location grids, and linked systems.

T8D Digital modes as functions

Match digital names to the job they perform.

After the last group, use the official-pool anchors to identify weak spots. Do randomized exam practice outside this book.

Modes and bandwidth

Match FM, SSB, CW, and television signals to typical bandwidth and use.

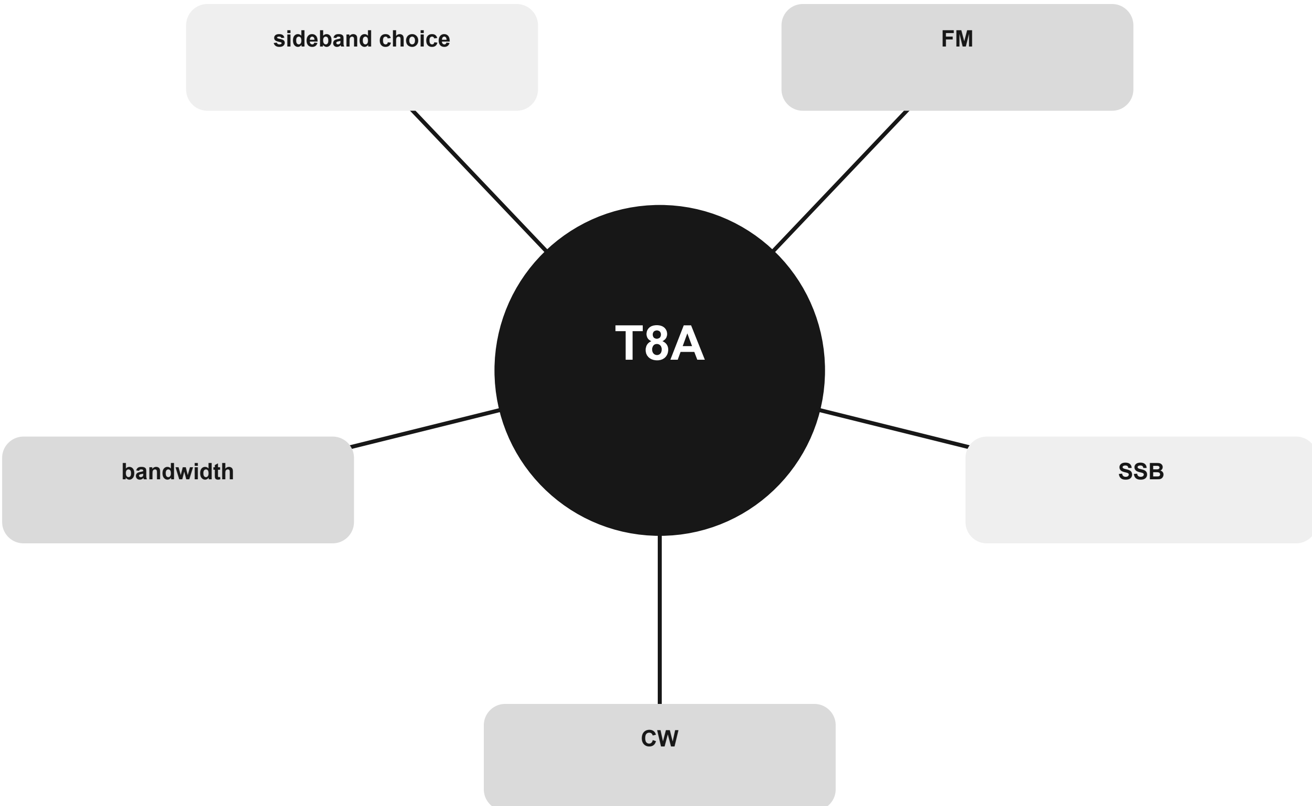
THE MENTAL MODEL

Different emission types occupy different widths of spectrum. CW is very narrow, SSB is narrower than FM voice, and fast-scan television is much wider.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are CW, SSB, FM, sideband choice, bandwidth.

Modes and bandwidth concept map



Read from the center outward. Each node is a cue that should lead back to the same working model.

Recognize, then connect

RECOGNIZE

CW

Use the relationship, not an isolated word.

SSB

Use the relationship, not an isolated word.

FM

Use the relationship, not an isolated word.

CONNECT

sideband choice

Use the relationship, not an isolated word.

bandwidth

Use the relationship, not an isolated word.

The exam often places related words close together. Do not choose by familiarity. Identify the requested job, path, unit, or rule first.

Official-pool anchors

T8A01 Which of the following is a form of amplitude modulation: Single sideband.

T8A02 What type of modulation is commonly used for VHF packet radio transmissions: FM or PM.

T8A03 Which type of voice mode is often used for long-distance (weak signal) contacts on the VHF and UHF bands: SSB.

T8A04 Which type of modulation is commonly used for VHF and UHF voice repeaters: FM or PM.

T8A05 Which of the following signal types has the narrowest bandwidth: CW.

T8A06 Which sideband is normally used for 10-meter HF, VHF, and UHF single-sideband communications: Upper sideband.

T8A07 What is one characteristic of single sideband (SSB) compared to FM: SSB signals have narrower bandwidth.

T8A08 What is the approximate bandwidth of a typical single sideband (SSB) voice signal: 3 kHz.

T8A09 What is the approximate bandwidth of an FM voice signal on VHF repeaters: Between 10 and 15 kHz.

T8A10 What is the approximate bandwidth of AM fast-scan TV transmissions: About 6 MHz.

T8A11 What is the approximate bandwidth required to transmit a CW signal: 150 Hz.

T8A12 Which of the following is a disadvantage of FM compared with single sideband: Only one signal can be received at a time.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

Satellite paths

Separate uplink, downlink, orbit data, Doppler shift, and power discipline.

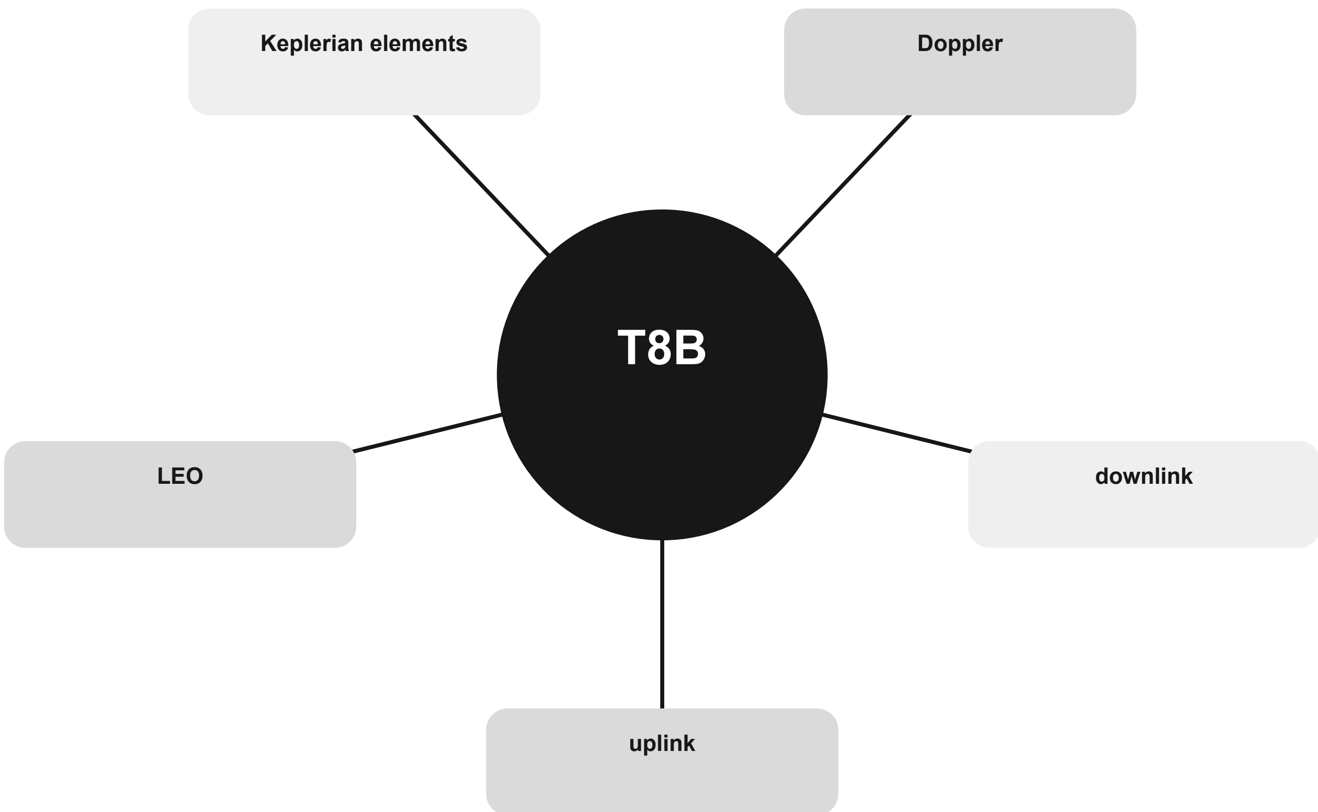
THE MENTAL MODEL

A satellite receives on its uplink and transmits on its downlink. Relative motion shifts the apparent frequency. Tracking data predicts the pass; excessive uplink power blocks other users.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are uplink, downlink, Doppler, Keplerian elements, LEO.

Satellite paths concept map



Read from the center outward. Each node is a cue that should lead back to the same working model.

Recognize, then connect

RECOGNIZE

uplink

Use the relationship, not an isolated word.

downlink

Use the relationship, not an isolated word.

Doppler

Use the relationship, not an isolated word.

CONNECT

Keplerian elements

Use the relationship, not an isolated word.

LEO

Use the relationship, not an isolated word.

The exam often places related words close together. Do not choose by familiarity. Identify the requested job, path, unit, or rule first.

Official-pool anchors

T8B01 What telemetry information is typically transmitted by satellite beacons: Health and status of the satellite.

T8B02 What is the impact of using excessive effective radiated power on a satellite uplink: Blocking access by other users.

T8B03 Which of the following are provided by satellite tracking programs: All these choices are correct.

T8B04 What mode of transmission is commonly used by amateur radio satellites: All these choices are correct.

T8B05 What is a satellite beacon: A transmission from a satellite that contains status information.

T8B06 Which of the following are inputs to a satellite tracking program: The Keplerian elements.

T8B07 What is Doppler shift in reference to satellite communications: An observed change in signal frequency caused by relative motion between the satellite and Earth station.

T8B08 What does it mean if a satellite is operating in U/V mode: The satellite uplink is in the 70-centimeter band and the downlink is in the 2-meter band.

T8B09 What causes spin fading of satellite signals: Rotation of the satellite and its antennas.

T8B10 What does the term LEO mean in reference to communication satellites: Low Earth Orbit, which has a period of around 100 minutes.

T8B11 Who is permitted to receive telemetry from an amateur radio satellite: Anyone.

T8B12 Which of the following is a way to determine whether your satellite uplink power into a linear transponder satellite is neither too low nor too high: Your signal strength on the downlink should be about the same as the beacon.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

Operating activities and internet linking

Recognize direction finding, contest exchange, location grids, and linked systems.

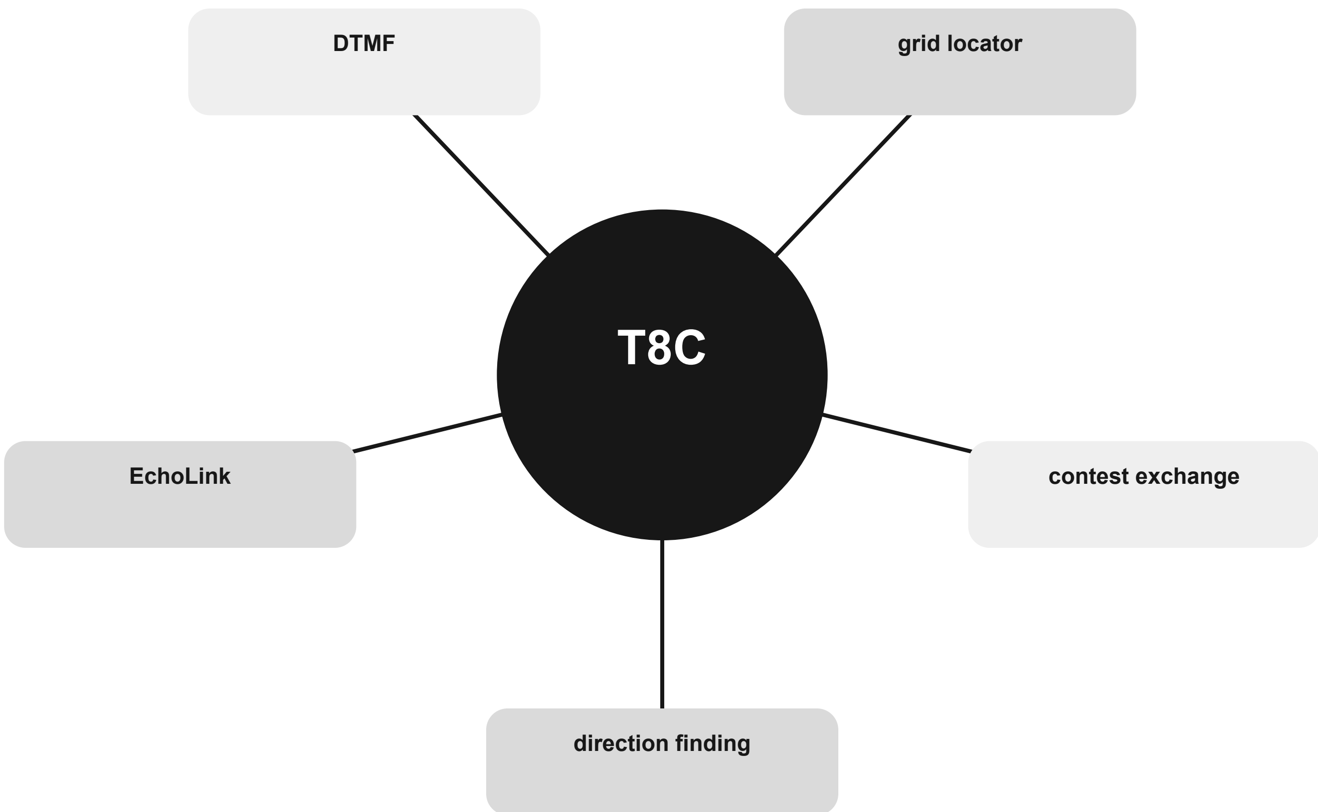
THE MENTAL MODEL

These activities use radio in different ways: direction finding locates a source, grid locators describe position, and internet-linked gateways extend voice paths between systems.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are direction finding, contest exchange, grid locator, DTMF, EchoLink.

Operating activities and internet linking concept map



Read from the center outward. Each node is a cue that should lead back to the same working model.

Recognize, then connect

RECOGNIZE

direction finding

Use the relationship, not an isolated word.

contest exchange

Use the relationship, not an isolated word.

grid locator

Use the relationship, not an isolated word.

CONNECT

DTMF

Use the relationship, not an isolated word.

EchoLink

Use the relationship, not an isolated word.

The exam often places related words close together. Do not choose by familiarity. Identify the requested job, path, unit, or rule first.

Official-pool anchors

T8C01 Which of the following methods is used to locate sources of noise interference or jamming: Radio direction finding.

T8C02 Which of these items would be useful for a hidden transmitter hunt: A directional antenna.

T8C03 What operating activity involves contacting as many stations as possible during a specified period: Contesting.

T8C04 Which of the following is good practice when contacting another station in a contest: Sending only the minimum information needed for proper identification and the contest exchange.

T8C05 What is a grid locator: A letter-number designator assigned to a geographic location.

T8C06 How is over the air access to Internet Radio Linking Project (IRLP) nodes accomplished: By using Dual-Tone Multi-Frequency (DTMF) signals.

T8C07 What is Voice Over Internet Protocol (VoIP): A method of delivering voice communications over the internet using digital techniques.

T8C08 What is the Internet Radio Linking Project (IRLP): A technique to connect amateur radio systems, such as repeaters, via the internet.

T8C09 Which of the following protocols enables an amateur station to transmit through a repeater without using a radio to initiate the transmission: EchoLink.

T8C10 What is required before using the EchoLink system: Register your call sign and provide proof of license.

T8C11 What is an amateur radio station that connects other amateur stations to the internet: A gateway.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

Digital modes as functions

Match digital names to the job they perform.

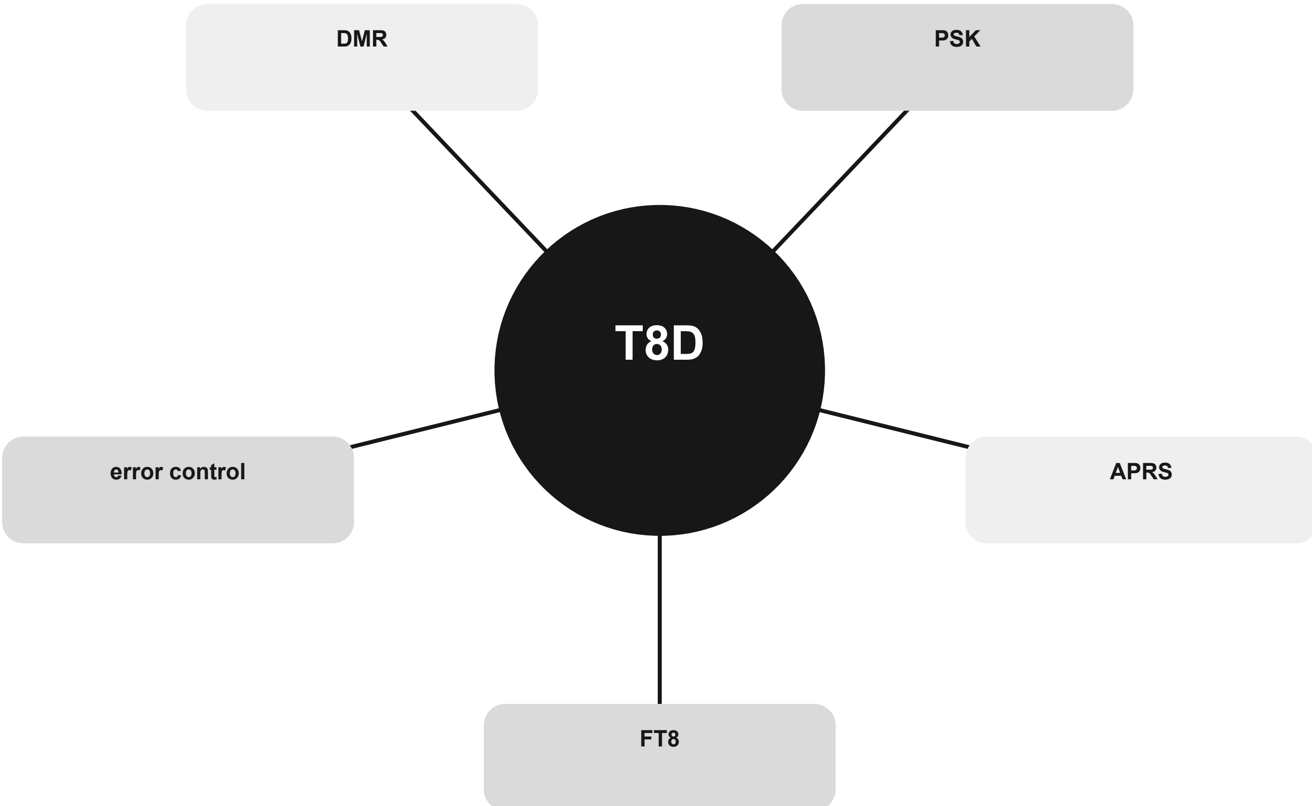
THE MENTAL MODEL

Some modes send text or position, some carry digital voice, and some work at very low signal levels. Error-control systems either correct data or request retransmission.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are FT8, APRS, PSK, DMR, error control.

Digital modes as functions concept map



Read from the center outward. Each node is a cue that should lead back to the same working model.

Recognize, then connect

RECOGNIZE

FT8

Use the relationship, not an isolated word.

APRS

Use the relationship, not an isolated word.

PSK

Use the relationship, not an isolated word.

CONNECT

DMR

Use the relationship, not an isolated word.

error control

Use the relationship, not an isolated word.

The exam often places related words close together. Do not choose by familiarity. Identify the requested job, path, unit, or rule first.

Official-pool anchors

T8D01 Which of the following is a digital communications mode: All these choices are correct.

T8D02 What is FT8: A digital mode capable of low signal-to-noise operation.

T8D03 What kind of data can be transmitted by APRS: All these choices are correct.

T8D04 What is meant by the term "NTSC?": An analog fast-scan color TV signal.

T8D05 Which of the following is an application of APRS: Providing real-time tactical digital communications in conjunction with a map showing the locations of stations.

T8D06 What does the abbreviation "PSK" mean: Phase Shift Keying.

T8D07 Which of the following describes DMR: A technique for time-multiplexing two digital voice signals on a single 12.5 kHz repeater channel.

T8D08 Which of the following is included in packet radio transmissions: All these choices are correct.

T8D09 What is CW: Another name for a Morse code transmission.

T8D10 Which of the following operating activities is supported by digital mode software in the WSJT-X software suite: All these choices are correct.

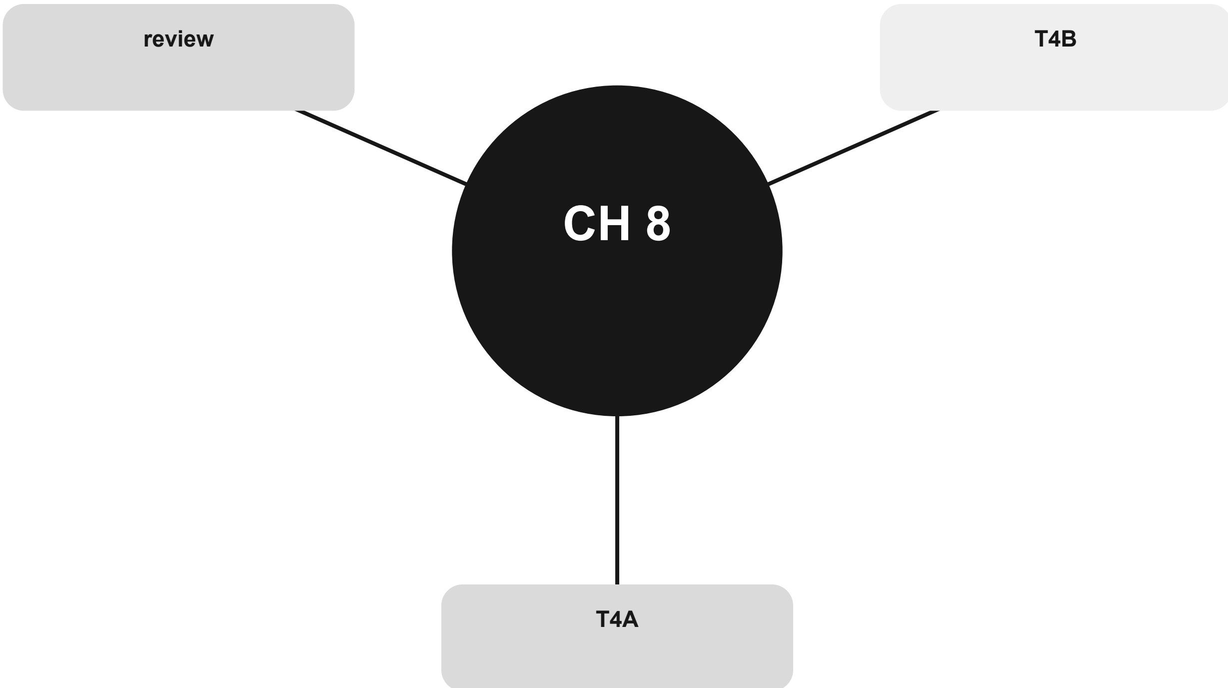
T8D11 What is the role of ARQ in a transmission system: An error correction method in which the receiving station detects errors and sends a request for retransmission.

T8D12 Which of the following best describes an amateur radio mesh network: An amateur-radio data network using commercial Wi-Fi equipment with modified firmware.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

First station and operating controls

Connect the station correctly and know which control changes which symptom.



Official groups: T4A, T4B

Your route through this chapter

T4A Connect the station

Build the power, audio, data, RF, and grounding paths.

T4B Controls change symptoms

Choose the control that acts on the problem you can hear or see.

After the last group, use the official-pool anchors to identify weak spots. Do randomized exam practice outside this book.

Connect the station

Build the power, audio, data, RF, and grounding paths.

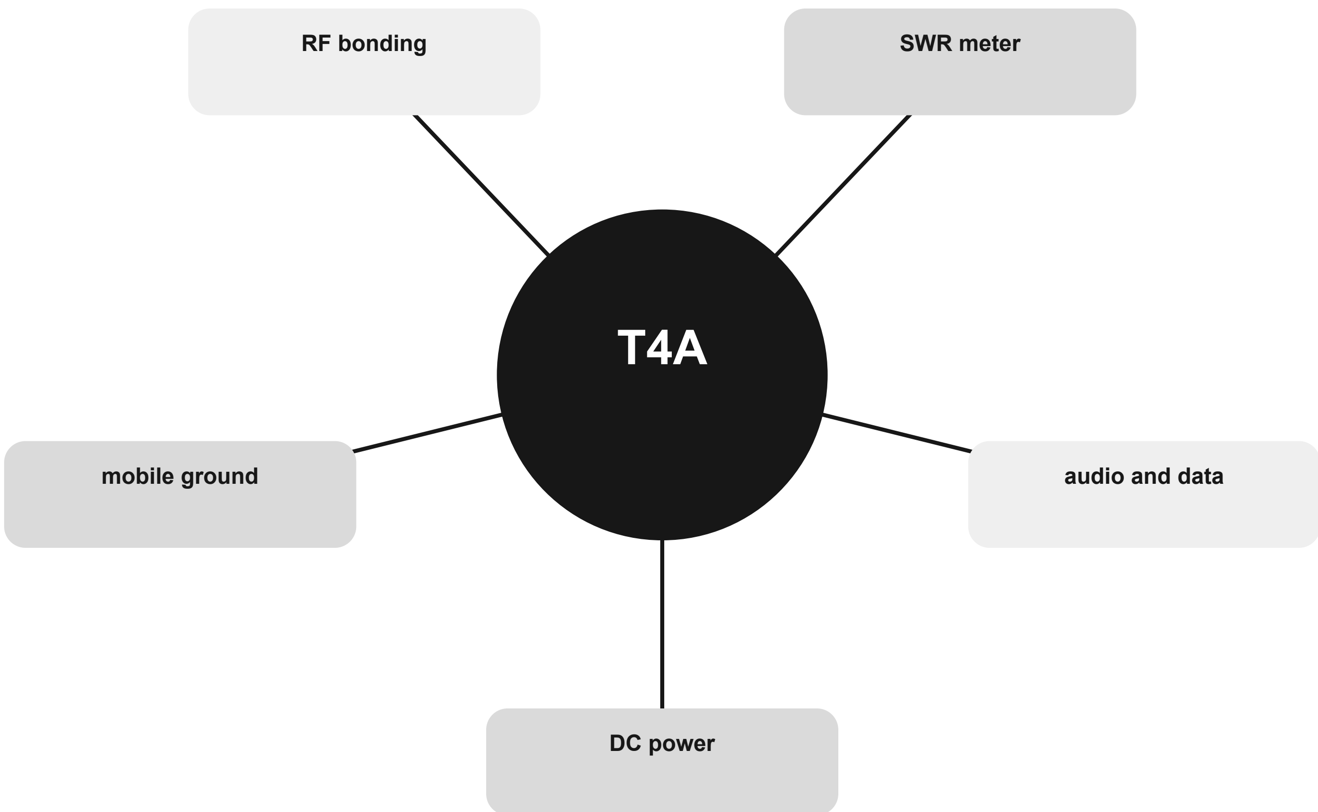
THE MENTAL MODEL

Treat the station as several separate paths. DC power must handle transmit current with low voltage drop. Audio/data interfaces connect the radio and computer; RF instruments sit in the feed line.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are DC power, audio and data, SWR meter, RF bonding, mobile ground.

Connect the station concept map



Read from the center outward. Each node is a cue that should lead back to the same working model.

Recognize, then connect

RECOGNIZE

DC power

Use the relationship, not an isolated word.

audio and data

Use the relationship, not an isolated word.

SWR meter

Use the relationship, not an isolated word.

CONNECT

RF bonding

Use the relationship, not an isolated word.

mobile ground

Use the relationship, not an isolated word.

The exam often places related words close together. Do not choose by familiarity. Identify the requested job, path, unit, or rule first.

Official-pool anchors

T4A01 Which of the following is an appropriate power supply rating for a typical 50-watt output mobile FM transceiver: 13.8 volts at 12 amperes.

T4A02 Which of the following should be considered when selecting an accessory SWR meter: The frequency and power level at which the measurements will be made.

T4A03 Why are short, heavy-gauge wires used for a transceiver’s DC power connection: To minimize voltage drop when transmitting.

T4A04 How are the audio input and output of a transceiver connected in a station configured to operate using FT8: To the audio output and input of a computer running FT8 software.

T4A05 Where should an RF power meter be installed: In the feed line, between the transmitter and antenna.

T4A06 What signals are used in a computer-radio interface for digital mode operation: Receive audio, transmit audio, and transmitter keying.

T4A07 Which of the following is one of the connections required between a computer and a transceiver to operate digital modes: Computer “line in” to transceiver speaker connector.

T4A08 Which of the following conductors is preferred for bonding at RF: Flat copper strap.

T4A09 How can you determine the length of time that equipment can be powered from a battery: Divide the battery ampere-hour rating by the average current draw of the equipment.

T4A10 What function does a digital mode hotspot perform for nearby transceivers: Communication with a digital voice or data network.

T4A11 Where should the negative power return of a mobile transceiver be connected in a vehicle: At the 12-volt battery chassis ground.

T4A12 What is an electronic keyer: A device that assists in manual sending of Morse code.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

Controls change symptoms

Choose the control that acts on the problem you can hear or see.

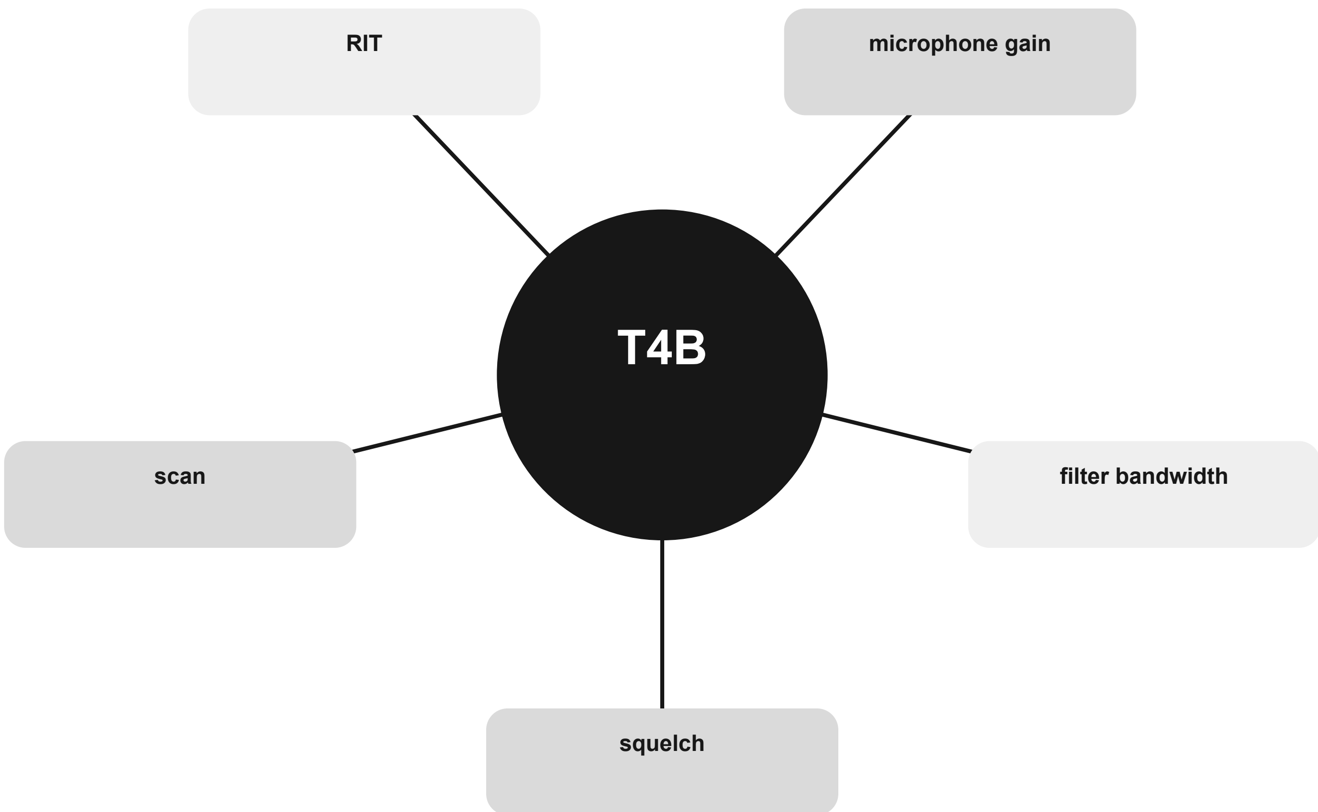
THE MENTAL MODEL

Squelch controls when receiver audio opens, filters control received bandwidth, microphone gain affects transmit audio, RIT shifts receive tuning, and scanning searches memories or frequencies.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are squelch, filter bandwidth, microphone gain, RIT, scan.

Controls change symptoms concept map



Read from the center outward. Each node is a cue that should lead back to the same working model.

Recognize, then connect

RECOGNIZE

squelch

Use the relationship, not an isolated word.

filter bandwidth

Use the relationship, not an isolated word.

microphone gain

Use the relationship, not an isolated word.

CONNECT

RIT

Use the relationship, not an isolated word.

scan

Use the relationship, not an isolated word.

The exam often places related words close together. Do not choose by familiarity. Identify the requested job, path, unit, or rule first.

Official-pool anchors

T4B01 What is the effect of excessive microphone gain on SSB transmissions: Distorted transmitted audio.

T4B02 Which of the following can be used to enter a transceiver’s operating frequency: The keypad or VFO knob.

T4B03 How is squelch adjusted so that a weak FM signal can be heard: Set the squelch threshold so that receiver output audio is on all the time.

T4B04 What does an FM signal sound like when received slightly off frequency: The audio becomes distorted.

T4B05 What does the scanning function of an FM transceiver do: Tunes through a range of frequencies to check for activity.

T4B06 Which of the following controls could be used if the voice pitch of a single-sideband signal returning to your CQ call seems too high or low: The RIT or Clarifier.

T4B07 What is a DMR “code plug”: Configuration data loaded onto your radio to access repeaters and talkgroups.

T4B08 What is the advantage of having a choice of receiver filter bandwidths in a multimode transceiver: Permits noise or interference reduction by selecting a bandwidth matching the mode.

T4B09 How is a specific group of stations selected on a DMR digital voice transceiver: By entering the group’s identification code.

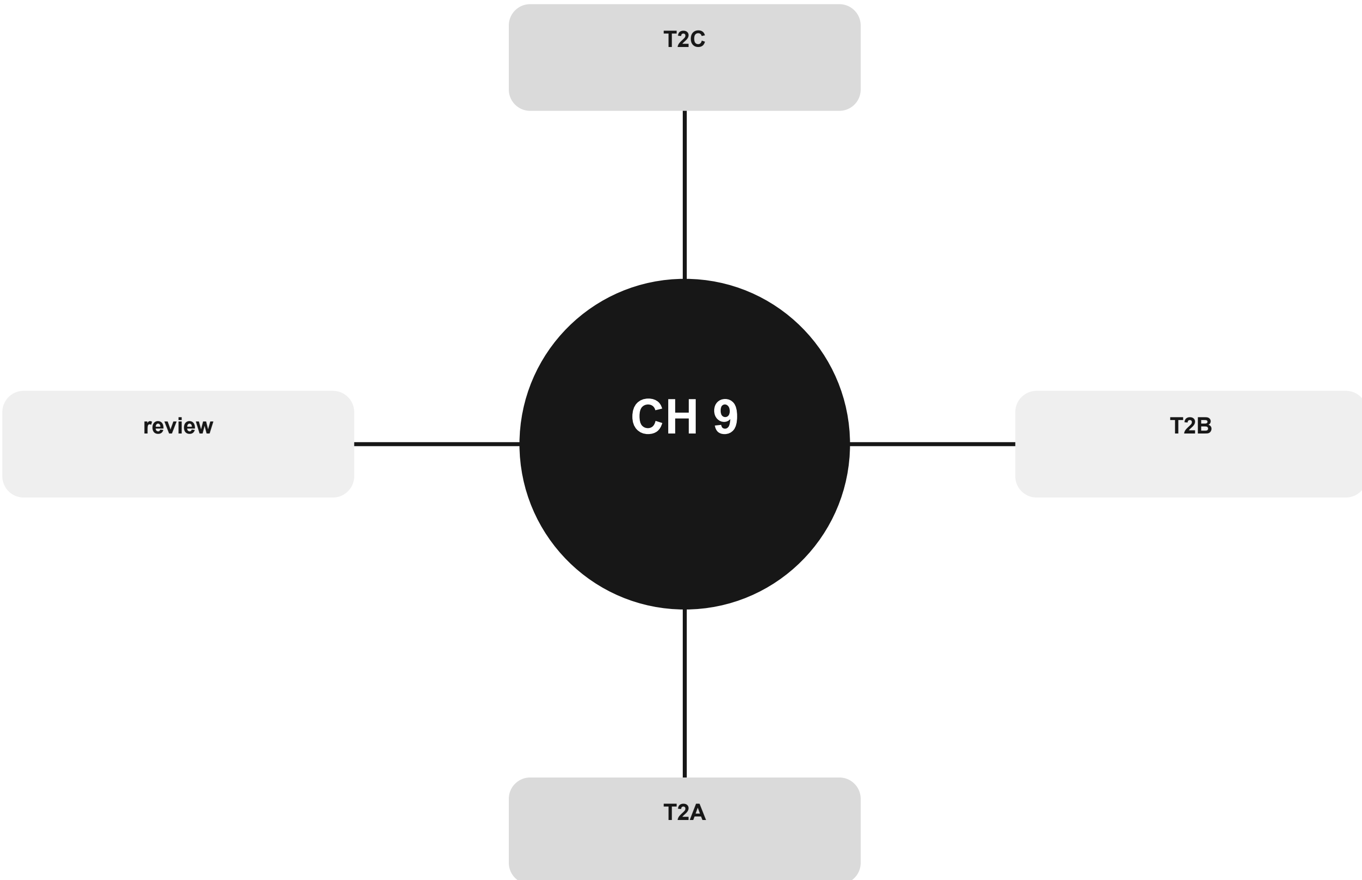
T4B10 Which of the following receiver filter bandwidths provides the best signal-to-noise ratio for SSB reception: 2400 Hz.

T4B11 Which of the following must be programmed into a D-STAR digital transceiver before transmitting: Your call sign.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

Getting on the air

Move from listening to a clear contact, repeater use, and disciplined net operation.



Official groups: T2A, T2B, T2C

Your route through this chapter

T2A Calling, listening, and frequency use

Make a contact and distinguish simplex, repeater offset, and band-plan guidance.

T2B Repeater access and shared channels

Understand tones, DMR identifiers, linked repeaters, and courteous frequency sharing.

T2C Nets and public service

Follow net control, message handling, and emergency boundaries.

After the last group, use the official-pool anchors to identify weak spots. Do randomized exam practice outside this book.

Calling, listening, and frequency use

Make a contact and distinguish simplex, repeater offset, and band-plan guidance.

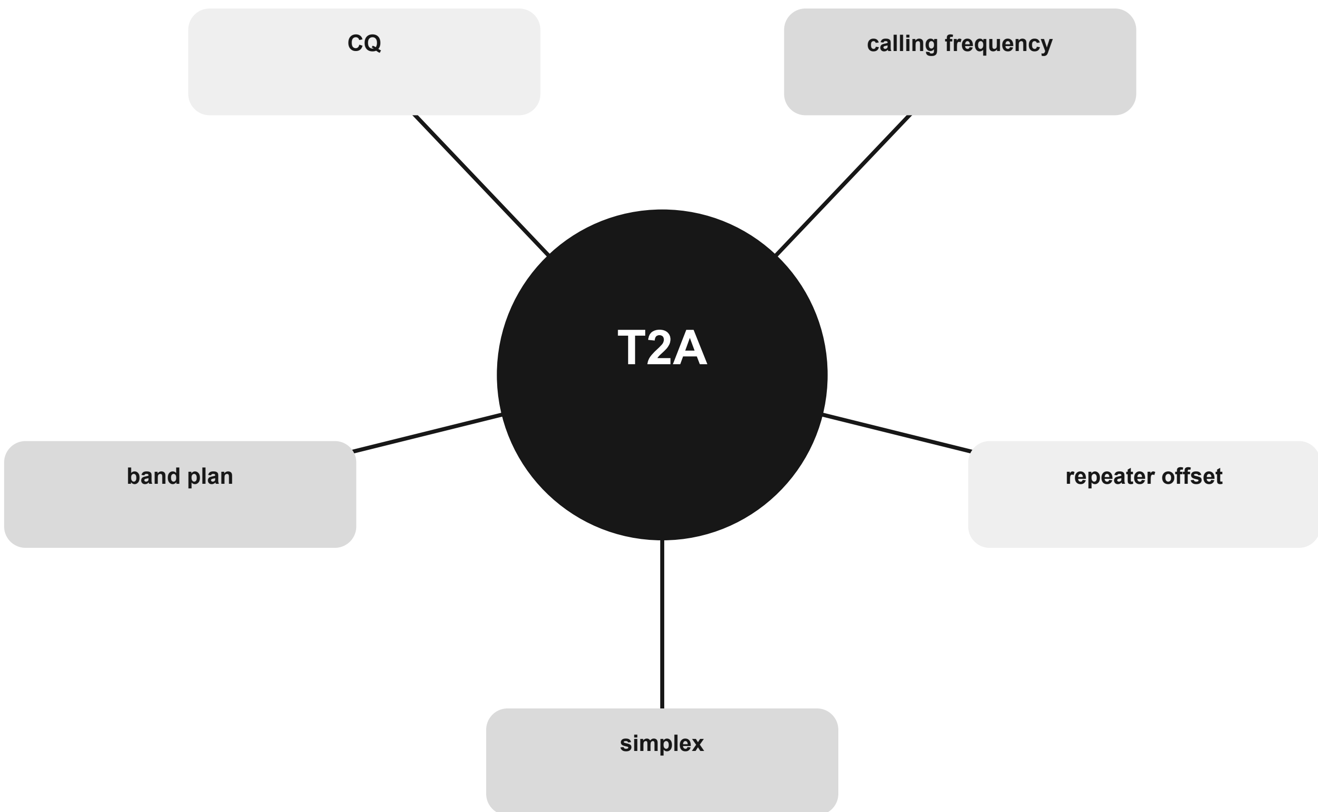
THE MENTAL MODEL

Listen first. On simplex, both stations use the same frequency. A repeater uses separate input and output frequencies; the difference is the offset. Band plans are voluntary coordination tools.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are simplex, repeater offset, calling frequency, CQ, band plan.

Calling, listening, and frequency use concept map



Read from the center outward. Each node is a cue that should lead back to the same working model.

Recognize, then connect

RECOGNIZE

CONNECT

simplex

Use the relationship, not an isolated word.

CQ

Use the relationship, not an isolated word.

repeater offset

Use the relationship, not an isolated word.

band plan

Use the relationship, not an isolated word.

calling frequency

Use the relationship, not an isolated word.

The exam often places related words close together. Do not choose by familiarity. Identify the requested job, path, unit, or rule first.

Official-pool anchors

T2A01 What is a common repeater frequency offset in the 2-meter band: Plus or minus 600 kHz.

T2A02 What is the national calling frequency for FM simplex operations in the 2-meter band: 146.520 MHz.

T2A03 What is a common repeater frequency offset in the 70-centimeter band: Plus or minus 5 MHz.

T2A04 What is an appropriate way to call another station on a repeater if you know the other station's call sign: Say the station's call sign, then identify with your call sign.

T2A05 How should you respond to a station calling CQ: Transmit the other station’s call sign followed by your call sign.

T2A06 What is an effective way to seek a call from any phone station when not using a repeater: Repeat “CQ” a few times, followed by “this is,” and your call sign, then pause to listen; repeat as necessary.

T2A07 What does the term "repeater offset" mean: The difference between a repeater’s transmit and receive frequencies.

T2A08 What is the meaning of the procedural signal “CQ”: Calling any station.

T2A09 Which of the following is a customary way to indicate a station is listening on a repeater and looking for a contact: The station’s call sign followed by the word “listening”.

T2A10 What is a band plan, beyond the privileges established by the FCC: A voluntary guideline for using different modes or activities within an amateur band.

T2A11 What term describes an amateur station that is transmitting and receiving on the same frequency: Simplex.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

Repeater access and shared channels

Understand tones, DMR identifiers, linked repeaters, and courteous frequency sharing.

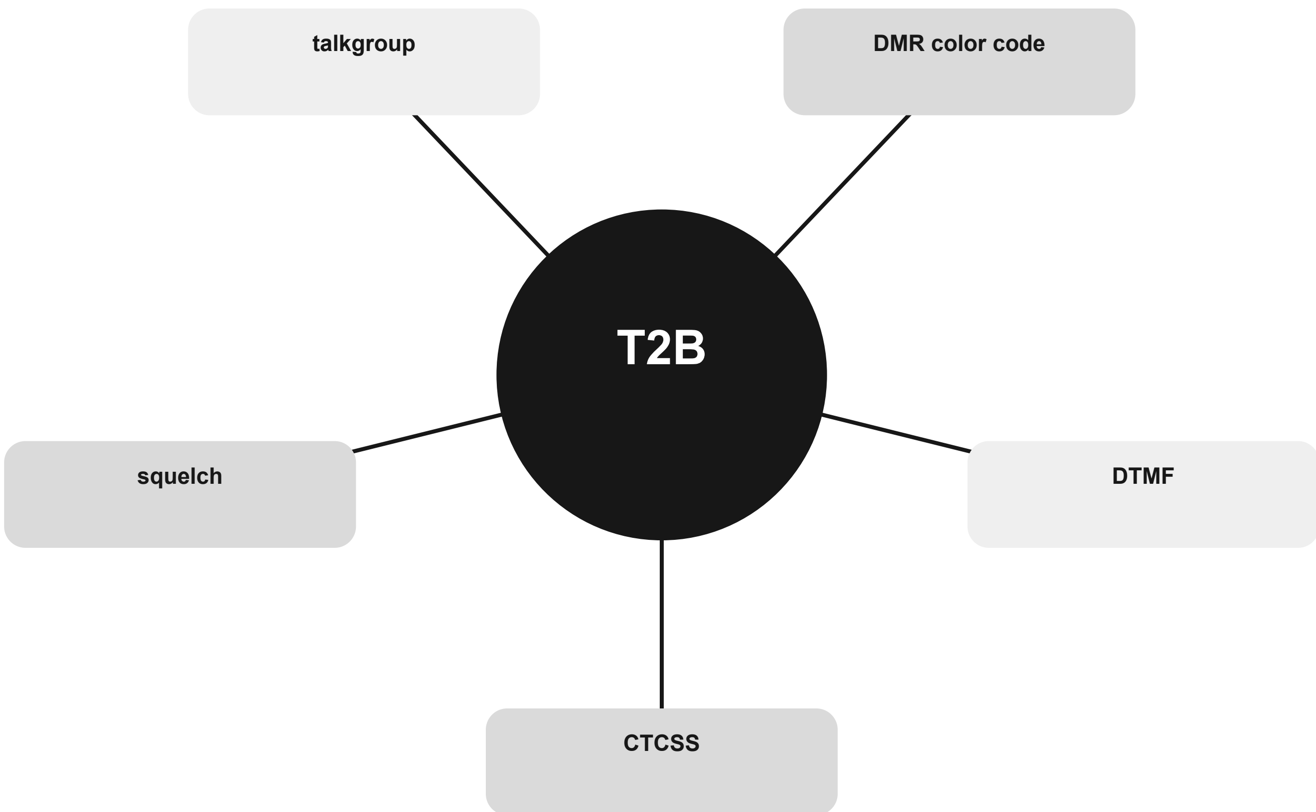
THE MENTAL MODEL

Analog repeaters may require a CTCSS tone; control functions may use DTMF. DMR organizes access with identifiers, color codes, and talkgroups. Squelch mutes idle noise.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are CTCSS, DTMF, DMR color code, talkgroup, squelch.

Repeater access and shared channels concept map



Read from the center outward. Each node is a cue that should lead back to the same working model.

Recognize, then connect

RECOGNIZE

CTCSS

Use the relationship, not an isolated word.

DTMF

Use the relationship, not an isolated word.

DMR color code

Use the relationship, not an isolated word.

CONNECT

talkgroup

Use the relationship, not an isolated word.

squelch

Use the relationship, not an isolated word.

The exam often places related words close together. Do not choose by familiarity. Identify the requested job, path, unit, or rule first.

Official-pool anchors

T2B01 What is the purpose of the reverse function on a VHF/UHF transceiver: To listen on a repeater's input frequency.

T2B02 What term describes the use of a sub-audible tone transmitted along with normal voice audio to open the squelch of a receiver: CTCSS.

T2B03 Which of the following describes a linked repeater network: A network of repeaters in which signals received by one repeater are transmitted by all the repeaters in the network.

T2B04 Which of the following could be the reason you are unable to access a repeater whose output you can hear: All these choices are correct.

T2B05 Which of the following would cause your FM transmission audio to drop out on voice peaks: You are talking too loudly.

T2B06 What type of signaling to a repeater uses two simultaneous audio tones: DTMF.

T2B07 How can you join a digital repeater's "talkgroup": Program your radio with the group's ID or code.

T2B08 Which of the following applies when two stations transmitting on the same frequency interfere with each other: The stations should negotiate continued use of the frequency.

T2B09 Why are simplex channels designated in the VHF/UHF band plans: So stations within range of each other can communicate without tying up a repeater.

T2B10 Which Q signal indicates that you are receiving interference from other stations: QRM.

T2B11 Which Q signal indicates that you are changing frequency: QSY.

T2B12 What is the digital color code used on DMR repeater systems: An access code which must be programmed into a DMR transmitter to access a specific repeater.

T2B13 What is the purpose of a squelch function: Mute the receiver audio when a signal is not present.

T2B14 Which of the following is a "talkgroup": An identifier used by DMR to organize radio traffic so that those who want to hear the group aren't bothered by other radio traffic.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

Nets and public service

Follow net control, message handling, and emergency boundaries.

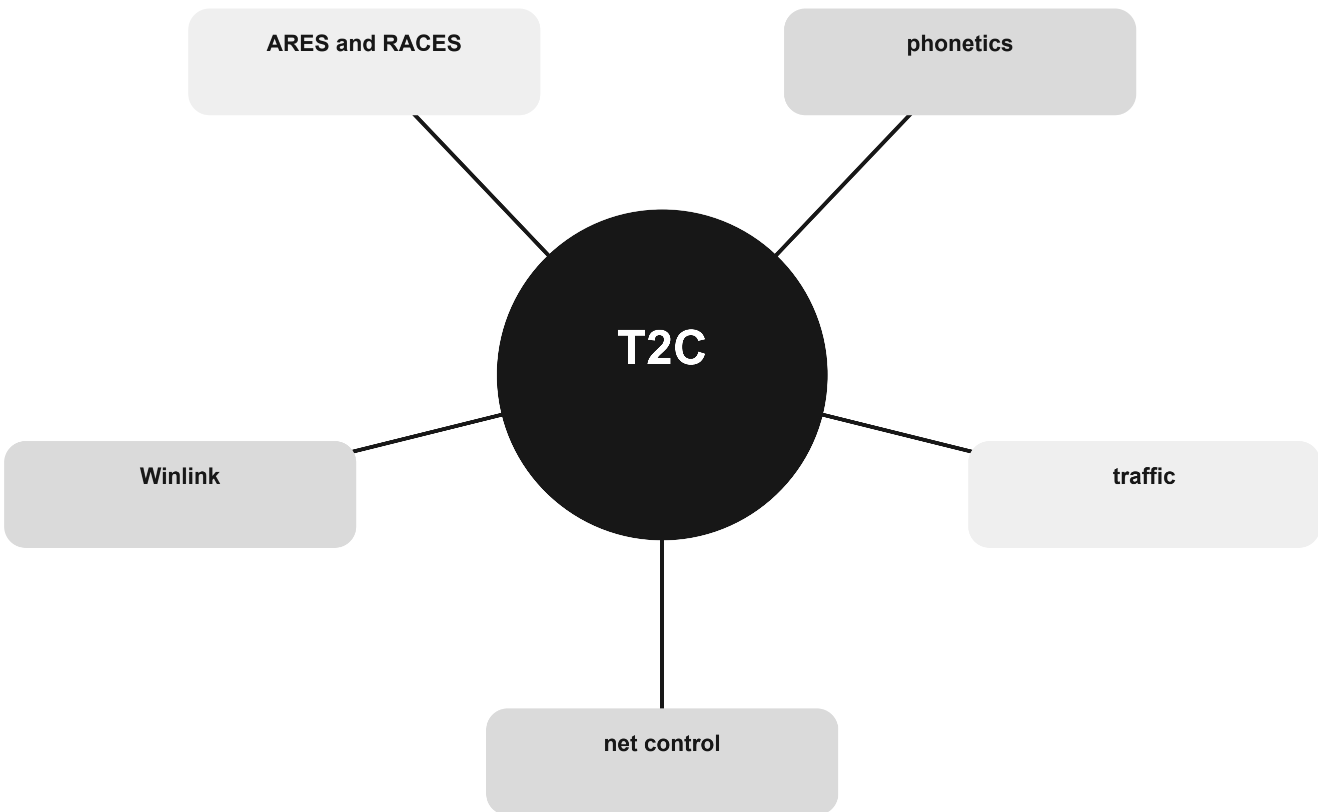
THE MENTAL MODEL

A directed net has a net control station coordinating traffic. Formal messages carry tracking information and a word count. Emergency circumstances do not erase FCC rules.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are net control, traffic, phonetics, ARES and RACES, Winlink.

Nets and public service concept map



Read from the center outward. Each node is a cue that should lead back to the same working model.

Recognize, then connect

RECOGNIZE

net control

Use the relationship, not an isolated word.

traffic

Use the relationship, not an isolated word.

phonetics

Use the relationship, not an isolated word.

CONNECT

ARES and RACES

Use the relationship, not an isolated word.

Winlink

Use the relationship, not an isolated word.

The exam often places related words close together. Do not choose by familiarity. Identify the requested job, path, unit, or rule first.

Official-pool anchors

T2C01 When do the FCC Part 97 Amateur Radio Service rules NOT apply to the operation of an amateur station: FCC rules always apply.

T2C02 Which of the following are typical duties of a Net Control Station: Call the net to order and direct communications between stations checking in.

T2C03 What technique is used to ensure that voice messages containing unusual words are received correctly: Spell the words using a standard phonetic alphabet.

T2C04 What is RACES: An FCC Part 97 amateur radio service for civil defense communications during national emergencies.

T2C05 What does the term “traffic” refer to in net operation: Formal messages exchanged by net stations.

T2C06 What is the Amateur Radio Emergency Service (ARES): A group of licensed amateurs who have voluntarily registered their qualifications and equipment for communications duty in the public service.

T2C07 Which of the following is standard practice when you participate in a net: Unless you are reporting an emergency, transmit only when directed by the net control station.

T2C08 Which of the following relays messages using email addresses based on amateur callsigns: Winlink.

T2C09 Are amateur station control operators ever permitted to operate outside the frequency privileges of their license class: Yes, but only in situations involving the immediate safety of human life or protection of property.

T2C10 What information is contained in the preamble of a formal traffic message: Information needed to track the message.

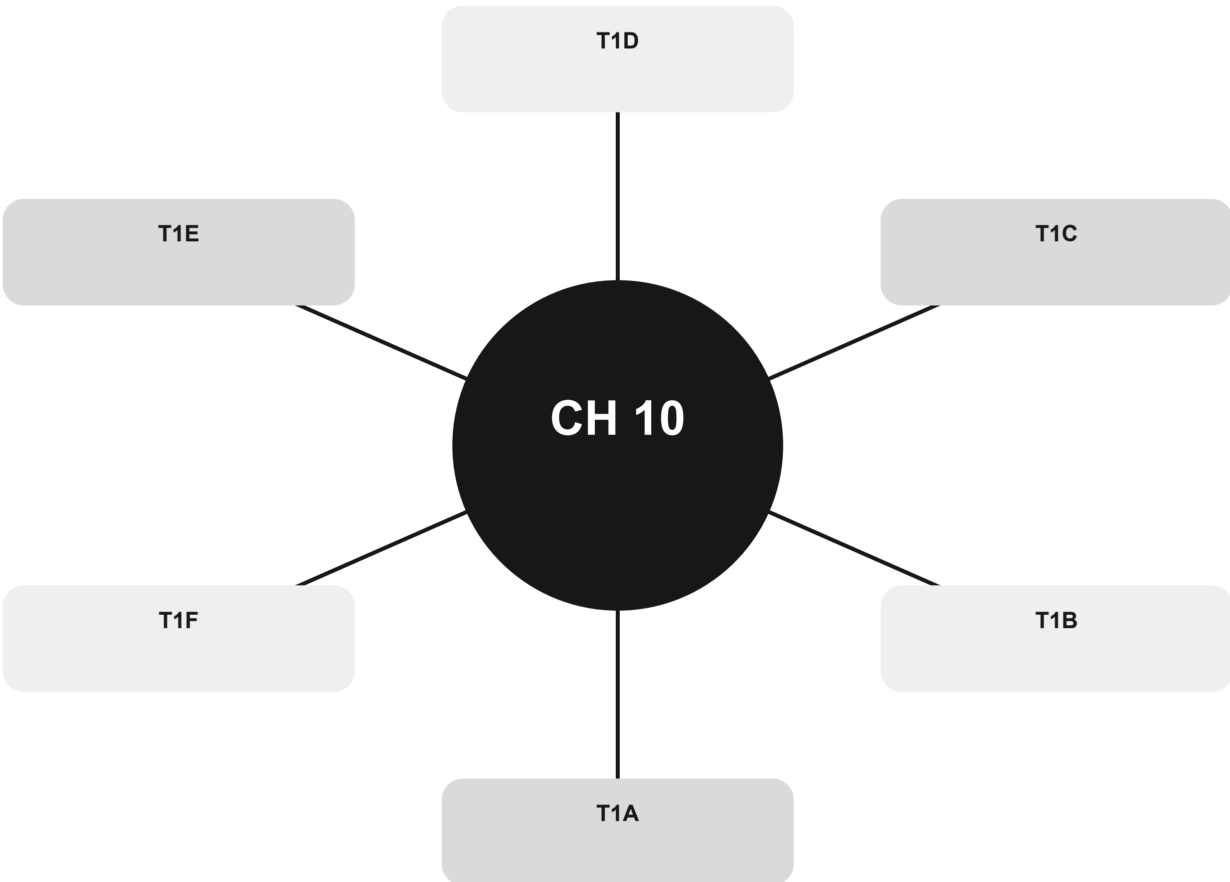
T2C11 What is meant by “check” in a radiogram header: The number of words or word equivalents in the text portion of the message.

T2C12 Which of the following requires certification by a civil defense agency: RACES.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

Rules as decision paths

Decide who may transmit, where, under whose control, and with what identification.



Official groups: T1A, T1B, T1C, T1D, T1E, T1F

Your route through this chapter

T1A Purpose, authority, and interference

Know who regulates amateur radio and why the service exists.

T1B Where a Technician may transmit

Read privileges as band, segment, mode, and power together.

T1C License lifecycle

Know when authority begins, how long it lasts, and what happens around renewal.

T1D Permitted and prohibited content

Classify a transmission by purpose rather than by convenience.

T1E Control operator responsibility

Separate the station licensee, control operator, control point, and control method.

T1F Identification and station types

Apply the call-sign clock and recognize repeater, club, and third-party roles.

After the last group, use the official-pool anchors to identify weak spots. Do randomized exam practice outside this book.

Purpose, authority, and interference

Know who regulates amateur radio and why the service exists.

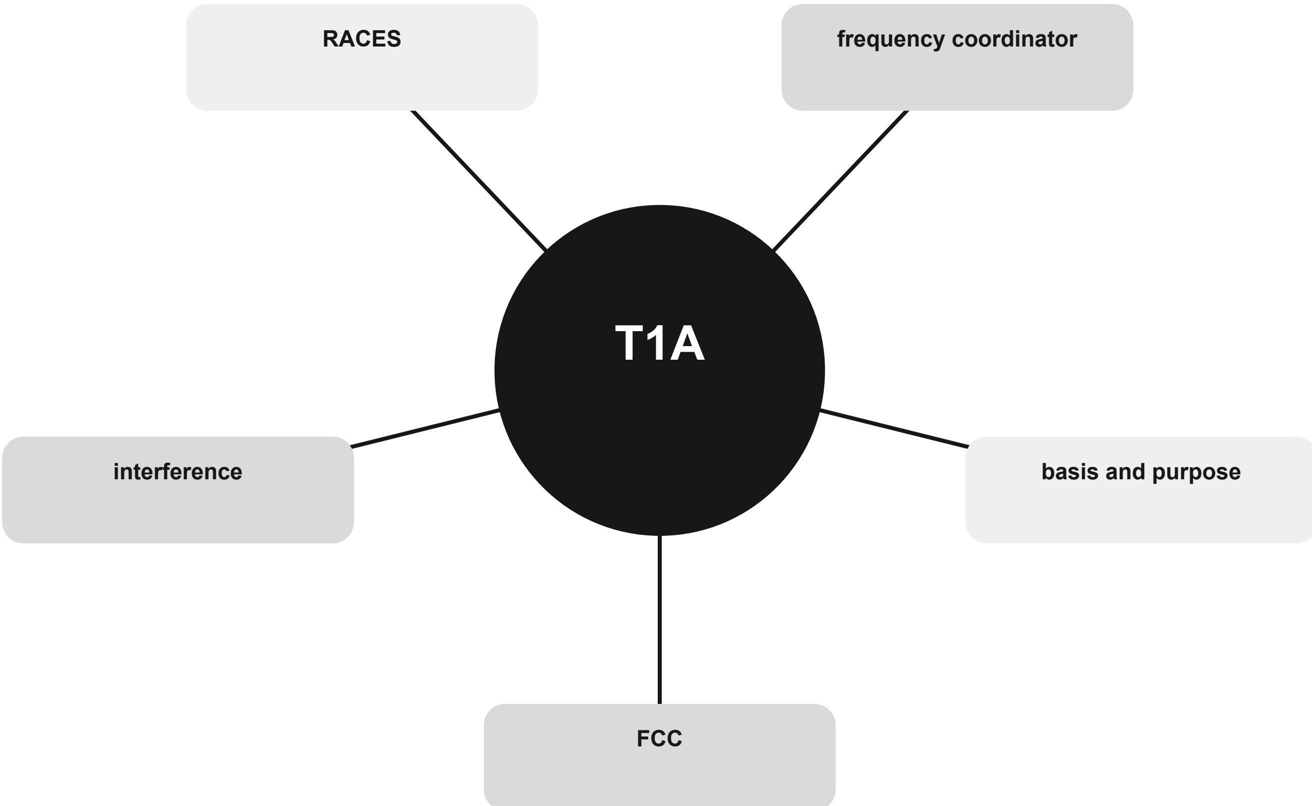
THE MENTAL MODEL

The FCC regulates the service. Its stated purposes include advancing radio skill and communication. Coordination is largely performed by amateurs, but willful interference remains prohibited.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are FCC, basis and purpose, frequency coordinator, RACES, interference.

Purpose, authority, and interference concept map



Read from the center outward. Each node is a cue that should lead back to the same working model.

Recognize, then connect

RECOGNIZE

CONNECT

FCC

Use the relationship, not an isolated word.

basis and purpose

Use the relationship, not an isolated word.

frequency coordinator

Use the relationship, not an isolated word.

RACES

Use the relationship, not an isolated word.

interference

Use the relationship, not an isolated word.

The exam often places related words close together. Do not choose by familiarity. Identify the requested job, path, unit, or rule first.

Official-pool anchors

T1A01 Which of the following is part of the Basis and Purpose of the Amateur Radio Service: Advancing skills in the technical and communication phases of the radio art.

T1A02 Which agency regulates and enforces the rules for the Amateur Radio Service in the United States: The FCC.

T1A03 What do the FCC rules state regarding the use of a phonetic alphabet for station identification in the Amateur Radio Service: It is encouraged when using phone emissions.

T1A04 How do you receive official notification of your new license and call sign after passing the exam: Email from the FCC with a link to download the license grant.

T1A05 What proves that the FCC has issued an operator/primary license grant: The license appears in the FCC ULS database.

T1A06 On which of the following HF frequencies can automatically controlled amateur propagation beacons be found: On ten meters, between 28.200 MHz and 28.300 MHz.

T1A07 What is the FCC Part 97 definition of a space station: An amateur station located more than 50 km above Earth's surface.

T1A08 Who recommends transmit/receive channels for repeater and auxiliary stations: A Volunteer Frequency Coordinator recognized by local amateurs.

T1A09 Who selects a Frequency Coordinator: Amateur operators in a local or regional area whose stations are eligible to be repeater or auxiliary stations.

T1A10 Besides an FCC-issued amateur operator license, what is required to be the control operator of a Radio Amateur Civil Emergency Service (RACES) Station: Certification of current enrollment by a civil defense organization.

T1A11 Which of the following is prohibited: Willful or malicious interference.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

Where a Technician may transmit

Read privileges as band, segment, mode, and power together.

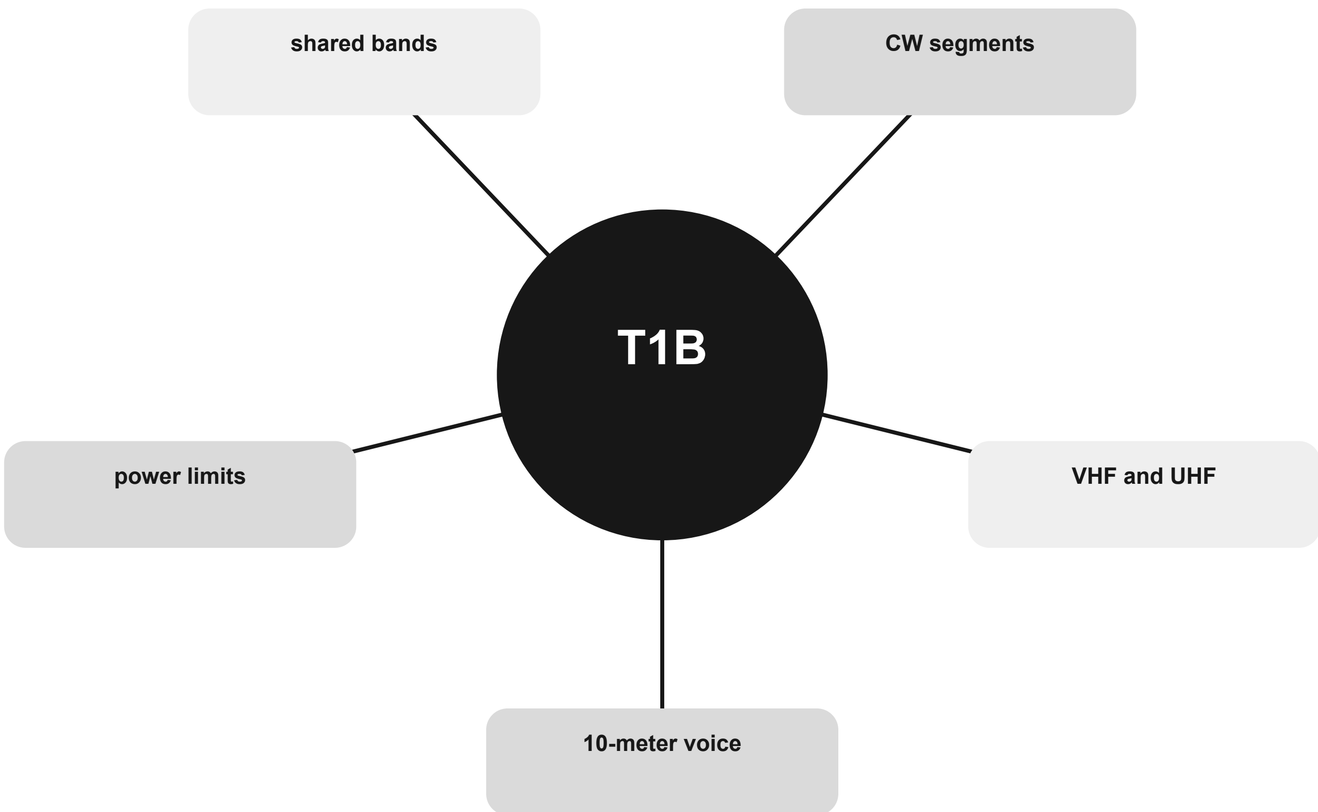
THE MENTAL MODEL

A license class does not grant one undivided block of spectrum. Check the band, the permitted segment, the emission, and the power limit. Stay clear of band edges and shared users.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are 10-meter voice, VHF and UHF, CW segments, shared bands, power limits.

Where a Technician may transmit concept map



Read from the center outward. Each node is a cue that should lead back to the same working model.

Recognize, then connect

RECOGNIZE

10-meter voice

Use the relationship, not an isolated word.

VHF and UHF

Use the relationship, not an isolated word.

CW segments

Use the relationship, not an isolated word.

CONNECT

shared bands

Use the relationship, not an isolated word.

power limits

Use the relationship, not an isolated word.

The exam often places related words close together. Do not choose by familiarity. Identify the requested job, path, unit, or rule first.

Official-pool anchors

T1B01 Which of the following frequency ranges are available for phone operation by Technician licensees: 28.300 MHz to 28.500 MHz.

T1B02 Which of the following U.S. amateur radio operators are allowed to contact the International Space Station (ISS) on VHF bands: Any amateur with a Technician class or higher license.

T1B03 Which frequency is in the 6-meter amateur band: 52.525 MHz.

T1B04 Which amateur band includes 146.52 MHz: 2 meters.

T1B05 Which of the following bands include frequencies where Technicians are authorized to use digital modes such as FT8: All these choices are correct.

T1B06 On which HF bands does a Technician class operator have phone privileges: 10-meter band only.

T1B07 Which of the following VHF/UHF band segments are limited to CW only: 50.0 MHz to 50.1 MHz and 144.0 MHz to 144.1 MHz.

T1B08 How are US amateurs restricted in segments of bands where the Amateur Radio Service is secondary: U.S. amateurs may find non-amateur stations in those segments, and must avoid interfering with them.

T1B09 Why should you not set your transmit frequency to be exactly at the edge of an amateur band or sub-band: All these choices are correct.

T1B10 Where may SSB phone be used in amateur bands above 50 MHz: In at least some segment of all these bands.

T1B11 What is the maximum peak envelope power output for Technician class operators in their HF band segments: 200 watts.

T1B12 Except for some specific restrictions, what is the maximum peak envelope power output for Technician class operators using frequencies above 30 MHz: 1500 watts.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

License lifecycle

Know when authority begins, how long it lasts, and what happens around renewal.

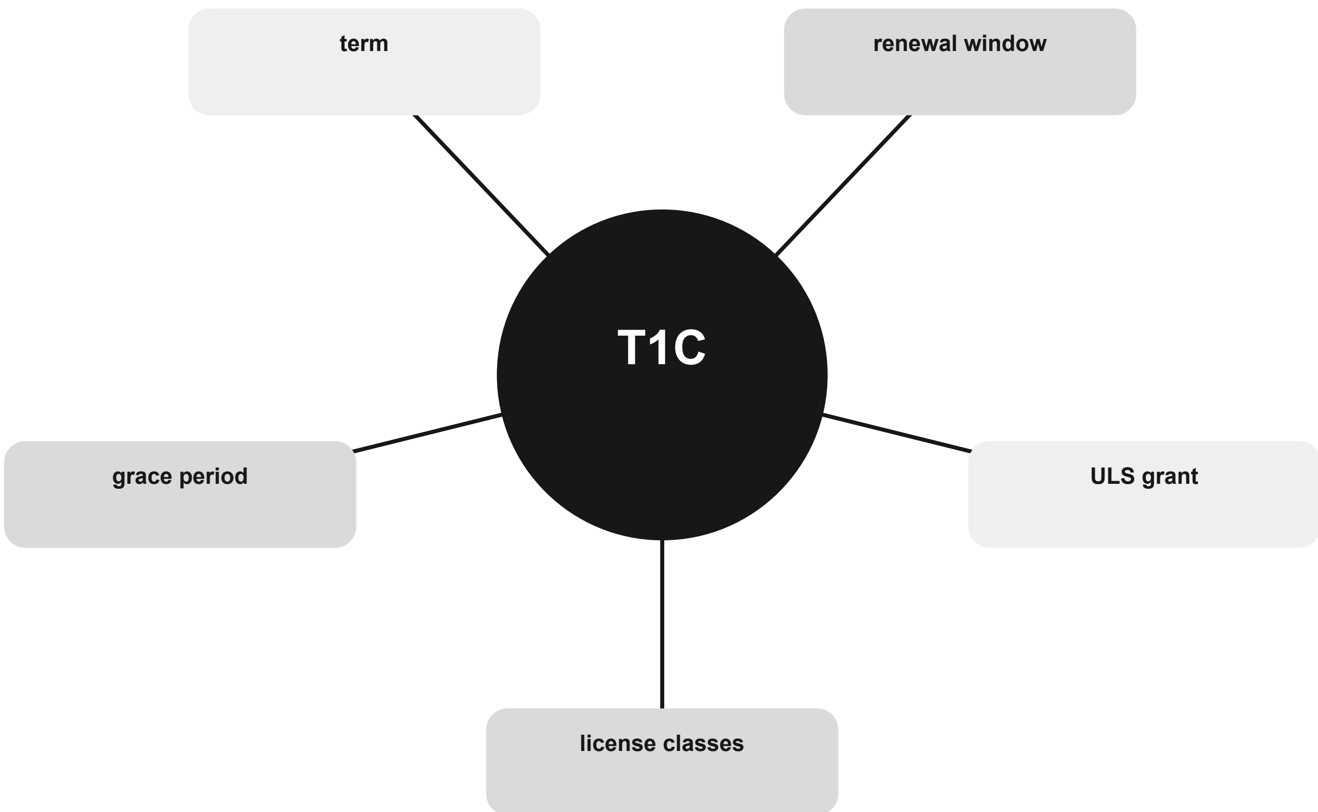
THE MENTAL MODEL

Operating authority begins when the grant appears in the FCC database. Licenses have a term, an advance renewal window, and a grace period with no operating authority after expiration until renewal.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are license classes, ULS grant, renewal window, term, grace period.

License lifecycle concept map



Read from the center outward. Each node is a cue that should lead back to the same working model.

Recognize, then connect

RECOGNIZE	CONNECT
license classes Use the relationship, not an isolated word.	**term** Use the relationship, not an isolated word.
ULS grant Use the relationship, not an isolated word.	**grace period** Use the relationship, not an isolated word.
renewal window Use the relationship, not an isolated word.	

The exam often places related words close together. Do not choose by familiarity. Identify the requested job, path, unit, or rule first.

Official-pool anchors

T1C01 For which classes of amateur radio licenses does the FCC currently issue new licenses: Technician, General, Amateur Extra.

T1C02 Who may select a desired call sign under the vanity call sign rules: Any licensed amateur.

T1C03 What types of international communications are FCC-licensed amateur radio stations permitted to make: Communications incidental to the purposes of the Amateur Radio Service and remarks of a personal character.

T1C04 What may happen if the FCC is unable to reach you by email: Revocation of the station license or suspension of the operator license.

T1C05 Which of the following is a valid Group D call sign format for Technician class: KF1XXX.

T1C06 Which of the following statements is true about using your amateur radio license when operating aboard a vessel or craft in international waters: You may operate from a US-documented vessel with the master's permission.

T1C07 How long before the expiration date may an amateur radio license renewal be requested: 90 days.

T1C08 What is the normal term for an FCC-issued amateur radio license: Ten years.

T1C09 What is the grace period for renewal if an amateur license expires: Two years.

T1C10 How soon after passing the examination for your first amateur radio license may you transmit on the amateur radio bands: As soon as your operator/station license grant appears in the FCC’s license database.

T1C11 If your license has expired and is still within the allowable grace period, may you continue to transmit on the amateur radio bands: No, you must wait until the license has been renewed.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

Permitted and prohibited content

Classify a transmission by purpose rather than by convenience.

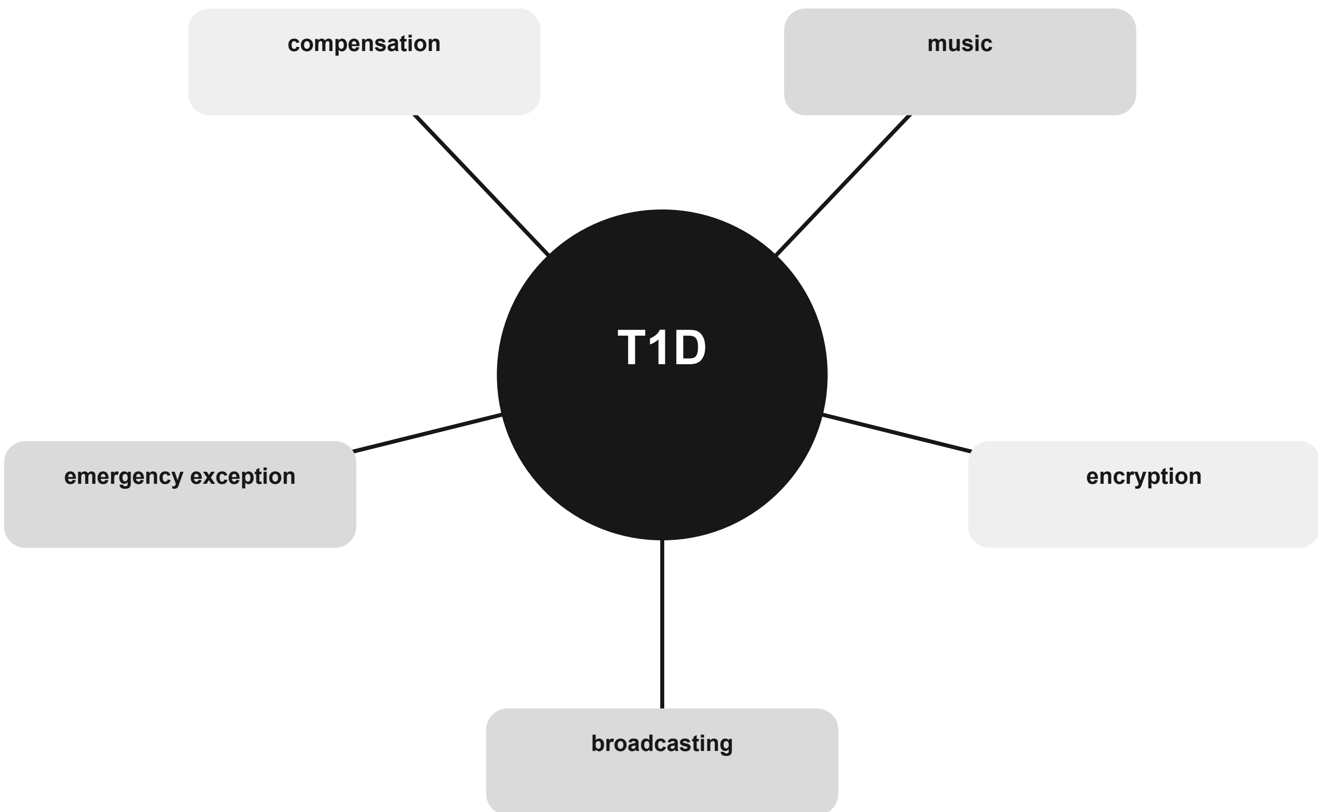
THE MENTAL MODEL

Amateur radio is not broadcasting. Encryption, music, compensation, one-way transmissions, and third-party material have narrow rules and exceptions. Immediate safety can change what is permitted.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are broadcasting, encryption, music, compensation, emergency exception.

Permitted and prohibited content concept map



Read from the center outward. Each node is a cue that should lead back to the same working model.

Recognize, then connect

RECOGNIZE

broadcasting

Use the relationship, not an isolated word.

encryption

Use the relationship, not an isolated word.

music

Use the relationship, not an isolated word.

CONNECT

compensation

Use the relationship, not an isolated word.

emergency exception

Use the relationship, not an isolated word.

The exam often places related words close together. Do not choose by familiarity. Identify the requested job, path, unit, or rule first.

Official-pool anchors

T1D01 With which countries are FCC-licensed amateur radio stations prohibited from exchanging communications: Any country whose administration has notified the International Telecommunication Union (ITU) that it objects to such communications.

T1D02 Under which of the following circumstances are one-way transmissions by an amateur station prohibited: Broadcasting.

T1D03 When is it permissible to transmit messages encoded to obscure their meaning: Only when transmitting control commands to space stations or model craft.

T1D04 Under what conditions is an amateur station authorized to transmit music using a phone emission: When incidental to an authorized retransmission of manned spacecraft communications.

T1D05 When may amateur radio operators use their stations to notify other amateurs of the availability of equipment for sale or trade: When selling amateur radio equipment and not on a regular basis.

T1D06 What, if any, are the restrictions concerning transmission of language that may be considered indecent or obscene: Any such language is prohibited.

T1D07 Which of the following is an example of an auxiliary station: A station sending one-way transmissions between a remote repeater receiver and the main repeater transmitter.

T1D08 In which of the following circumstances may the control operator of an amateur station receive compensation for operating that station: When the communication is a part of classroom instruction at an educational institution.

T1D09 When may amateur stations transmit information in support of broadcasting, program production, or news gathering, assuming no other means is available: When such communications are directly related to the immediate safety of human life or protection of property.

T1D10 How does the FCC define broadcasting for the Amateur Radio Service: Transmissions intended for reception by the general public.

T1D11 When may an amateur station transmit without identifying on the air: When transmitting signals to control model craft.

T1D12 Which of the following is required when making on-the-air test transmissions: Identify the transmitting station.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

Control operator responsibility

Separate the station licensee, control operator, control point, and control method.

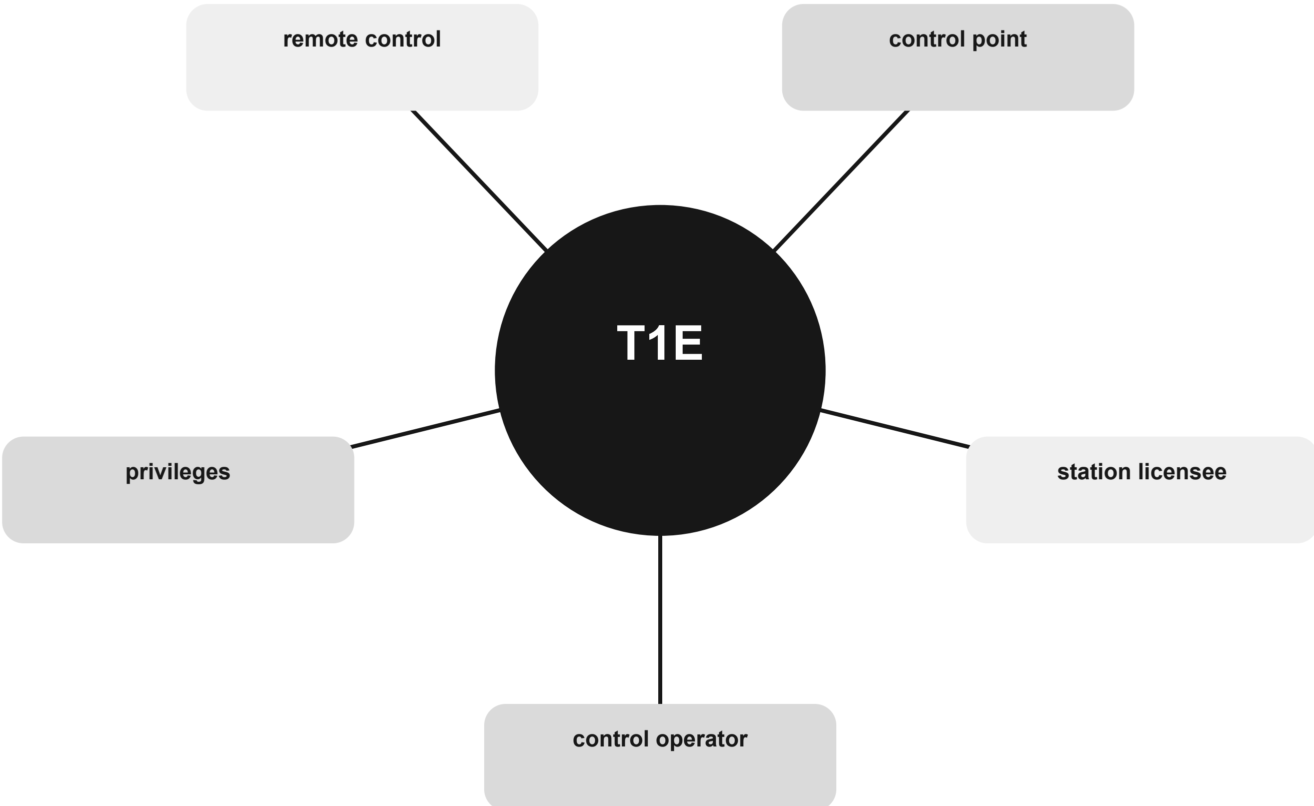
THE MENTAL MODEL

Every station transmission has a designated control operator whose license determines privileges. The station licensee and control operator share responsibility for proper operation.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are control operator, station licensee, control point, remote control, privileges.

Control operator responsibility concept map



Read from the center outward. Each node is a cue that should lead back to the same working model.

Recognize, then connect

RECOGNIZE	CONNECT
control operator Use the relationship, not an isolated word.	**remote control** Use the relationship, not an isolated word.
station licensee Use the relationship, not an isolated word.	**privileges** Use the relationship, not an isolated word.
control point Use the relationship, not an isolated word.	

The exam often places related words close together. Do not choose by familiarity. Identify the requested job, path, unit, or rule first.

Official-pool anchors

T1E01 When may an amateur station transmit without a control operator: Never.

T1E02 Who may be the control operator of a station communicating through an amateur satellite or space station: Any amateur allowed to transmit on the satellite uplink frequency.

T1E03 Who must designate the station control operator: The station licensee.

T1E04 What determines the transmitting frequency privileges of an amateur station: The class of operator license held by the control operator.

T1E05 What is an amateur station's control point: The location at which the control operator function is performed.

T1E06 When, other than during an emergency, may a Technician class licensee be the control operator of a station operating in an Amateur Extra Class band segment: At no time.

T1E07 When the control operator is not the station licensee, who is responsible for the proper operation of the station: The control operator and the station licensee.

T1E08 Which of the following is an example of automatic control: Repeater operation.

T1E09 Which amateur stations may be remotely controlled: Any station.

T1E10 Which of the following is an example of remote control as defined in Part 97: Operating the station over the internet.

T1E11 What is a control operator as defined in Part 97: An amateur operator designated by the licensee of a station to be responsible for transmissions and FCC rules compliance at that station.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

Identification and station types

Apply the call-sign clock and recognize repeater, club, and third-party roles.

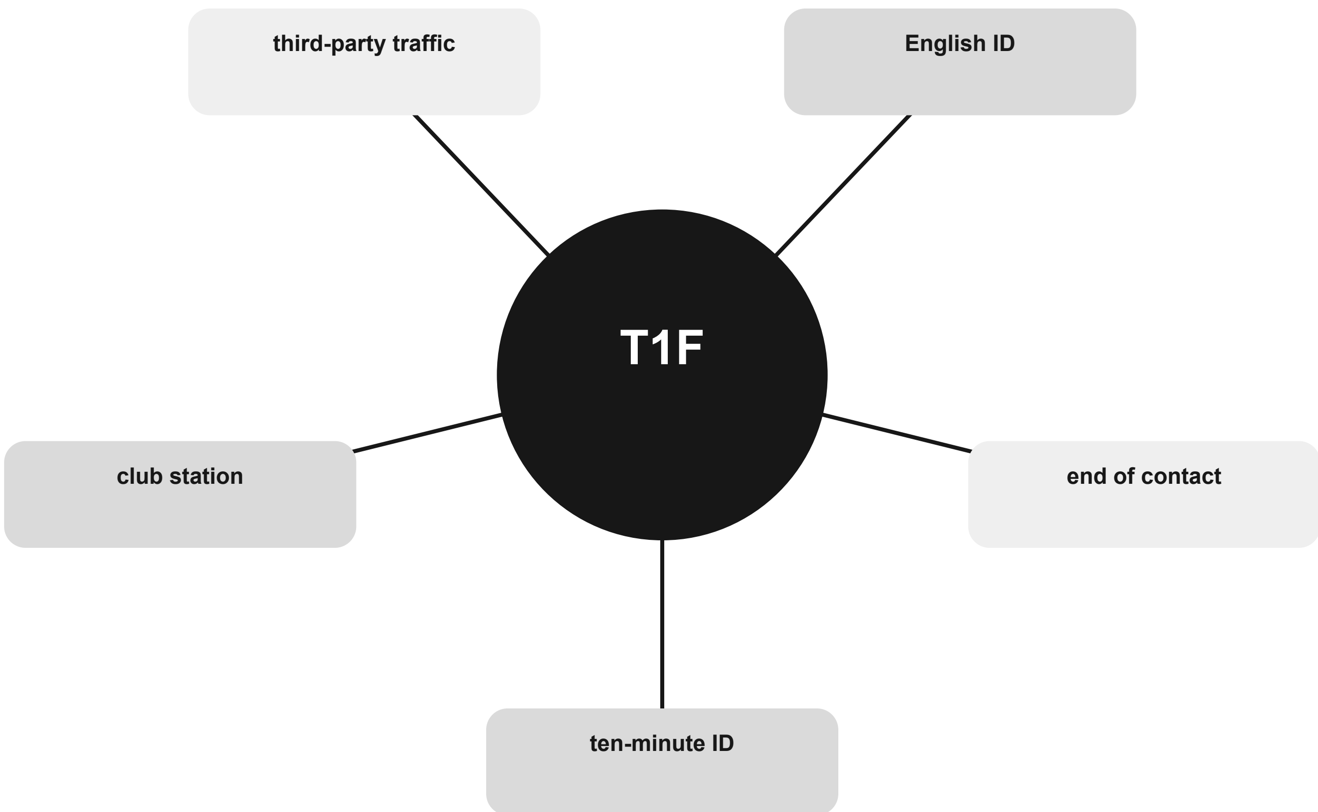
THE MENTAL MODEL

Identify at least every ten minutes and at the end of a communication. Use an allowed emission and English-language identification. Station type determines who is responsible for identification.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are ten-minute ID, end of contact, English ID, third-party traffic, club station.

Identification and station types concept map



Read from the center outward. Each node is a cue that should lead back to the same working model.

Recognize, then connect

RECOGNIZE

ten-minute ID

Use the relationship, not an isolated word.

end of contact

Use the relationship, not an isolated word.

English ID

Use the relationship, not an isolated word.

CONNECT

third-party traffic

Use the relationship, not an isolated word.

club station

Use the relationship, not an isolated word.

The exam often places related words close together. Do not choose by familiarity. Identify the requested job, path, unit, or rule first.

Official-pool anchors

T1F01 When must the station licensee make the station and the station records available for inspection: At any time upon request by an FCC representative.

T1F02 How often must you identify with your FCC-assigned call sign when using tactical call signs such as “Race Headquarters”: At least every 10 minutes during and at the end of a communication.

T1F03 When are you required to transmit your assigned call sign: At least every 10 minutes during and at the end of a communication.

T1F04 What language must you use for identification when using a phone emission: English.

T1F05 What method of call sign identification is required for a station transmitting phone signals: Send the call sign using a CW or phone emission.

T1F06 Which of the following self-assigned indicators are acceptable when using a phone transmission: All these choices are correct.

T1F07 Which of the following restrictions apply when a non-licensed person speaks to a foreign amateur radio station via a station under the control of an FCC-licensed amateur radio operator: The foreign station must be in a country with which the U.S. has a third-party agreement.

T1F08 What is the definition of third-party communications: A message from a control operator to another amateur station control operator on behalf of another person.

T1F09 What type of amateur station simultaneously retransmits the signal of another amateur station on a different channel or channels: Repeater station.

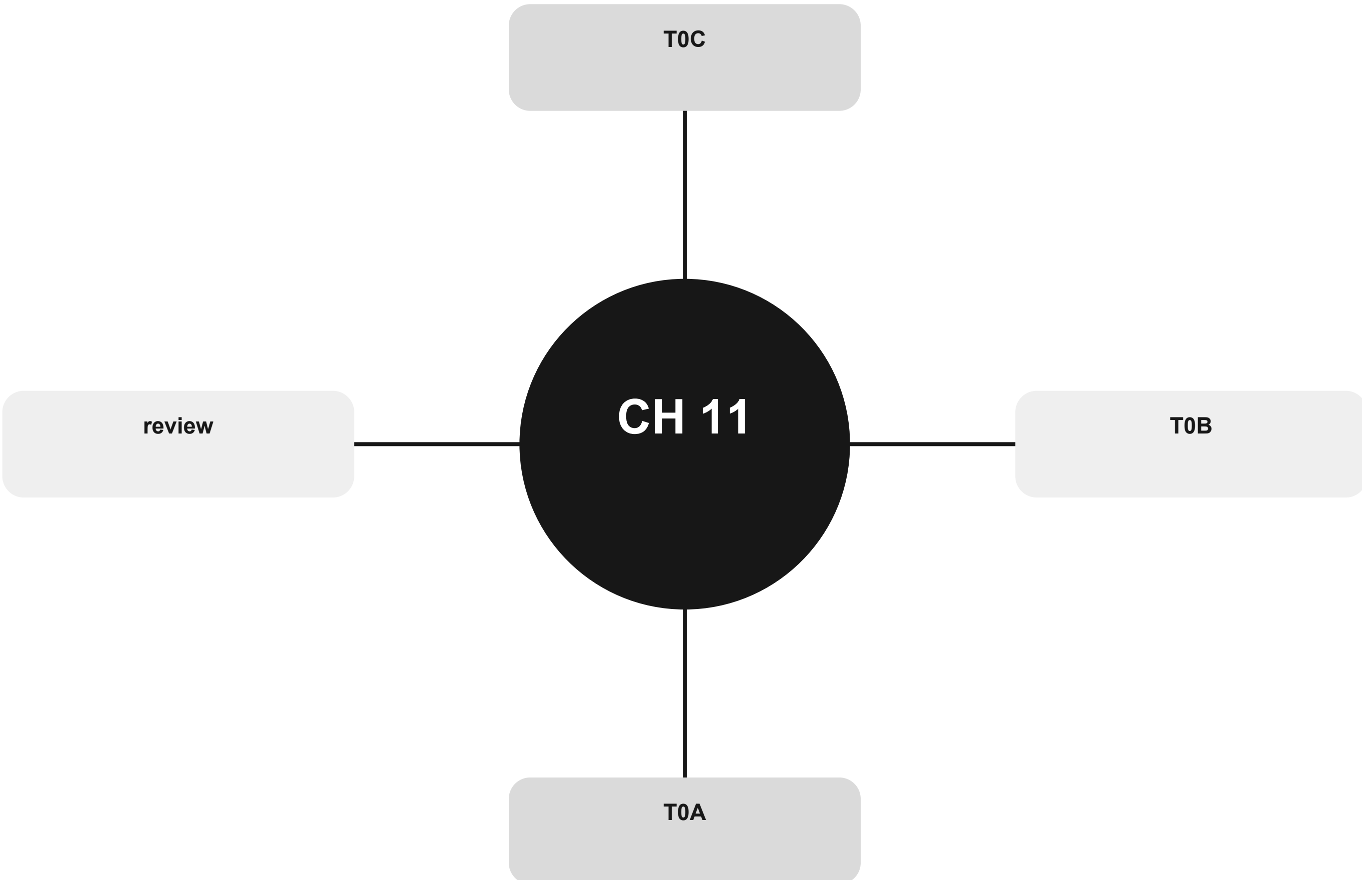
T1F10 Who is accountable if a repeater inadvertently retransmits communications that violate the FCC rules: The control operator of the originating station.

T1F11 Which of the following is a requirement for the issuance of a club station license grant: The club must have at least four members.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

Safety before shortcuts

Recognize the hazard first; choose the safe action before the technical shortcut.



Official groups: T0A, T0B, T0C

Your route through this chapter

T0A Electrical, battery, and lightning hazards

Choose isolation, protection, bonding, and correct ratings.

T0B Antenna and tower safety

Keep structures and every possible fall path clear of power lines.

T0C RF exposure

Distinguish non-ionizing RF from ionizing radiation and control exposure.

After the last group, use the official-pool anchors to identify weak spots. Do randomized exam practice outside this book.

Electrical, battery, and lightning hazards

Choose isolation, protection, bonding, and correct ratings.

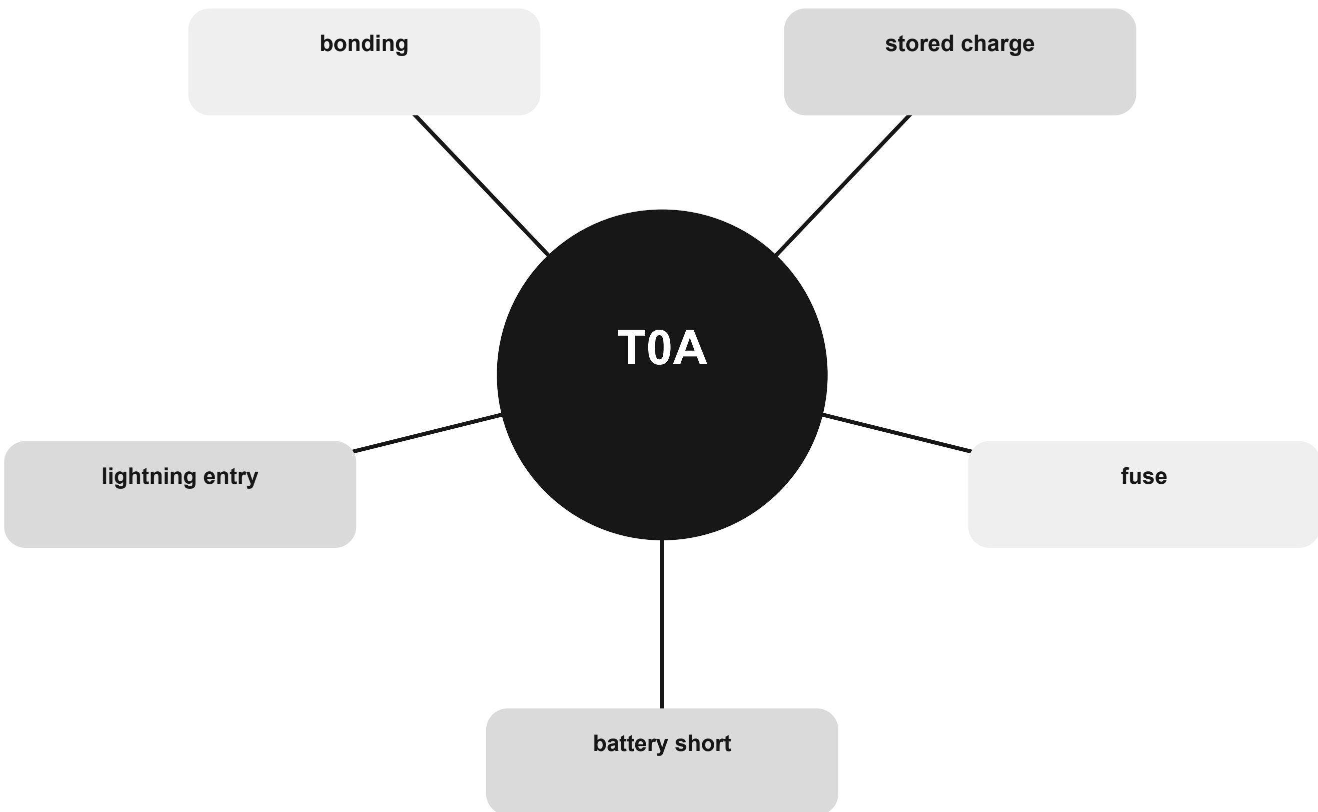
THE MENTAL MODEL

High current can start a fire even at low voltage. Fuses belong in the hot conductor. Capacitors can store dangerous energy after power is removed. Lightning grounds should be short, direct, and bonded.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are battery short, fuse, stored charge, bonding, lightning entry.

Electrical, battery, and lightning hazards concept map



Read from the center outward. Each node is a cue that should lead back to the same working model.

Recognize, then connect

RECOGNIZE

battery short

Use the relationship, not an isolated word.

fuse

Use the relationship, not an isolated word.

stored charge

Use the relationship, not an isolated word.

CONNECT

bonding

Use the relationship, not an isolated word.

lightning entry

Use the relationship, not an isolated word.

The exam often places related words close together. Do not choose by familiarity. Identify the requested job, path, unit, or rule first.

Official-pool anchors

T0A01 Which of the following is a safety hazard of a 12-volt storage battery that lacks internal protection circuitry: Shorting the terminals can cause burns, fire, or an explosion.

T0A02 What health hazard is posed by electrical current flowing through the body: All these choices are correct.

T0A03 In the United States, what circuit does black wire insulation indicate in a three-wire 120 V AC cable: Hot.

T0A04 What is the purpose of a fuse in an electrical circuit: To remove power in case of an overload.

T0A05 Why should a 5-ampere fuse never be replaced with a 20-ampere fuse: Excessive current could cause a fire.

T0A06 What is a good way to guard against electrical shock at your station: All these choices are correct.

T0A07 Where should a lightning arrester be installed in a coaxial feed line: On a grounded panel near where feed lines enter the building.

T0A08 Where should a fuse or circuit breaker be installed in a 120V AC power circuit: In series with the hot conductor only.

T0A09 What should be done to all external ground rods or earth connections: Bond them together with heavy wire or conductive strap.

T0A10 What hazard exists when rapidly charging or discharging an unprotected battery: Overheating or out-gassing.

T0A11 What hazard exists in a power supply immediately after turning it off: Charge stored in filter capacitors.

T0A12 Which of the following precautions should be taken when measuring high voltages with a voltmeter: Ensure that the voltmeter and its leads are rated for use at the voltages being measured.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

Antenna and tower safety

Keep structures and every possible fall path clear of power lines.

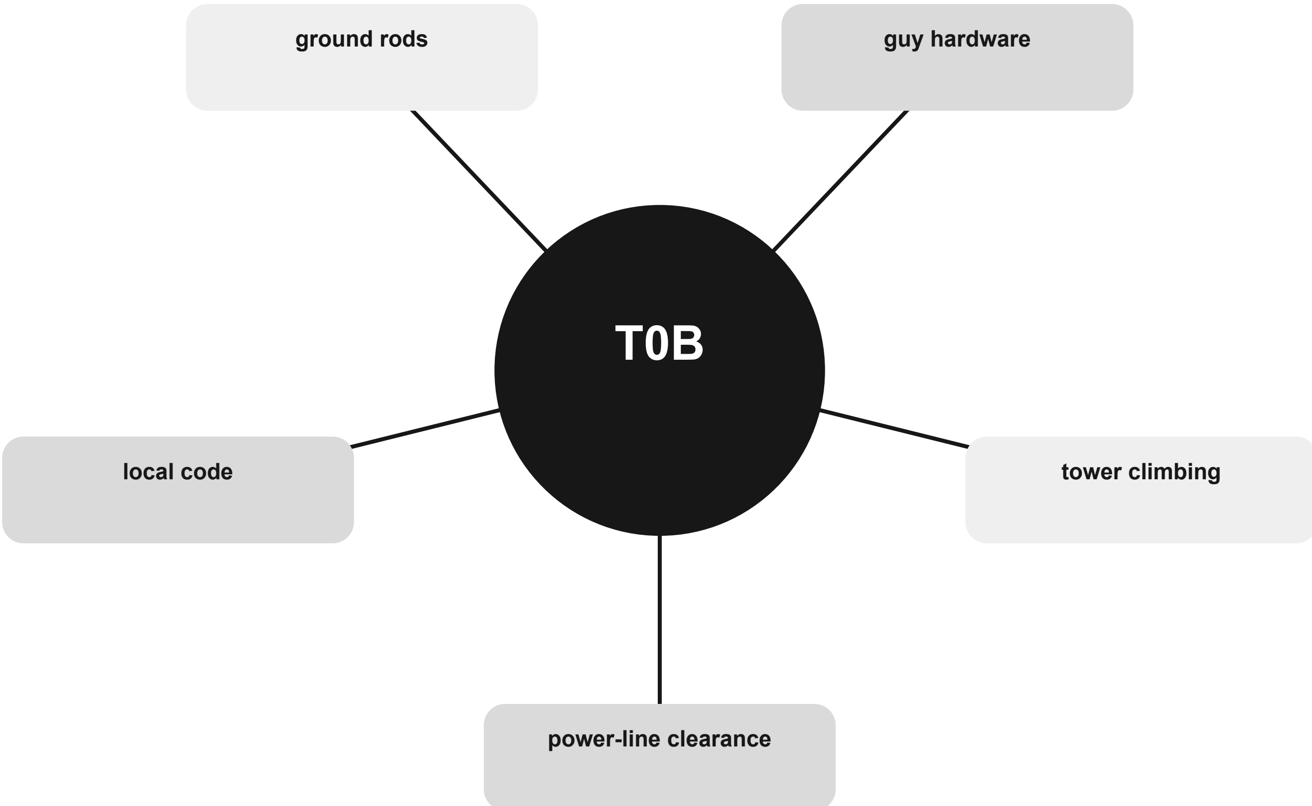
THE MENTAL MODEL

The first antenna-site question is not signal strength; it is whether any part can reach an energized line. Use proper climbing controls, grounding, and local electrical-code requirements.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are power-line clearance, tower climbing, guy hardware, ground rods, local code.

Antenna and tower safety concept map



Read from the center outward. Each node is a cue that should lead back to the same working model.

Recognize, then connect

RECOGNIZE

power-line clearance

Use the relationship, not an isolated word.

tower climbing

Use the relationship, not an isolated word.

guy hardware

Use the relationship, not an isolated word.

CONNECT

ground rods

Use the relationship, not an isolated word.

local code

Use the relationship, not an isolated word.

The exam often places related words close together. Do not choose by familiarity. Identify the requested job, path, unit, or rule first.

Official-pool anchors

T0B01 Which of the following is good practice when installing ground wires on a tower for lightning protection: Ensure that connections are short and direct.

T0B02 What is required when climbing an antenna tower: All these choices are correct.

T0B03 Under what circumstances is it safe to climb a tower without a helper or observer: Never.

T0B04 Which of the following is an important safety precaution to observe when putting up an antenna tower: Look for and stay clear of any overhead electrical wires.

T0B05 What is the purpose of a safety wire through a turnbuckle used to tension guy lines: Prevent loosening of the turnbuckle from vibration.

T0B06 What is the minimum safe distance from a power line to allow when installing an antenna: Enough so that if the antenna falls, no part of it can come within 10 feet of the power wires.

T0B07 Which of the following is an important safety rule to remember when using a crank-up tower: This type of tower must not be climbed unless it is retracted, or mechanical safety locking devices have been installed.

T0B08 Which is a proper grounding method for a tower: Separate eight-foot ground rods for each tower leg, bonded to the tower and each other.

T0B09 Why should you avoid attaching an antenna to a utility pole: The antenna could contact high-voltage power lines.

T0B10 Which of the following is true when installing grounding conductors used for lightning protection: Sharp bends must be avoided.

T0B11 Which of the following establishes grounding requirements for an amateur radio tower or antenna: Local electrical codes.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

RF exposure

Distinguish non-ionizing RF from ionizing radiation and control exposure.

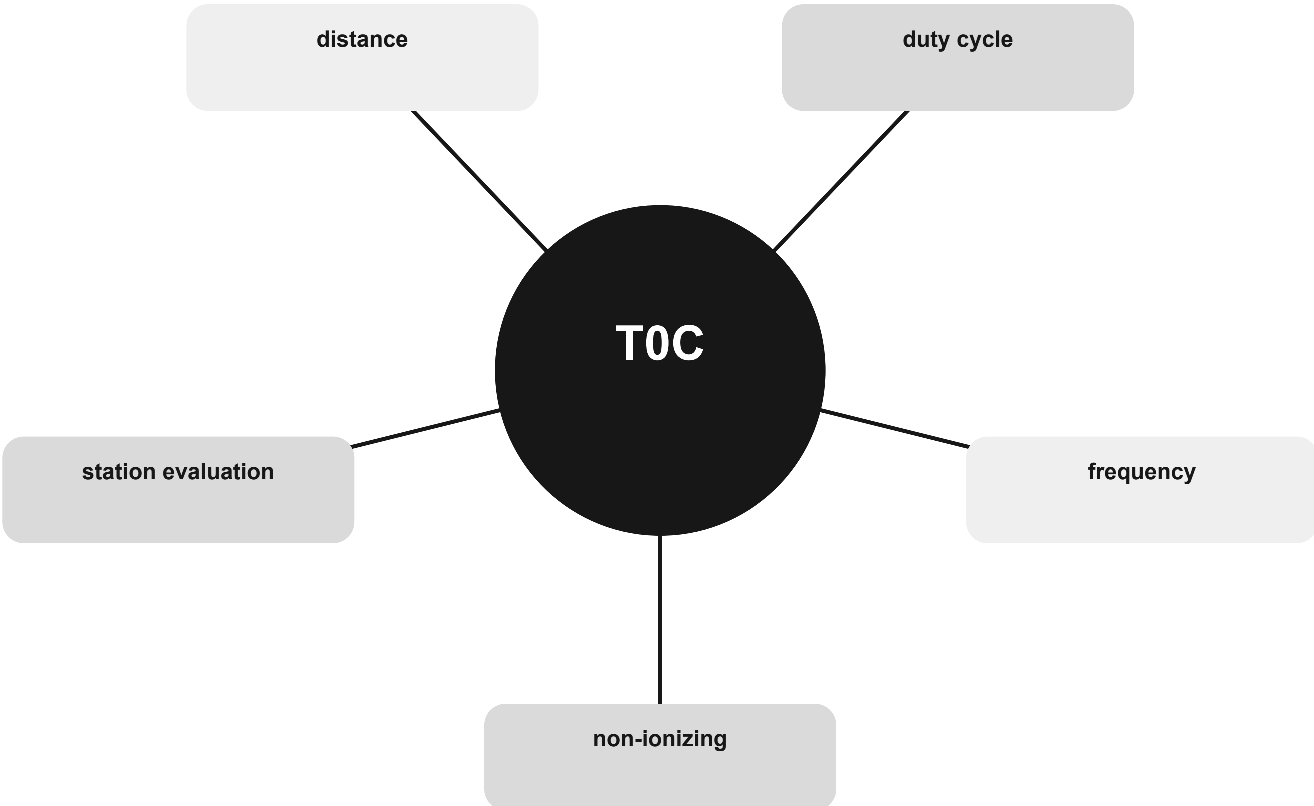
THE MENTAL MODEL

RF exposure depends on frequency, power, duty cycle, antenna gain, and distance. Evaluate the station and repeat the evaluation when the transmitter or antenna system changes.

Exam reading move

Name what is changing, what stays fixed, and which unit, path, role, or rule the question is testing. The useful cues in this group are non-ionizing, frequency, duty cycle, distance, station evaluation.

RF exposure concept map



Read from the center outward. Each node is a cue that should lead back to the same working model.

Recognize, then connect

RECOGNIZE

CONNECT

non-ionizing

Use the relationship, not an isolated word.

frequency

Use the relationship, not an isolated word.

duty cycle

Use the relationship, not an isolated word.

distance

Use the relationship, not an isolated word.

station evaluation

Use the relationship, not an isolated word.

The exam often places related words close together. Do not choose by familiarity. Identify the requested job, path, unit, or rule first.

Official-pool anchors

T0C01 What type of radiation are radio signals: Non-ionizing radiation.

T0C02 Which of the following bands has the lowest maximum permissible exposure for RF safety: 50 MHz.

T0C03 How does the allowable power density for RF safety change if duty cycle changes from 100 percent to 50 percent: It increases by a factor of 2.

T0C04 What factors affect the RF exposure of people near an amateur station antenna: All these choices are correct.

T0C05 Why do exposure limits vary with frequency: The human body absorbs more RF energy at some frequencies than at others.

T0C06 Which of the following is an acceptable method to determine whether your station complies with FCC RF exposure regulations: All these choices are correct.

T0C07 What hazard is created by touching an antenna during a transmission: RF burn to skin.

T0C08 Which of the following actions can reduce exposure to RF radiation: Relocate antennas.

T0C09 How can you make sure your station stays in compliance with RF safety regulations: By re-evaluating the station whenever an item in the transmitter or antenna system is changed.

T0C10 Why is duty cycle one of the factors used to determine safe RF radiation exposure levels: It affects the average exposure to radiation.

T0C11 What is the definition of duty cycle during the averaging time for RF exposure: The percentage of time that a transmitter is transmitting.

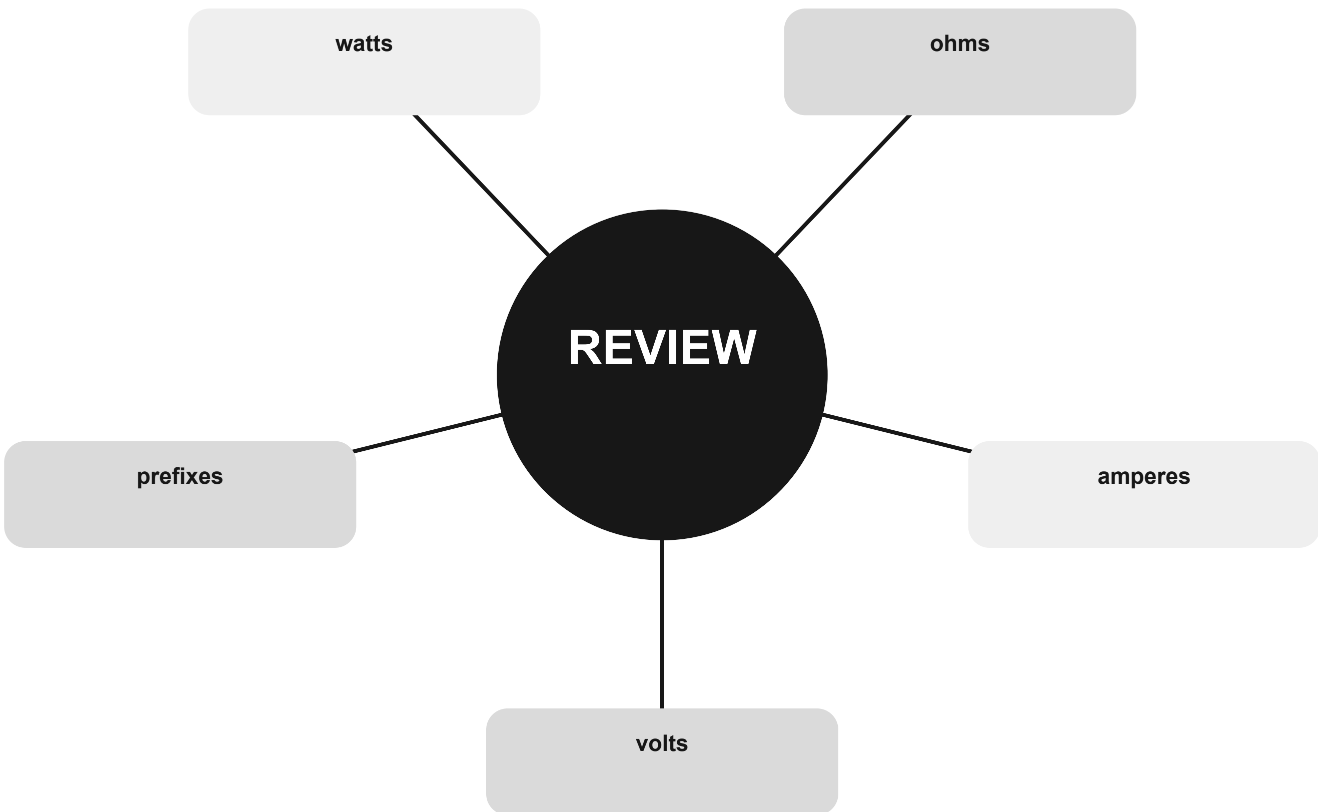
T0C12 How does RF radiation differ from ionizing radiation (radioactivity): RF radiation does not have sufficient energy to cause chemical changes in cells and damage DNA.

T0C13 Who is responsible for ensuring that no person is exposed to RF energy above the FCC exposure limits: The station licensee.

Quick check: pick two IDs. State the clue, the relationship, and the answer without rereading the stem.

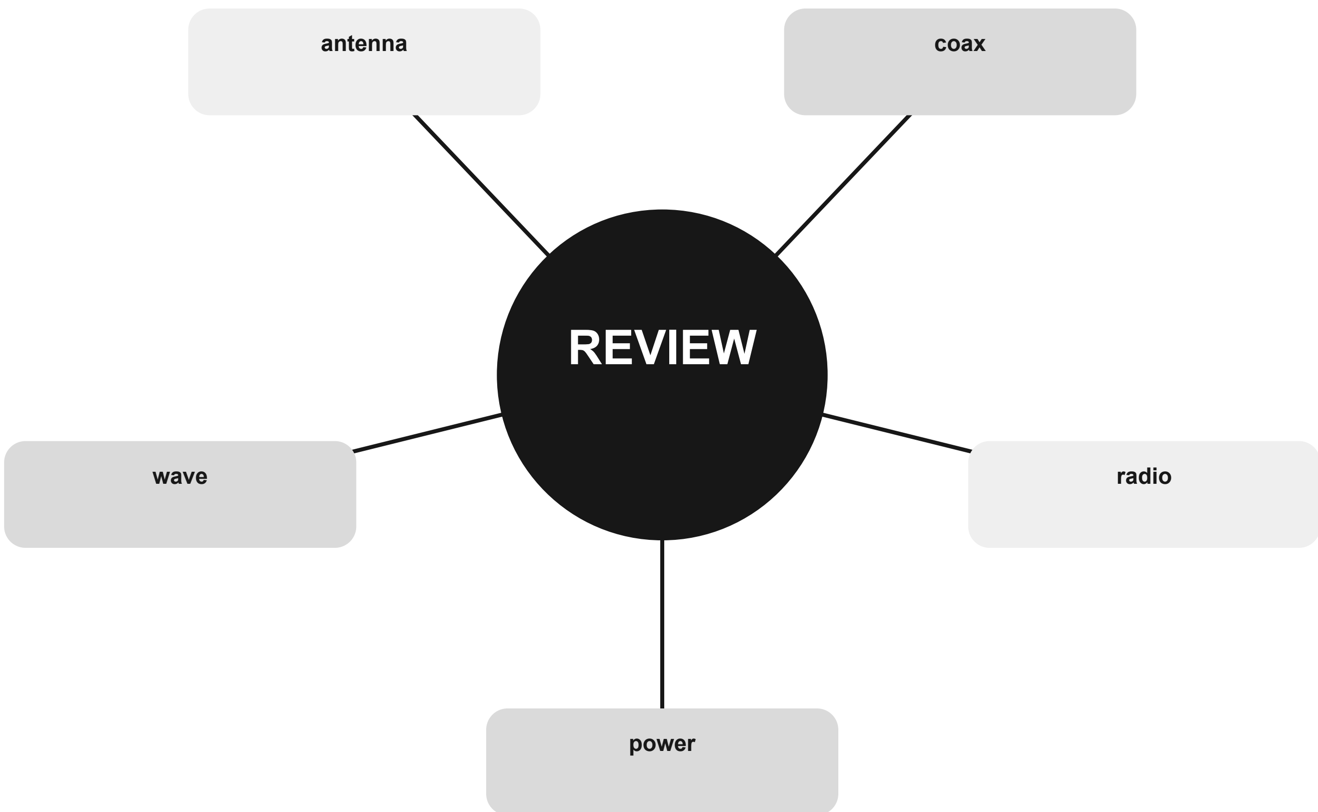
Formula route

Requested unit → formula → substitute → sanity check



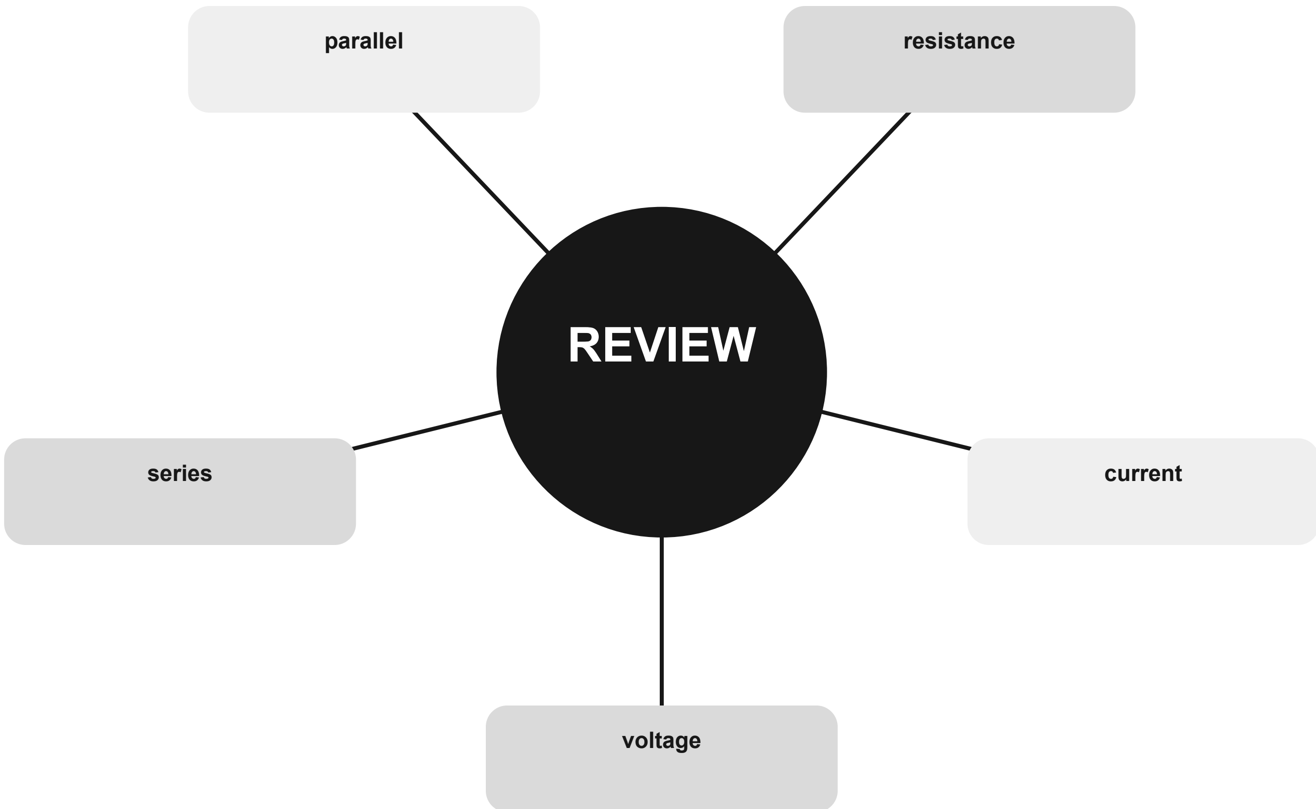
Path route

source → equipment → feed line → antenna → propagation



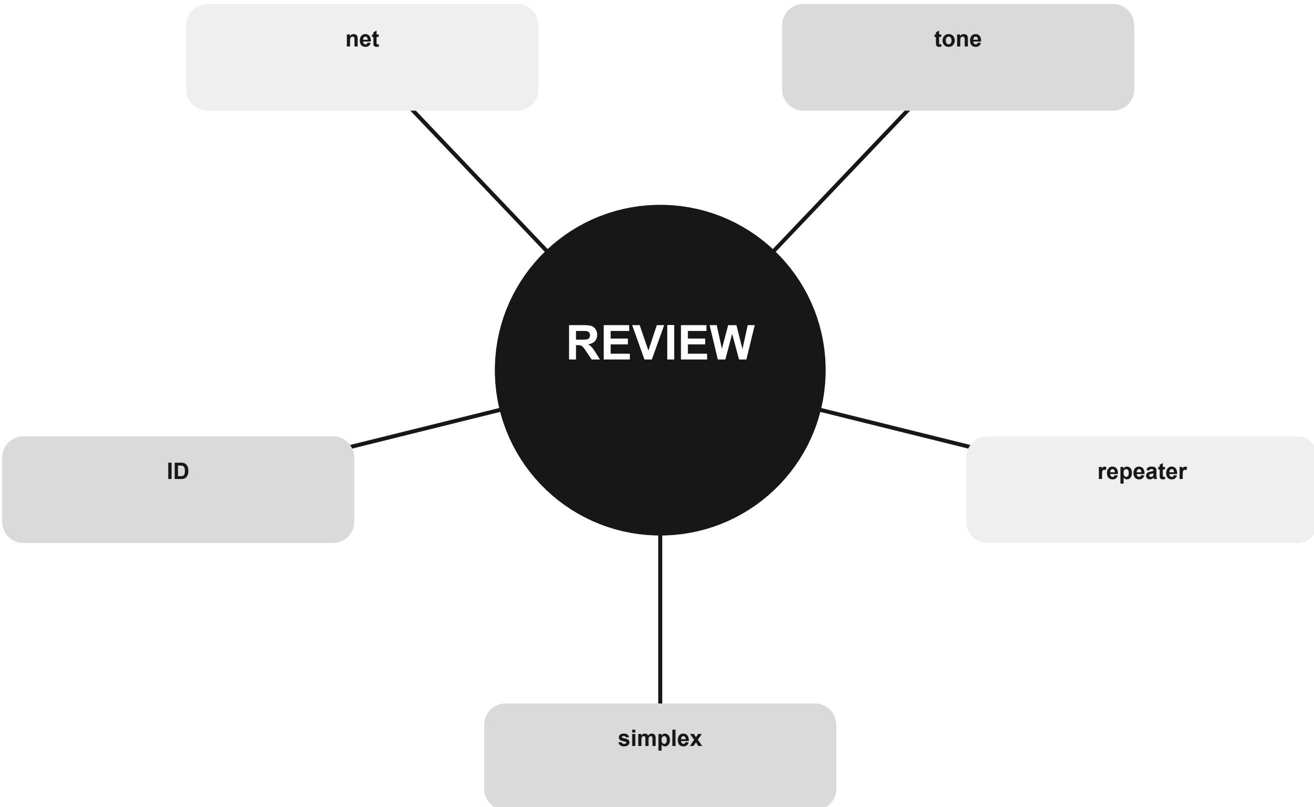
Measurement route

quantity → connection → range → power state → reading



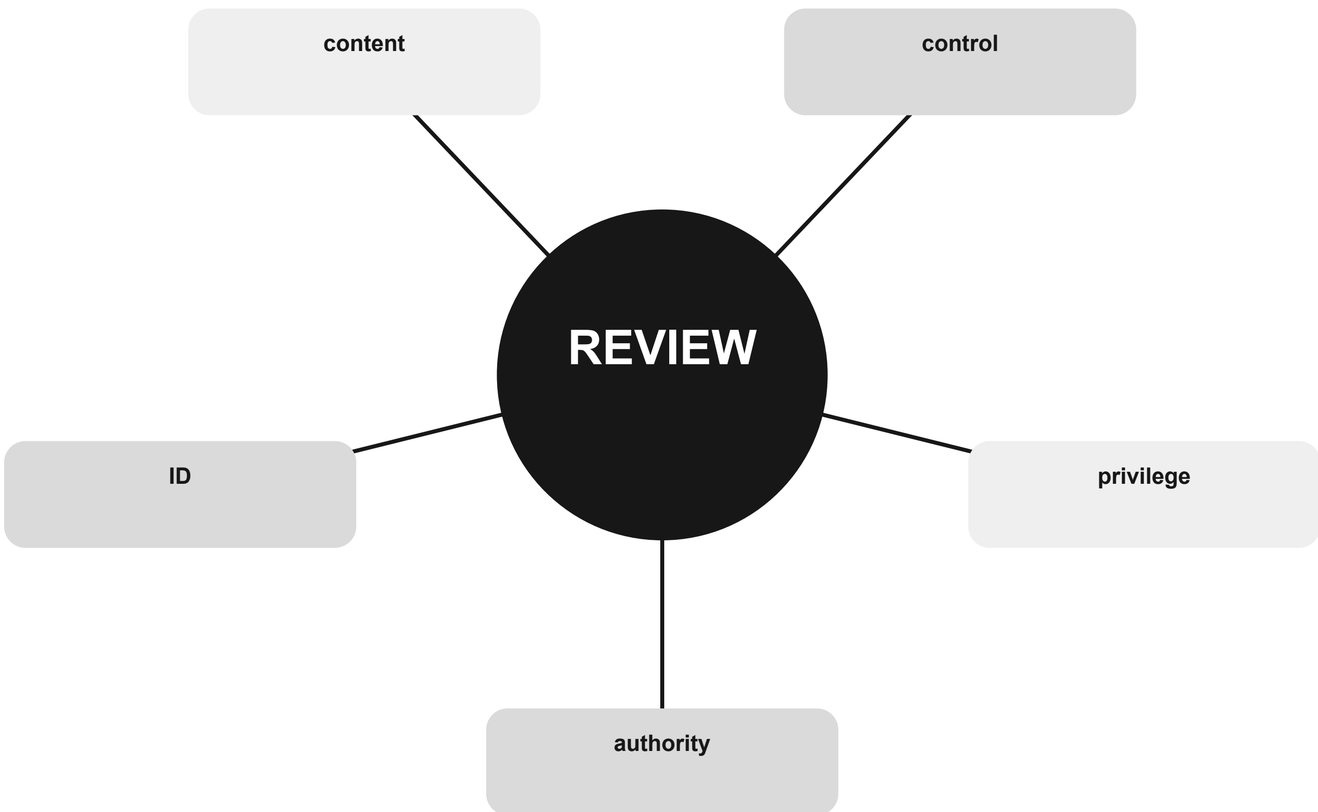
Operating route

listen → identify → contact → share → log or pass traffic



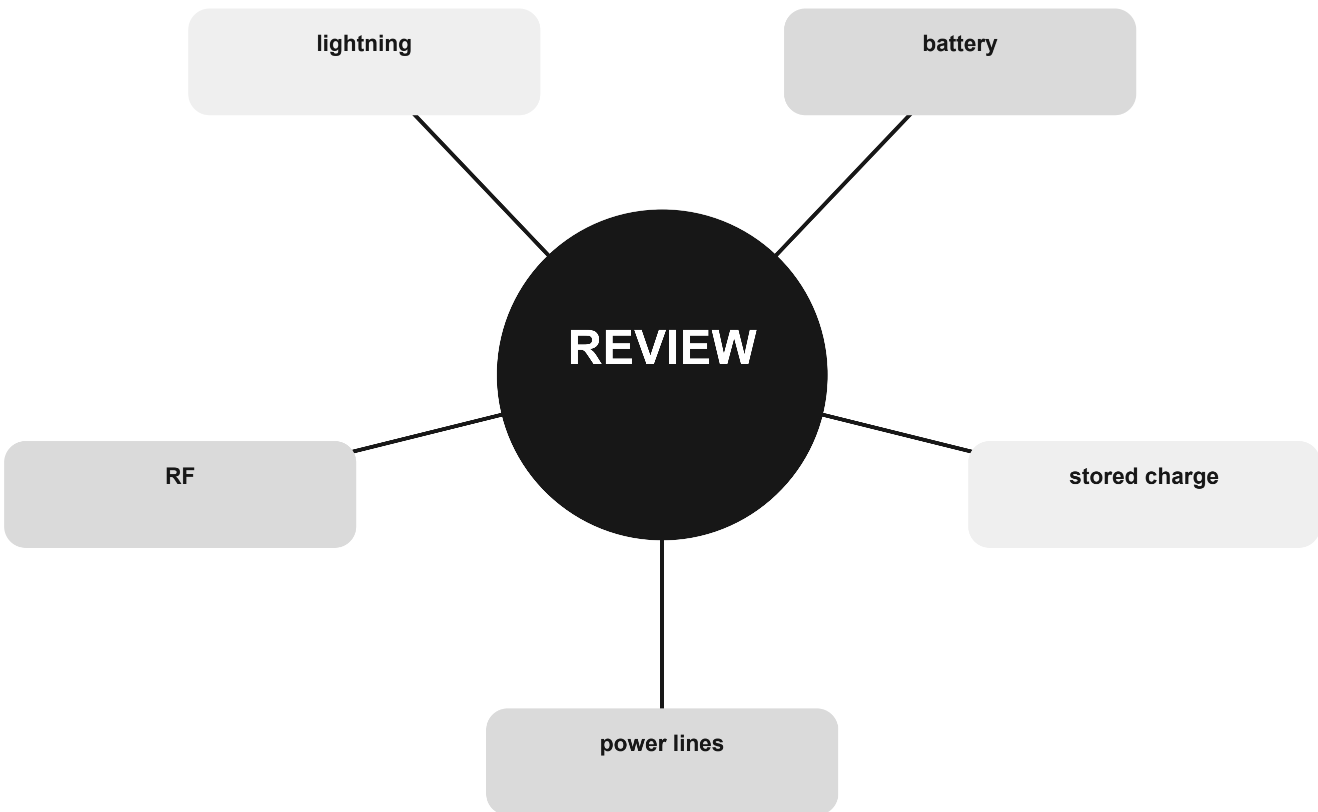
Rules route

license → band → mode → power → content → identification



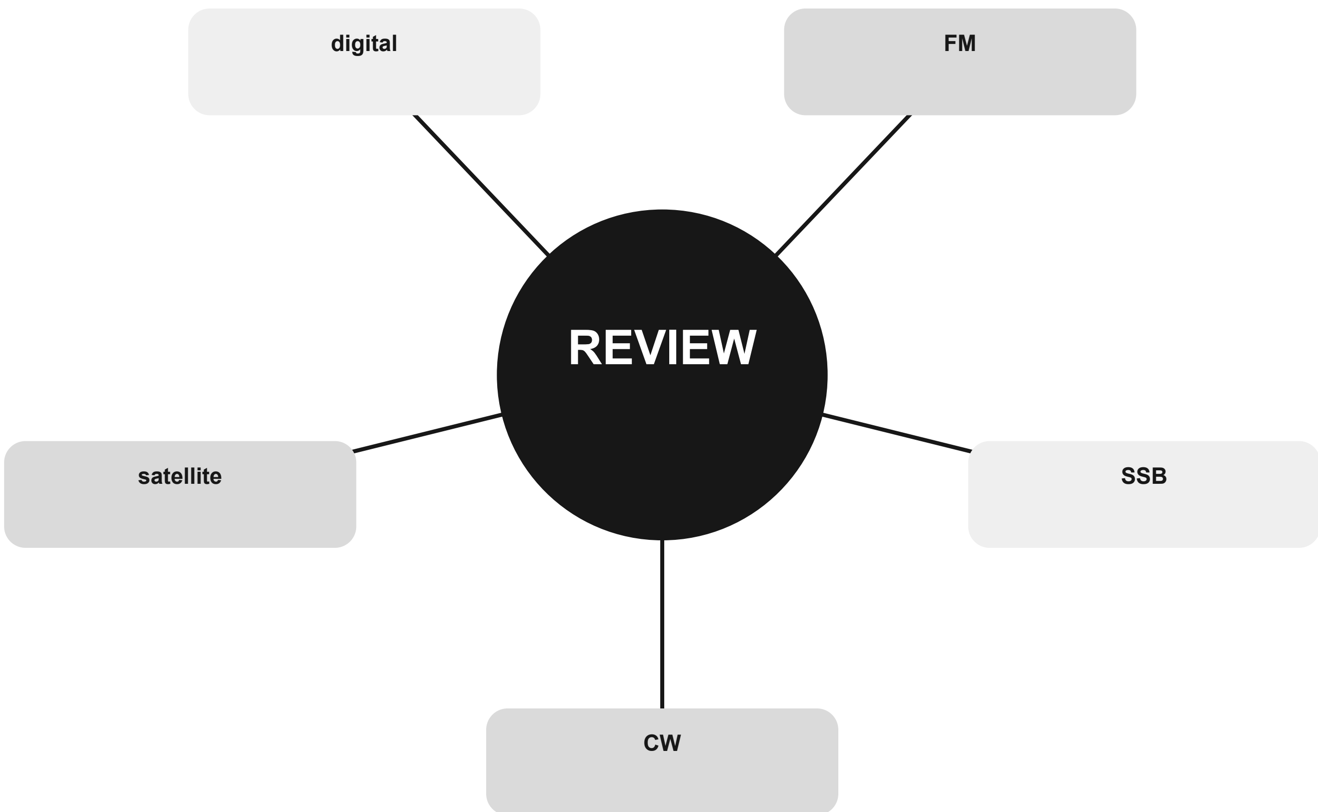
Safety route

hazard → distance/isolation → protection → verify → proceed



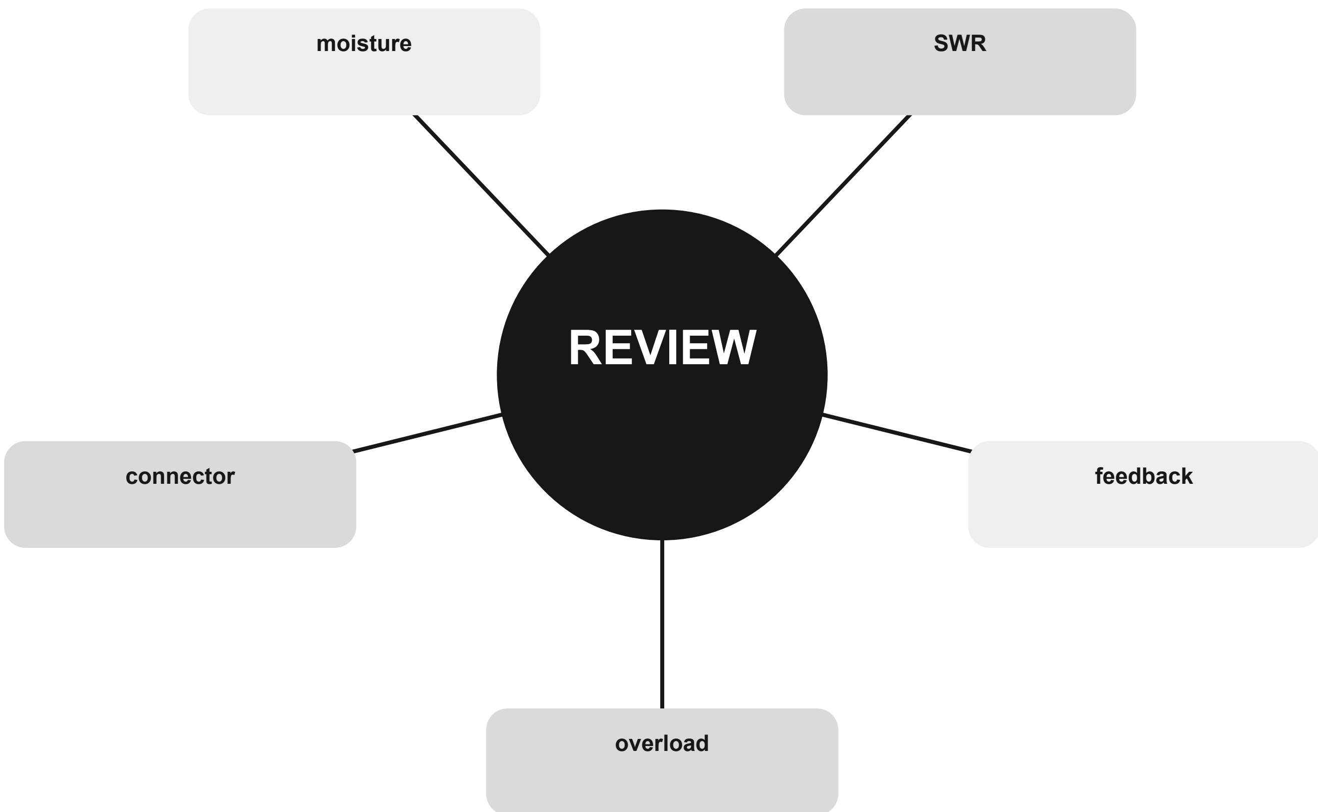
Signals route

information → modulation → bandwidth → path → receiver



Troubleshooting route

symptom → simplest cause → safe test → one change



How to diagnose a miss

Find the official ID. Name its group. Return to the concept page. Explain the visual. Retest with a different randomized item.

THE PRACTICE LOOP

Find the official ID. Name its group. Return to the concept page. Explain the visual. Retest with a different randomized item.

The book is a diagnostic map. Randomized practice remains the better test simulator.

Last-minute boundary

Do not learn new folklore on exam morning. Review the compact maps, verify current pool status, and use the official ID system for anything uncertain.

THE PRACTICE LOOP

Do not learn new folklore on exam morning. Review the compact maps, verify current pool status, and use the official ID system for anything uncertain.

> **The book is a diagnostic map. Randomized practice remains the better test simulator.**

Question routes 1 of 20

T1A01	Chapter 10 · T1A · primary concept page 133	Direct Recall
T1A02	Chapter 10 · T1A · primary concept page 133	Direct Recall
T1A03	Chapter 10 · T1A · primary concept page 133	Direct Recall
T1A04	Chapter 10 · T1A · primary concept page 133	Direct Recall
T1A05	Chapter 10 · T1A · primary concept page 133	Direct Recall
T1A06	Chapter 10 · T1A · primary concept page 133	Direct Recall
T1A07	Chapter 10 · T1A · primary concept page 133	Direct Recall
T1A08	Chapter 10 · T1A · primary concept page 133	Direct Recall
T1A09	Chapter 10 · T1A · primary concept page 133	Direct Recall
T1A10	Chapter 10 · T1A · primary concept page 133	Direct Recall
T1A11	Chapter 10 · T1A · primary concept page 133	Direct Recall
T1B01	Chapter 10 · T1B · primary concept page 137	Direct Recall
T1B02	Chapter 10 · T1B · primary concept page 137	Direct Recall
T1B03	Chapter 10 · T1B · primary concept page 137	Direct Recall
T1B04	Chapter 10 · T1B · primary concept page 137	Direct Recall
T1B05	Chapter 10 · T1B · primary concept page 137	Direct Recall
T1B06	Chapter 10 · T1B · primary concept page 137	Direct Recall
T1B07	Chapter 10 · T1B · primary concept page 137	Direct Recall
T1B08	Chapter 10 · T1B · primary concept page 137	Direct Recall
T1B09	Chapter 10 · T1B · primary concept page 137	Direct Recall
T1B10	Chapter 10 · T1B · primary concept page 137	Direct Recall

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T1B11	Chapter 10 · T1B · primary concept page 137	Direct Recall
T1B12	Chapter 10 · T1B · primary concept page 137	Direct Recall
T1C01	Chapter 10 · T1C · primary concept page 141	Direct Recall
T1C02	Chapter 10 · T1C · primary concept page 141	Direct Recall
T1C03	Chapter 10 · T1C · primary concept page 141	Direct Recall
T1C04	Chapter 10 · T1C · primary concept page 141	Direct Recall
T1C05	Chapter 10 · T1C · primary concept page 141	Direct Recall
T1C06	Chapter 10 · T1C · primary concept page 141	Direct Recall
T1C07	Chapter 10 · T1C · primary concept page 141	Direct Recall
T1C08	Chapter 10 · T1C · primary concept page 141	Direct Recall
T1C09	Chapter 10 · T1C · primary concept page 141	Direct Recall
T1C10	Chapter 10 · T1C · primary concept page 141	Direct Recall
T1C11	Chapter 10 · T1C · primary concept page 141	Direct Recall
T1D01	Chapter 10 · T1D · primary concept page 145	Direct Recall
T1D02	Chapter 10 · T1D · primary concept page 145	Direct Recall
T1D03	Chapter 10 · T1D · primary concept page 145	Direct Recall
T1D04	Chapter 10 · T1D · primary concept page 145	Direct Recall
T1D05	Chapter 10 · T1D · primary concept page 145	Direct Recall
T1D06	Chapter 10 · T1D · primary concept page 145	Direct Recall
T1D07	Chapter 10 · T1D · primary concept page 145	Direct Recall
T1D08	Chapter 10 · T1D · primary concept page 145	Direct Recall

Question routes 3 of 20

T1D09	Chapter 10 · T1D · primary concept page 145	Direct Recall
T1D10	Chapter 10 · T1D · primary concept page 145	Direct Recall
T1D11	Chapter 10 · T1D · primary concept page 145	Direct Recall
T1D12	Chapter 10 · T1D · primary concept page 145	Direct Recall
T1E01	Chapter 10 · T1E · primary concept page 149	Direct Recall
T1E02	Chapter 10 · T1E · primary concept page 149	Direct Recall
T1E03	Chapter 10 · T1E · primary concept page 149	Direct Recall
T1E04	Chapter 10 · T1E · primary concept page 149	Direct Recall
T1E05	Chapter 10 · T1E · primary concept page 149	Direct Recall
T1E06	Chapter 10 · T1E · primary concept page 149	Direct Recall
T1E07	Chapter 10 · T1E · primary concept page 149	Direct Recall
T1E08	Chapter 10 · T1E · primary concept page 149	Direct Recall
T1E09	Chapter 10 · T1E · primary concept page 149	Direct Recall
T1E10	Chapter 10 · T1E · primary concept page 149	Direct Recall
T1E11	Chapter 10 · T1E · primary concept page 149	Direct Recall
T1F01	Chapter 10 · T1F · primary concept page 153	Direct Recall
T1F02	Chapter 10 · T1F · primary concept page 153	Direct Recall
T1F03	Chapter 10 · T1F · primary concept page 153	Direct Recall
T1F04	Chapter 10 · T1F · primary concept page 153	Direct Recall
T1F05	Chapter 10 · T1F · primary concept page 153	Direct Recall
T1F06	Chapter 10 · T1F · primary concept page 153	Direct Recall

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T1F07	Chapter 10 · T1F · primary concept page 153	Direct Recall
T1F08	Chapter 10 · T1F · primary concept page 153	Direct Recall
T1F09	Chapter 10 · T1F · primary concept page 153	Direct Recall
T1F10	Chapter 10 · T1F · primary concept page 153	Direct Recall
T1F11	Chapter 10 · T1F · primary concept page 153	Direct Recall
T2A01	Chapter 9 · T2A · primary concept page 119	Direct Recall
T2A02	Chapter 9 · T2A · primary concept page 119	Direct Recall
T2A03	Chapter 9 · T2A · primary concept page 119	Direct Recall
T2A04	Chapter 9 · T2A · primary concept page 119	Direct Recall
T2A05	Chapter 9 · T2A · primary concept page 119	Direct Recall
T2A06	Chapter 9 · T2A · primary concept page 119	Direct Recall
T2A07	Chapter 9 · T2A · primary concept page 119	Direct Recall
T2A08	Chapter 9 · T2A · primary concept page 119	Direct Recall
T2A09	Chapter 9 · T2A · primary concept page 119	Direct Recall
T2A10	Chapter 9 · T2A · primary concept page 119	Direct Recall
T2A11	Chapter 9 · T2A · primary concept page 119	Direct Recall
T2B01	Chapter 9 · T2B · primary concept page 123	Direct Recall
T2B02	Chapter 9 · T2B · primary concept page 123	Direct Recall
T2B03	Chapter 9 · T2B · primary concept page 123	Direct Recall
T2B04	Chapter 9 · T2B · primary concept page 123	Direct Recall
T2B05	Chapter 9 · T2B · primary concept page 123	Direct Recall

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T2B06	Chapter 9 · T2B · primary concept page 123	Direct Recall
T2B07	Chapter 9 · T2B · primary concept page 123	Direct Recall
T2B08	Chapter 9 · T2B · primary concept page 123	Direct Recall
T2B09	Chapter 9 · T2B · primary concept page 123	Direct Recall
T2B10	Chapter 9 · T2B · primary concept page 123	Direct Recall
T2B11	Chapter 9 · T2B · primary concept page 123	Direct Recall
T2B12	Chapter 9 · T2B · primary concept page 123	Direct Recall
T2B13	Chapter 9 · T2B · primary concept page 123	Direct Recall
T2B14	Chapter 9 · T2B · primary concept page 123	Direct Recall
T2C01	Chapter 9 · T2C · primary concept page 127	Direct Recall
T2C02	Chapter 9 · T2C · primary concept page 127	Direct Recall
T2C03	Chapter 9 · T2C · primary concept page 127	Direct Recall
T2C04	Chapter 9 · T2C · primary concept page 127	Direct Recall
T2C05	Chapter 9 · T2C · primary concept page 127	Direct Recall
T2C06	Chapter 9 · T2C · primary concept page 127	Direct Recall
T2C07	Chapter 9 · T2C · primary concept page 127	Direct Recall
T2C08	Chapter 9 · T2C · primary concept page 127	Direct Recall
T2C09	Chapter 9 · T2C · primary concept page 127	Direct Recall
T2C10	Chapter 9 · T2C · primary concept page 127	Direct Recall
T2C11	Chapter 9 · T2C · primary concept page 127	Direct Recall
T2C12	Chapter 9 · T2C · primary concept page 127	Direct Recall

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T3A01	Chapter 5 · T3A · primary concept page 67	Taught
T3A02	Chapter 5 · T3A · primary concept page 67	Taught
T3A03	Chapter 5 · T3A · primary concept page 67	Taught
T3A04	Chapter 5 · T3A · primary concept page 67	Taught
T3A05	Chapter 5 · T3A · primary concept page 67	Taught
T3A06	Chapter 5 · T3A · primary concept page 67	Taught
T3A07	Chapter 5 · T3A · primary concept page 67	Taught
T3A08	Chapter 5 · T3A · primary concept page 67	Taught
T3A09	Chapter 5 · T3A · primary concept page 67	Taught
T3A10	Chapter 5 · T3A · primary concept page 67	Taught
T3A11	Chapter 5 · T3A · primary concept page 67	Taught
T3A12	Chapter 5 · T3A · primary concept page 67	Taught
T3B01	Chapter 5 · T3B · primary concept page 71	Taught
T3B02	Chapter 5 · T3B · primary concept page 71	Taught
T3B03	Chapter 5 · T3B · primary concept page 71	Taught
T3B04	Chapter 5 · T3B · primary concept page 71	Taught
T3B05	Chapter 5 · T3B · primary concept page 71	Taught
T3B06	Chapter 5 · T3B · primary concept page 71	Taught
T3B07	Chapter 5 · T3B · primary concept page 71	Taught
T3B08	Chapter 5 · T3B · primary concept page 71	Taught
T3B09	Chapter 5 · T3B · primary concept page 71	Taught

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T3B10	Chapter 5 · T3B · primary concept page 71	Taught
T3B11	Chapter 5 · T3B · primary concept page 71	Taught
T3B12	Chapter 5 · T3B · primary concept page 71	Taught
T3C01	Chapter 5 · T3C · primary concept page 75	Taught
T3C02	Chapter 5 · T3C · primary concept page 75	Taught
T3C03	Chapter 5 · T3C · primary concept page 75	Taught
T3C04	Chapter 5 · T3C · primary concept page 75	Taught
T3C05	Chapter 5 · T3C · primary concept page 75	Taught
T3C06	Chapter 5 · T3C · primary concept page 75	Taught
T3C07	Chapter 5 · T3C · primary concept page 75	Taught
T3C08	Chapter 5 · T3C · primary concept page 75	Taught
T3C09	Chapter 5 · T3C · primary concept page 75	Taught
T3C10	Chapter 5 · T3C · primary concept page 75	Taught
T3C11	Chapter 5 · T3C · primary concept page 75	Taught
T4A01	Chapter 8 · T4A · primary concept page 109	Direct Recall
T4A02	Chapter 8 · T4A · primary concept page 109	Direct Recall
T4A03	Chapter 8 · T4A · primary concept page 109	Direct Recall
T4A04	Chapter 8 · T4A · primary concept page 109	Direct Recall
T4A05	Chapter 8 · T4A · primary concept page 109	Direct Recall
T4A06	Chapter 8 · T4A · primary concept page 109	Direct Recall
T4A07	Chapter 8 · T4A · primary concept page 109	Direct Recall

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T4A08	Chapter 8 · T4A · primary concept page 109	Direct Recall
T4A09	Chapter 8 · T4A · primary concept page 109	Direct Recall
T4A10	Chapter 8 · T4A · primary concept page 109	Direct Recall
T4A11	Chapter 8 · T4A · primary concept page 109	Direct Recall
T4A12	Chapter 8 · T4A · primary concept page 109	Direct Recall
T4B01	Chapter 8 · T4B · primary concept page 113	Direct Recall
T4B02	Chapter 8 · T4B · primary concept page 113	Direct Recall
T4B03	Chapter 8 · T4B · primary concept page 113	Direct Recall
T4B04	Chapter 8 · T4B · primary concept page 113	Direct Recall
T4B05	Chapter 8 · T4B · primary concept page 113	Direct Recall
T4B06	Chapter 8 · T4B · primary concept page 113	Direct Recall
T4B07	Chapter 8 · T4B · primary concept page 113	Direct Recall
T4B08	Chapter 8 · T4B · primary concept page 113	Direct Recall
T4B09	Chapter 8 · T4B · primary concept page 113	Direct Recall
T4B10	Chapter 8 · T4B · primary concept page 113	Direct Recall
T4B11	Chapter 8 · T4B · primary concept page 113	Direct Recall
T5A01	Chapter 1 · T5A · primary concept page 11	Taught
T5A02	Chapter 1 · T5A · primary concept page 11	Taught
T5A03	Chapter 1 · T5A · primary concept page 11	Taught
T5A04	Chapter 1 · T5A · primary concept page 11	Taught
T5A05	Chapter 1 · T5A · primary concept page 11	Taught

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T5A06	Chapter 1 · T5A · primary concept page 11	Taught
T5A07	Chapter 1 · T5A · primary concept page 11	Taught
T5A08	Chapter 1 · T5A · primary concept page 11	Taught
T5A09	Chapter 1 · T5A · primary concept page 11	Taught
T5A10	Chapter 1 · T5A · primary concept page 11	Taught
T5A11	Chapter 1 · T5A · primary concept page 11	Taught
T5B01	Chapter 1 · T5B · primary concept page 15	Taught
T5B02	Chapter 1 · T5B · primary concept page 15	Taught
T5B03	Chapter 1 · T5B · primary concept page 15	Taught
T5B04	Chapter 1 · T5B · primary concept page 15	Taught
T5B05	Chapter 1 · T5B · primary concept page 15	Taught
T5B06	Chapter 1 · T5B · primary concept page 15	Taught
T5B07	Chapter 1 · T5B · primary concept page 15	Taught
T5B08	Chapter 1 · T5B · primary concept page 15	Taught
T5B09	Chapter 1 · T5B · primary concept page 15	Taught
T5B10	Chapter 1 · T5B · primary concept page 15	Taught
T5B11	Chapter 1 · T5B · primary concept page 15	Taught
T5B12	Chapter 1 · T5B · primary concept page 15	Taught
T5B13	Chapter 1 · T5B · primary concept page 15	Taught
T5C01	Chapter 1 · T5C · primary concept page 19	Taught
T5C02	Chapter 1 · T5C · primary concept page 19	Taught

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T5C03	Chapter 1 · T5C · primary concept page 19	Taught
T5C04	Chapter 1 · T5C · primary concept page 19	Taught
T5C05	Chapter 1 · T5C · primary concept page 19	Taught
T5C06	Chapter 1 · T5C · primary concept page 19	Taught
T5C07	Chapter 1 · T5C · primary concept page 19	Taught
T5C08	Chapter 1 · T5C · primary concept page 19	Taught
T5C09	Chapter 1 · T5C · primary concept page 19	Taught
T5C10	Chapter 1 · T5C · primary concept page 19	Taught
T5C11	Chapter 1 · T5C · primary concept page 19	Taught
T5C12	Chapter 1 · T5C · primary concept page 19	Taught
T5D01	Chapter 1 · T5D · primary concept page 23	Taught
T5D02	Chapter 1 · T5D · primary concept page 23	Taught
T5D03	Chapter 1 · T5D · primary concept page 23	Taught
T5D04	Chapter 1 · T5D · primary concept page 23	Taught
T5D05	Chapter 1 · T5D · primary concept page 23	Taught
T5D06	Chapter 1 · T5D · primary concept page 23	Taught
T5D07	Chapter 1 · T5D · primary concept page 23	Taught
T5D08	Chapter 1 · T5D · primary concept page 23	Taught
T5D09	Chapter 1 · T5D · primary concept page 23	Taught
T5D10	Chapter 1 · T5D · primary concept page 23	Taught
T5D11	Chapter 1 · T5D · primary concept page 23	Taught

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T5D12	Chapter 1 · T5D · primary concept page 23	Taught
T5D13	Chapter 1 · T5D · primary concept page 23	Taught
T5D14	Chapter 1 · T5D · primary concept page 23	Taught
T6A01	Chapter 2 · T6A · primary concept page 29	Taught
T6A02	Chapter 2 · T6A · primary concept page 29	Taught
T6A03	Chapter 2 · T6A · primary concept page 29	Taught
T6A04	Chapter 2 · T6A · primary concept page 29	Taught
T6A05	Chapter 2 · T6A · primary concept page 29	Taught
T6A06	Chapter 2 · T6A · primary concept page 29	Taught
T6A07	Chapter 2 · T6A · primary concept page 29	Taught
T6A08	Chapter 2 · T6A · primary concept page 29	Taught
T6A09	Chapter 2 · T6A · primary concept page 29	Taught
T6A10	Chapter 2 · T6A · primary concept page 29	Taught
T6A11	Chapter 2 · T6A · primary concept page 29	Taught
T6B01	Chapter 2 · T6B · primary concept page 33	Taught
T6B02	Chapter 2 · T6B · primary concept page 33	Taught
T6B03	Chapter 2 · T6B · primary concept page 33	Taught
T6B04	Chapter 2 · T6B · primary concept page 33	Taught
T6B05	Chapter 2 · T6B · primary concept page 33	Taught
T6B06	Chapter 2 · T6B · primary concept page 33	Taught
T6B07	Chapter 2 · T6B · primary concept page 33	Taught

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T6B08	Chapter 2 · T6B · primary concept page 33	Taught
T6B09	Chapter 2 · T6B · primary concept page 33	Taught
T6B10	Chapter 2 · T6B · primary concept page 33	Taught
T6B11	Chapter 2 · T6B · primary concept page 33	Taught
T6B12	Chapter 2 · T6B · primary concept page 33	Taught
T6C01	Chapter 2 · T6C · primary concept page 37	Taught
T6C02	Chapter 2 · T6C · primary concept page 37	Taught
T6C03	Chapter 2 · T6C · primary concept page 37	Taught
T6C04	Chapter 2 · T6C · primary concept page 37	Taught
T6C05	Chapter 2 · T6C · primary concept page 37	Taught
T6C06	Chapter 2 · T6C · primary concept page 37	Taught
T6C07	Chapter 2 · T6C · primary concept page 37	Taught
T6C08	Chapter 2 · T6C · primary concept page 37	Taught
T6C09	Chapter 2 · T6C · primary concept page 37	Taught
T6C10	Chapter 2 · T6C · primary concept page 37	Taught
T6C11	Chapter 2 · T6C · primary concept page 37	Taught
T6C12	Chapter 2 · T6C · primary concept page 37	Taught
T6D01	Chapter 2 · T6D · primary concept page 41	Taught
T6D02	Chapter 2 · T6D · primary concept page 41	Taught
T6D03	Chapter 2 · T6D · primary concept page 41	Taught
T6D04	Chapter 2 · T6D · primary concept page 41	Taught

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T6D05	Chapter 2 · T6D · primary concept page 41	Taught
T6D06	Chapter 2 · T6D · primary concept page 41	Taught
T6D07	Chapter 2 · T6D · primary concept page 41	Taught
T6D08	Chapter 2 · T6D · primary concept page 41	Taught
T6D09	Chapter 2 · T6D · primary concept page 41	Taught
T6D10	Chapter 2 · T6D · primary concept page 41	Taught
T6D11	Chapter 2 · T6D · primary concept page 41	Taught
T7A01	Chapter 3 · T7A · primary concept page 47	Taught
T7A02	Chapter 3 · T7A · primary concept page 47	Taught
T7A03	Chapter 3 · T7A · primary concept page 47	Taught
T7A04	Chapter 3 · T7A · primary concept page 47	Taught
T7A05	Chapter 3 · T7A · primary concept page 47	Taught
T7A06	Chapter 3 · T7A · primary concept page 47	Taught
T7A07	Chapter 3 · T7A · primary concept page 47	Taught
T7A08	Chapter 3 · T7A · primary concept page 47	Taught
T7A09	Chapter 3 · T7A · primary concept page 47	Taught
T7A10	Chapter 3 · T7A · primary concept page 47	Taught
T7A11	Chapter 3 · T7A · primary concept page 47	Taught
T7B01	Chapter 4 · T7B · primary concept page 57	Taught
T7B02	Chapter 4 · T7B · primary concept page 57	Taught
T7B03	Chapter 4 · T7B · primary concept page 57	Taught

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T7B04	Chapter 4 · T7B · primary concept page 57	Taught
T7B05	Chapter 4 · T7B · primary concept page 57	Taught
T7B06	Chapter 4 · T7B · primary concept page 57	Taught
T7B07	Chapter 4 · T7B · primary concept page 57	Taught
T7B08	Chapter 4 · T7B · primary concept page 57	Taught
T7B09	Chapter 4 · T7B · primary concept page 57	Taught
T7B10	Chapter 4 · T7B · primary concept page 57	Taught
T7B11	Chapter 4 · T7B · primary concept page 57	Taught
T7C01	Chapter 4 · T7C · primary concept page 61	Taught
T7C02	Chapter 4 · T7C · primary concept page 61	Taught
T7C03	Chapter 4 · T7C · primary concept page 61	Taught
T7C04	Chapter 4 · T7C · primary concept page 61	Taught
T7C05	Chapter 4 · T7C · primary concept page 61	Taught
T7C06	Chapter 4 · T7C · primary concept page 61	Taught
T7C07	Chapter 4 · T7C · primary concept page 61	Taught
T7C08	Chapter 4 · T7C · primary concept page 61	Taught
T7C09	Chapter 4 · T7C · primary concept page 61	Taught
T7C10	Chapter 4 · T7C · primary concept page 61	Taught
T7C11	Chapter 4 · T7C · primary concept page 61	Taught
T7D01	Chapter 3 · T7D · primary concept page 51	Taught
T7D02	Chapter 3 · T7D · primary concept page 51	Taught

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T7D03	Chapter 3 · T7D · primary concept page 51	Taught
T7D04	Chapter 3 · T7D · primary concept page 51	Taught
T7D05	Chapter 3 · T7D · primary concept page 51	Taught
T7D06	Chapter 3 · T7D · primary concept page 51	Taught
T7D07	Chapter 3 · T7D · primary concept page 51	Taught
T7D08	Chapter 3 · T7D · primary concept page 51	Taught
T7D09	Chapter 3 · T7D · primary concept page 51	Taught
T7D10	Chapter 3 · T7D · primary concept page 51	Taught
T7D11	Chapter 3 · T7D · primary concept page 51	Taught
T8A01	Chapter 7 · T8A · primary concept page 91	Direct Recall
T8A02	Chapter 7 · T8A · primary concept page 91	Direct Recall
T8A03	Chapter 7 · T8A · primary concept page 91	Direct Recall
T8A04	Chapter 7 · T8A · primary concept page 91	Direct Recall
T8A05	Chapter 7 · T8A · primary concept page 91	Direct Recall
T8A06	Chapter 7 · T8A · primary concept page 91	Direct Recall
T8A07	Chapter 7 · T8A · primary concept page 91	Direct Recall
T8A08	Chapter 7 · T8A · primary concept page 91	Direct Recall
T8A09	Chapter 7 · T8A · primary concept page 91	Direct Recall
T8A10	Chapter 7 · T8A · primary concept page 91	Direct Recall
T8A11	Chapter 7 · T8A · primary concept page 91	Direct Recall
T8A12	Chapter 7 · T8A · primary concept page 91	Direct Recall

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T8B01	Chapter 7 · T8B · primary concept page 95	Direct Recall
T8B02	Chapter 7 · T8B · primary concept page 95	Direct Recall
T8B03	Chapter 7 · T8B · primary concept page 95	Direct Recall
T8B04	Chapter 7 · T8B · primary concept page 95	Direct Recall
T8B05	Chapter 7 · T8B · primary concept page 95	Direct Recall
T8B06	Chapter 7 · T8B · primary concept page 95	Direct Recall
T8B07	Chapter 7 · T8B · primary concept page 95	Direct Recall
T8B08	Chapter 7 · T8B · primary concept page 95	Direct Recall
T8B09	Chapter 7 · T8B · primary concept page 95	Direct Recall
T8B10	Chapter 7 · T8B · primary concept page 95	Direct Recall
T8B11	Chapter 7 · T8B · primary concept page 95	Direct Recall
T8B12	Chapter 7 · T8B · primary concept page 95	Direct Recall
T8C01	Chapter 7 · T8C · primary concept page 99	Direct Recall
T8C02	Chapter 7 · T8C · primary concept page 99	Direct Recall
T8C03	Chapter 7 · T8C · primary concept page 99	Direct Recall
T8C04	Chapter 7 · T8C · primary concept page 99	Direct Recall
T8C05	Chapter 7 · T8C · primary concept page 99	Direct Recall
T8C06	Chapter 7 · T8C · primary concept page 99	Direct Recall
T8C07	Chapter 7 · T8C · primary concept page 99	Direct Recall
T8C08	Chapter 7 · T8C · primary concept page 99	Direct Recall
T8C09	Chapter 7 · T8C · primary concept page 99	Direct Recall

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T8C10	Chapter 7 · T8C · primary concept page 99	Direct Recall
T8C11	Chapter 7 · T8C · primary concept page 99	Direct Recall
T8D01	Chapter 7 · T8D · primary concept page 103	Direct Recall
T8D02	Chapter 7 · T8D · primary concept page 103	Direct Recall
T8D03	Chapter 7 · T8D · primary concept page 103	Direct Recall
T8D04	Chapter 7 · T8D · primary concept page 103	Direct Recall
T8D05	Chapter 7 · T8D · primary concept page 103	Direct Recall
T8D06	Chapter 7 · T8D · primary concept page 103	Direct Recall
T8D07	Chapter 7 · T8D · primary concept page 103	Direct Recall
T8D08	Chapter 7 · T8D · primary concept page 103	Direct Recall
T8D09	Chapter 7 · T8D · primary concept page 103	Direct Recall
T8D10	Chapter 7 · T8D · primary concept page 103	Direct Recall
T8D11	Chapter 7 · T8D · primary concept page 103	Direct Recall
T8D12	Chapter 7 · T8D · primary concept page 103	Direct Recall
T9A01	Chapter 6 · T9A · primary concept page 81	Taught
T9A02	Chapter 6 · T9A · primary concept page 81	Taught
T9A03	Chapter 6 · T9A · primary concept page 81	Taught
T9A04	Chapter 6 · T9A · primary concept page 81	Taught
T9A05	Chapter 6 · T9A · primary concept page 81	Taught
T9A06	Chapter 6 · T9A · primary concept page 81	Taught
T9A07	Chapter 6 · T9A · primary concept page 81	Taught

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T9A08	Chapter 6 · T9A · primary concept page 81	Taught
T9A09	Chapter 6 · T9A · primary concept page 81	Taught
T9A10	Chapter 6 · T9A · primary concept page 81	Taught
T9A11	Chapter 6 · T9A · primary concept page 81	Taught
T9B01	Chapter 6 · T9B · primary concept page 85	Taught
T9B02	Chapter 6 · T9B · primary concept page 85	Taught
T9B03	Chapter 6 · T9B · primary concept page 85	Taught
T9B04	Chapter 6 · T9B · primary concept page 85	Taught
T9B05	Chapter 6 · T9B · primary concept page 85	Taught
T9B06	Chapter 6 · T9B · primary concept page 85	Taught
T9B07	Chapter 6 · T9B · primary concept page 85	Taught
T9B08	Chapter 6 · T9B · primary concept page 85	Taught
T9B09	Chapter 6 · T9B · primary concept page 85	Taught
T9B10	Chapter 6 · T9B · primary concept page 85	Taught
T9B11	Chapter 6 · T9B · primary concept page 85	Taught
T9B12	Chapter 6 · T9B · primary concept page 85	Taught
T0A01	Chapter 11 · T0A · primary concept page 159	Direct Recall
T0A02	Chapter 11 · T0A · primary concept page 159	Direct Recall
T0A03	Chapter 11 · T0A · primary concept page 159	Direct Recall
T0A04	Chapter 11 · T0A · primary concept page 159	Direct Recall
T0A05	Chapter 11 · T0A · primary concept page 159	Direct Recall

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T0A06	Chapter 11 · T0A · primary concept page 159	Direct Recall
T0A07	Chapter 11 · T0A · primary concept page 159	Direct Recall
T0A08	Chapter 11 · T0A · primary concept page 159	Direct Recall
T0A09	Chapter 11 · T0A · primary concept page 159	Direct Recall
T0A10	Chapter 11 · T0A · primary concept page 159	Direct Recall
T0A11	Chapter 11 · T0A · primary concept page 159	Direct Recall
T0A12	Chapter 11 · T0A · primary concept page 159	Direct Recall
T0B01	Chapter 11 · T0B · primary concept page 163	Direct Recall
T0B02	Chapter 11 · T0B · primary concept page 163	Direct Recall
T0B03	Chapter 11 · T0B · primary concept page 163	Direct Recall
T0B04	Chapter 11 · T0B · primary concept page 163	Direct Recall
T0B05	Chapter 11 · T0B · primary concept page 163	Direct Recall
T0B06	Chapter 11 · T0B · primary concept page 163	Direct Recall
T0B07	Chapter 11 · T0B · primary concept page 163	Direct Recall
T0B08	Chapter 11 · T0B · primary concept page 163	Direct Recall
T0B09	Chapter 11 · T0B · primary concept page 163	Direct Recall
T0B10	Chapter 11 · T0B · primary concept page 163	Direct Recall
T0B11	Chapter 11 · T0B · primary concept page 163	Direct Recall
T0C01	Chapter 11 · T0C · primary concept page 167	Direct Recall
T0C02	Chapter 11 · T0C · primary concept page 167	Direct Recall
T0C03	Chapter 11 · T0C · primary concept page 167	Direct Recall

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T0C04	Chapter 11 · T0C · primary concept page 167	Direct Recall
T0C05	Chapter 11 · T0C · primary concept page 167	Direct Recall
T0C06	Chapter 11 · T0C · primary concept page 167	Direct Recall
T0C07	Chapter 11 · T0C · primary concept page 167	Direct Recall
T0C08	Chapter 11 · T0C · primary concept page 167	Direct Recall
T0C09	Chapter 11 · T0C · primary concept page 167	Direct Recall
T0C10	Chapter 11 · T0C · primary concept page 167	Direct Recall
T0C11	Chapter 11 · T0C · primary concept page 167	Direct Recall
T0C12	Chapter 11 · T0C · primary concept page 167	Direct Recall
T0C13	Chapter 11 · T0C · primary concept page 167	Direct Recall

Source hierarchy

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Essential glossary

AC: current that reverses direction. DC: current in one direction. Bandwidth: occupied frequency range. Feed line: cable or line carrying RF. Impedance: opposition to AC. Modulation: placing information on a carrier. Repeater: station that receives and retransmits. SWR: indication of load-to-line match.

Abbreviations

AGC automatic gain control · APRS Automatic Packet Reporting System · CTCSS continuous tone-coded squelch system · CW continuous wave/Morse · DMR Digital Mobile Radio · DTMF dual-tone multi-frequency · FCC Federal Communications Commission · RF radio frequency · RIT receiver incremental tuning · ULS Universal Licensing System.

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